

Multidimensional Models

The time series models and stochastic processes discussed in previous chapters consisted of real-valued random variables indexed by a subset of the real line $\mathbb{R}$. A number of applications require self-similar or long-range dependent *multidimensional* models indexed by a finite-dimensional space $\mathbb{R}^q$ or $\mathbb{Z}^q$, and taking values in a finite-dimensional space $\mathbb{R}^p$. That is, random processes X such that for fixed $\omega \in \Omega$,

$$\begin{array}{ccc} X(\omega, \cdot) : \mathbb{R}^q & \rightarrow & \mathbb{R}^p \\ t & \mapsto & X(\omega, t) \end{array} \quad \text{or} \quad \begin{array}{ccc} X(\omega, \cdot) : \mathbb{Z}^q & \rightarrow & \mathbb{R}^p \\ t & \mapsto & X(\omega, t). \end{array}$$

We shall discuss such models in this chapter by considering extensions of the notions of self-similarity, fractional Brownian motion and long-range dependence. We shall focus on the cases

- $p \geq 1, q = 1$ (Sections 9.3 and 9.4), and
- $q \geq 1, p = 1$ (Sections 9.5 and 9.6).

We shall use the following terminology when referring to multidimensional models $X = \{X(t)\}$. When the index t belongs to a subset of a finite-dimensional space $\mathbb{R}^q$ or $\mathbb{Z}^q$, we shall refer to X as a *random field* (or a *spatial process*). When the variables $X(t)$ take values in a subset of a finite-dimensional space $\mathbb{R}^p$, we shall refer to X as a *vector process* (*motion*, when X has stationary increments). For example, in Section 9.3 below, we consider (vector) operator fractional Brownian motions which are extensions of fractional Brownian motion to processes taking values in $\mathbb{R}^p$.

We start in Section 9.1 with the fundamental notions behind multidimensional models. In Section 9.2, we define self-similarity in the multidimensional setting $q \geq 1, p \geq 1$, through the notion of operator self-similarity. Sections 9.3 and 9.4 focus on (vector) operator fractional Brownian motions and vector long-range dependence in the case $p \geq 1, q = 1$. We also discuss fractional cointegration. Sections 9.5 and 9.6 concern operator fractional Brownian fields and spatial long-range dependence in the case $q \geq 1, p = 1$. We thus consider the vector and spatial contexts separately. This is done in part for clarity, in part to highlight the differences between the two contexts, and in part because multidimensional models are not as well developed as one-dimensional models. Moreover, the two contexts are often considered separately in applications.

As noted above, we extend in this chapter the notion of self-similarity and fractional Brownian motion to both the vector and spatial settings. These extensions concern non-stationary processes. We also consider multivariate stationary time series with long-range dependence. As noted in Section 9.6, it is possible to provide a general notion of vector

long-range dependence (see Section 9.4), but in the spatial context, there is no corresponding general definition, only particular cases. Of special importance are the isotropic case and some forms of anisotropic cases, described in Sections 9.6.1 and 9.6.2. One way to describe some of these stationary spatial time series is to use spherical harmonics.

For the convenience of the reader, we now list the various models considered.

In the case of *vector processes* ($p \geq 1, q = 1$):

- Vector operator FBM: vector *OFBM* (Section 9.3)
- Time reversible vector OFBM (Theorem 9.3.11)
- Vector FBM: *VFBM* (Section 9.3.3)
- Single-parameter vector OFBM (Example 9.3.23)
- Vector FARIMA(0, D , 0) (Section 9.4.2)
- Vector FGN (Section 9.4.3)

In the case of *random fields* ($q \geq 1, p = 1$):

- Operator fractional Brownian field: *OBF* (Section 9.5)
- Isotropic fractional Brownian field: *isotropic FBF* (Example 9.5.18)
- Anisotropic fractional Brownian field: *anisotropic FBF* (Example 9.5.19)
- Fractional Brownian sheet (Example 9.5.20)
- Spatial LRD fields with explicit autocovariances (Example 9.6.4)
- Spatial FARIMA (Example 9.6.5)

9.1 Fundamentals of Multidimensional Models

We recall here some fundamental notions behind multidimensional models, starting with basic facts for matrices (Section 9.1.1) and then distinguishing between models in the vector case (Section 9.1.2) and the spatial case (Section 9.1.3).

9.1.1 Basics of Matrix Analysis

We gather here some basic notions related to matrices as needed throughout this chapter. Our main reference is the excellent monograph on matrix analysis by Horn and Johnson [482]. The focus is on $p \times p$, $p \geq 1$, square matrices with complex-valued entries: that is, the elements of $\mathbb{C}^{p \times p}$. But the latter also include matrices with real-valued entries: that is, the elements of $\mathbb{R}^{p \times p}$. Matrices will usually be denoted in capitals such as A , B , etc.

For $A = (a_{jk})_{j,k=1,\dots,p} \in \mathbb{C}^{p \times p}$, A^* stands for the *Hermitian transpose* of A ; that is, the matrix $A^* = (\bar{a}_{kj})_{j,k=1,\dots,p}$ with its (j, k) element being $\bar{a}_{kj}$, where the bar indicates complex conjugation. For example,

$$A = \begin{pmatrix} 1 & i \\ 2 & 3 \end{pmatrix}, \quad A^* = \begin{pmatrix} 1 & 2 \\ -i & 3 \end{pmatrix}.$$

This definition naturally extends to non-square matrices, including vectors. For $A \in \mathbb{R}^{p \times p}$, we have only transposition, namely, $A^* = A'$. Similarly, we define $\bar{A} = (\bar{a}_{jk})_{j,k=1,\dots,p}$; that is, the matrix consisting of the complex conjugates of the elements of the matrix A . A matrix

$A \in \mathbb{C}^{p \times p}$ is *Hermitian symmetric* or simply *Hermitian* if $A^* = A$. For example, the 2×2 matrix

$$A = \begin{pmatrix} 1 & i \\ -i & 0 \end{pmatrix}$$

is Hermitian symmetric since $\bar{i} = -i$ and the diagonal entries 1 and 0 are real. A diagonal matrix will be denoted $\text{diag}(a_1, \dots, a_p)$ where $a_j \in \mathbb{C}$. In some instances, the a_j s will be square matrices themselves and the matrix $\text{diag}(a_1, \dots, a_p)$ will be block diagonal.

The matrix $A \in \mathbb{C}^{p \times p}$ is nonsingular if its inverse A^{-1} exists, or equivalently if its rows (or columns) are linearly independent. It is singular otherwise.

An *eigenvalue* $\eta \in \mathbb{C}$ of a matrix $A \in \mathbb{C}^{p \times p}$ satisfies $Au = \eta u$ for a (nonzero) *eigenvector* $u \in \mathbb{C}^p$. A matrix A has at most p distinct eigenvalues. By saying that the eigenvalues of A are $\eta_1, \dots, \eta_p$, we shall imply that these eigenvalues may not be all distinct. They appear in the so-called characteristic polynomial

$$\det(A - \eta I) = \prod_{k=1}^p (\eta_k - \eta),$$

and are obtained by setting $\det(A - \eta I) = 0$. We shall also use the so-called Jordan decomposition of $A \in \mathbb{C}^{p \times p}$ having the form

$$A = PJP^{-1}, \quad (9.1.1)$$

where P is a nonsingular $p \times p$ matrix, $J = \text{diag}(J_{\lambda_1}, \dots, J_{\lambda_N})$ is in Jordan canonical form with the Jordan blocks $J_{\lambda_1}, \dots, J_{\lambda_N}$ on the diagonal and $\lambda_1, \dots, \lambda_N$ are (possibly repeated) eigenvalues of A . A Jordan block J_λ is a matrix of the form

$$J_\lambda = \begin{pmatrix} \lambda & 0 & 0 & \dots & 0 \\ 1 & \lambda & 0 & \dots & 0 \\ 0 & 1 & \lambda & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 & \lambda \end{pmatrix}. \quad (9.1.2)$$

The matrix P is not unique but the Jordan block matrix J_λ is unique up to the ordering of the blocks.

A matrix $A \in \mathbb{R}^{p \times p}$ with real-valued entries can have both complex eigenvalues and complex eigenvectors. An Hermitian symmetric matrix A has real eigenvalues only. If the eigenvalues are nonnegative (positive, resp.), the Hermitian symmetric matrix A is called *positive semidefinite* (*positive definite*, resp.). In fact, positive semidefiniteness (positive definiteness, resp.) is well known to be equivalent to $u^*Au \geq 0$; that is, u^*Au is real and nonnegative for any $u \in \mathbb{C}$ ($u^*Au > 0$ for any nonzero $u \in \mathbb{C}$, resp.). For a positive semidefinite Hermitian symmetric matrix A , there is a unique positive semidefinite Hermitian symmetric matrix B such that $B^2 = A$ (Horn and Johnson [482], Theorem 7.2.6). One denotes $B = A^{1/2}$.

Note that $A = KK^*$ is always positive semidefinite since $u^*KK^*u = (u^*K)(u^*K)^* = |u^*K|^2 \geq 0$.

For $A \in \mathbb{C}^{p \times p}$, its matrix (operator) norm is defined as

$$\|A\| = \max_{\|u\|_2=1} \|Au\|_2, \quad (9.1.3)$$

where for a vector $x = (x_1, \dots, x_p)' \in \mathbb{C}^p$, $\|x\|_2 = (\sum_{j=1}^p |x_j|^2)^{1/2}$ is the usual Euclidean norm.

9.1.2 Vector Setting

In the vector setting and in discrete time, we shall work with vector-valued ($\mathbb{R}^p$ -valued),¹ second-order stationary time series $X = \{X_n\}_{n \in \mathbb{Z}}$. As in Section 2.1, such series have a constant mean vector $\mu_X = \mathbb{E}X_n \in \mathbb{R}^p$ and a $p \times p$ autocovariance matrix function

$$\gamma_X(m-n) = \mathbb{E}X_m X_n' - \mathbb{E}X_m \mathbb{E}X_n' = \mathbb{E}X_m X_n' - \mu_X \mu_X', \quad m, n \in \mathbb{Z}, \quad (9.1.4)$$

which depends only on the distance $m-n$. With $\gamma_X(\cdot) = (\gamma_{X,jk}(\cdot))_{j,k=1,\dots,p}$, $\mu_X = (\mu_{X,1}, \dots, \mu_{X,p})'$ and $X_n = (X_{1,n}, \dots, X_{p,n})'$, the last expression can be written component-wise as

$$\gamma_{X,jk}(m-n) = \mathbb{E}X_{j,m} X_{k,n} - \mathbb{E}X_{j,m} \mathbb{E}X_{k,n} = \mathbb{E}X_{j,m} X_{k,n} - \mu_{X,j} \mu_{X,k}, \quad j, k = 1, \dots, p.$$

Note that, by the second-order stationarity, the matrix $\gamma_X(n)$ satisfies

$$\gamma_X(n) = \mathbb{E}X_n X_0' - \mu_X \mu_X' = \mathbb{E}X_0 X_{-n}' - \mu_X \mu_X' = (\mathbb{E}X_{-n} X_0' - \mu_X \mu_X')' = \gamma_X(-n)', \quad (9.1.5)$$

and thus $\gamma_{X,jk}(n) = \gamma_{X,kj}(-n)$. The third equality in (9.1.5) follows by using transposition twice. The cross-covariances $\gamma_{X,jk}(n)$ need not be equal to $\gamma_{X,jk}(-n)$ when $j \neq k$. Therefore, in contrast to the univariate case, it is *not* true in general that

$$\gamma_X(n) = \gamma_X(-n), \quad n \in \mathbb{Z}, \quad (9.1.6)$$

or, equivalently, by using (9.1.5), that

$$\gamma_X(n) = \gamma_X(n)', \quad n \in \mathbb{Z}, \quad (9.1.7)$$

When (9.1.6)–(9.1.7) hold, the time series X is called *time reversible* (cf. (9.3.34)).

The spectral density $f_X(\lambda) = (f_{X,jk}(\lambda))_{j,k=1,\dots,p}$, $\lambda \in (-\pi, \pi]$, is now matrix-valued, if it exists. It satisfies

$$\int_{-\pi}^{\pi} e^{in\lambda} f_X(\lambda) d\lambda = \gamma_X(n), \quad n \in \mathbb{Z}. \quad (9.1.8)$$

While in the univariate case, the spectral density $f_X(\lambda)$ is real and in fact nonnegative, this is not the case anymore in the vector case. Since (9.1.6) may not hold, $f_X(\lambda)$ is, in general, complex-valued. For fixed $\lambda \in (-\pi, \pi]$, the matrix $f_X(\lambda)$ is Hermitian symmetric and positive semidefinite. The function $f_X(\lambda)$, $\lambda \in (-\pi, \pi]$, is Hermitian; that is,

$$f_X(-\lambda) = \overline{f_X(\lambda)}, \quad \lambda \in (-\pi, \pi].$$

¹ Since a $r \times c$ matrix has r rows and c columns, each vector X_n is a $p \times 1$ matrix and its transpose X_n' is a $1 \times p$ matrix.

The spectral representation of the process X is now written as $X_n = \int_{(-\pi, \pi]} e^{in\lambda} Z(d\lambda)$ for a vector complex-valued random measure $Z(d\lambda)$ with $\mathbb{E}Z(d\lambda)Z(d\lambda)^* = f_X(\lambda)d\lambda$ and, more generally, $\mathbb{E}Z(d\lambda)Z(d\lambda)^* = F_X(d\lambda)$ where F_X is called the *spectral distribution function* (cf. Section 1.3.4). For example, if $p = 2$,

$$f_X(\lambda) = \begin{pmatrix} f_{X,11}(\lambda) & f_{X,12}(\lambda) \\ f_{X,21}(\lambda) & f_{X,22}(\lambda) \end{pmatrix},$$

where $f_{X,12}(\lambda)$ and $f_{X,21}(\lambda)$ are called the *cross spectrum*. The ratio

$$\mathcal{H}_{12}^2(\lambda) = \frac{|f_{12}(\lambda)|^2}{f_{11}(\lambda)f_{22}(\lambda)}$$

is interpreted as the *coherence* or *squared coherence function*. We have $0 \leq \mathcal{H}_{12}^2(\lambda) \leq 1$ by the Cauchy-Schwarz inequality, and thus a value close to one suggests a strong linear relationship between the components $Z_1(d\lambda)$ and $Z_2(d\lambda)$ of the vector $Z(d\lambda) = (Z_1(d\lambda), Z_2(d\lambda))'$. See Hannan [450], Brillinger [185], Brockwell and Davis [186], Lütkepohl [654] for more information on vector-valued time series X .

Warning: Two definitions of an autocovariance function are used in the literature. One definition is given in (9.1.4). The other definition is to set

$$\tilde{\gamma}_X(m-n) = \mathbb{E}X_n X_m' - \mathbb{E}X_n \mathbb{E}X_m' = \mathbb{E}X_n X_m' - \mu_X \mu_X', \quad m, n \in \mathbb{Z} \quad (9.1.9)$$

(e.g., Hannan [450]). Note that $\tilde{\gamma}_X(m-n)$ is also a $p \times p$ matrix. The connection between (9.1.4) and (9.1.9) is $\tilde{\gamma}_X(n) = \gamma_X(n)' = \gamma_X(-n)$, $n \in \mathbb{Z}$, in view of (9.1.5). Since the two definitions are not equivalent, it is important to be aware of which convention is used in a given source. For example, several works behind Section 9.4 (Kechagias and Pipiras [551], Helgason et al. [463]) use, in fact, the convention (9.1.9) and not (9.1.4). We also note that the choice of the convention (9.1.4) or (9.1.9) has implications on how spectral representations of time series are expressed. If (9.1.8) relates the autocovariance function $\tilde{\gamma}_X$ and the spectral density f_X , the spectral representation of a time series X would be written $X_n = \int_{(-\pi, \pi]} e^{-in\lambda} Z(d\lambda)$ for a vector complex-valued random measure $Z(d\lambda)$ with $\mathbb{E}Z(d\lambda)Z(d\lambda)^* = \tilde{f}_X(\lambda)d\lambda$, since in this case $\tilde{\gamma}_X(m-n) = \mathbb{E}X_n X_m' = \int_{(-\pi, \pi]} e^{i(m-n)\lambda} \tilde{f}_X(\lambda)d\lambda$. On the other hand, under the convention (9.1.4) and as used in this book (see Section 1.3.4 and above), the spectral representation involves the deterministic kernel $e^{in\lambda}$, instead of $e^{-in\lambda}$.

Since $f_{X,jk}(\lambda)$, $j \neq k$, is complex-valued in general, it can be expressed in polar coordinates as

$$f_{X,jk}(\lambda) = \alpha_{X,jk}(\lambda)e^{i\phi_{X,jk}(\lambda)} \quad (9.1.10)$$

for real-valued functions $\alpha_{X,jk}(\lambda)$ and $\phi_{X,jk}(\lambda)$, known respectively as the amplitude spectrum and the phase spectrum of $\{X_{j,n}\}$ and $\{X_{k,n}\}$. In fact, there are two common but slightly different ways to represent the amplitude and phase spectra. On one hand, recall that a complex number $z = z_1 + iz_2 \in \mathbb{C}$, $z_1, z_2 \in \mathbb{R}$, can be represented as

$$z = |z|e^{i\text{Arg}(z)}, \quad (9.1.11)$$

where the so-called principal argument function $\text{Arg}(z) \in (-\pi, \pi]$ by convention. Assume now that $z_1 \neq 0$. Note that

$$\text{Arg}(z) = \arctan\left(\frac{z_2}{z_1}\right) + \pi \text{sign}(z_2)1_{\{z_1 < 0\}},$$

where by $\arctan$ we mean the principal value,² namely, the angle in $(-\pi/2, \pi/2)$. Indeed, if z_1 and z_2 have the same sign, then $\text{Arg}(z) = \arctan(z_2/z_1)$, but if $z_1 < 0$, then Arg and $\arctan$ differ by $+\pi$ if $z_2 > 0$ and by $-\pi$ if $z_2 < 0$. Since $e^0 = 1$ and $e^{\pm i\pi} = -1$, we can then also write

$$\begin{aligned} z &= \sqrt{z_1^2 + z_2^2} \exp \left\{ i \left(\arctan\left(\frac{z_2}{z_1}\right) + \pi \text{sign}(z_2)1_{\{z_1 < 0\}} \right) \right\} \\ &= \text{sign}(z_1) \sqrt{z_1^2 + z_2^2} \exp \left\{ i \arctan\left(\frac{z_2}{z_1}\right) \right\} \\ &= z_1 \sqrt{1 + \frac{z_2^2}{z_1^2}} \exp \left\{ i \arctan\left(\frac{z_2}{z_1}\right) \right\} = z_1 \sqrt{1 + \tan^2 \left(\arctan \frac{z_2}{z_1} \right)} \exp \left\{ i \arctan\left(\frac{z_2}{z_1}\right) \right\} \\ &= z_1 \sqrt{1 + \frac{\sin^2 \phi}{\cos^2 \phi}} \exp \{ i\phi \} = \frac{z_1}{\cos \phi} \exp \{ i\phi \}, \quad \text{with } \phi = \arctan \left(\frac{z_2}{z_1} \right) \in \left(-\frac{\pi}{2}, \frac{\pi}{2} \right). \end{aligned} \quad (9.1.12)$$

We thus get two specifications of the amplitude $\alpha_{X,jk}(\lambda)$ and the phase $\phi_{X,jk}(\lambda)$ of $f_{j,k}(\lambda)$ in (9.1.10). The first corresponds to (9.1.11) (e.g., Brockwell and Davis [186], p. 422), namely,

$$\alpha_{X,jk}(\lambda) = |f_{X,jk}(\lambda)|, \quad \phi_{X,jk}(\lambda) = \text{Arg}(f_{X,jk}(\lambda)). \quad (9.1.13)$$

The second specification is obtained by using (9.1.12) (e.g., Hannan [450], pp. 43–44):

$$\alpha_{X,jk}(\lambda) = \frac{\Re f_{X,jk}(\lambda)}{\cos \phi_{X,jk}(\lambda)}, \quad \phi_{X,jk}(\lambda) = \arctan \frac{\Im f_{X,jk}(\lambda)}{\Re f_{X,jk}(\lambda)}. \quad (9.1.14)$$

Note that, in the case (9.1.13), $\phi_{j,k}(\lambda) \in (-\pi, \pi]$ since this is the range of the Arg function, but in the case (9.1.14), $\phi_{X,jk}(\lambda) \in (-\pi/2, \pi/2)$ since this is the range of the $\arctan$ function.

9.1.3 Spatial Setting

In the spatial and discrete (lattice) setting, we focus on second-order stationary fields $X = \{X_n\}_{n \in \mathbb{Z}^q}$ taking values in $\mathbb{R}$. The autocovariance function of X is defined as

$$\gamma_X(m - n) = \mathbb{E}X_m X_n - \mathbb{E}X_m \mathbb{E}X_n = \mathbb{E}X_m X_n - \mu_X^2, \quad m, n \in \mathbb{Z}^q, \quad (9.1.15)$$

and depend only on $|m - n|$. The spectral density function $f_X(\lambda)$, $\lambda \in (-\pi, \pi]^q$, if it exists, satisfies

$$\int_{(-\pi, \pi]^q} e^{i\langle n, \lambda \rangle} f_X(\lambda) d\lambda = \gamma_X(n), \quad n \in \mathbb{Z}^q, \quad (9.1.16)$$

where $\langle \cdot, \cdot \rangle$ indicates the inner product in $\mathbb{R}^q$.

² The principal value is often denoted Arctan .

Remark 9.1.1 In the spatial setting, it is also common to consider second-order stationary random fields $X = \{X_u\}$ indexed by $u \in \mathbb{R}^q$. Such fields may be more appropriate (than those indexed by $\mathbb{Z}^q$) in modeling spatial phenomenon when data is collected at non-lattice locations. The results and discussion in the spatial setting found below also apply in this context with the following standard difference. The spectral density function $f_X(\lambda)$, if it exists, is now a function of $\lambda \in \mathbb{R}^q$ and satisfies

$$\int_{\mathbb{R}^q} e^{i\langle u, \lambda \rangle} f_X(\lambda) d\lambda = \gamma_X(u), \quad u \in \mathbb{R}^q.$$

See also Remark 1.3.8.

9.2 Operator Self-Similarity

Let us consider for a moment the general multidimensional setting

$$\begin{aligned} X(\omega, \cdot) : \mathbb{R}^q &\rightarrow \mathbb{R}^p \\ t &\mapsto X(\omega, t) \end{aligned}$$

and ask what one means by self-similarity in the multidimensional setting. The following general definition will be useful.

Definition 9.2.1 A vector-valued ($\mathbb{R}^p$ -valued) random field $\{X(t)\}_{t \in \mathbb{R}^q}$ is called *operator self-similar* with matrix exponents $E \in \mathbb{R}^{q \times q}$ and $H \in \mathbb{R}^{p \times p}$ if, for any $c > 0$,

$$\{X(c^E t)\}_{t \in \mathbb{R}^q} \stackrel{d}{=} \{c^H X(t)\}_{t \in \mathbb{R}^q}. \quad (9.2.1)$$

The relation (9.2.1) involves matrix exponentials $c^E = e^{(\log c)E}$ and $c^H = e^{(\log c)H}$ defined as follows. For $c > 0$ and general matrices $A, B \in \mathbb{C}^{m \times m}$, one sets

$$e^A := \sum_{k=0}^{\infty} \frac{A^k}{k!}, \quad (9.2.2)$$

$$c^B := e^{(\log c)B} := \sum_{k=0}^{\infty} \frac{(\log c)^k B^k}{k!}. \quad (9.2.3)$$

Note that e^A and c^B are matrices in $\mathbb{C}^{m \times m}$! Therefore in (9.2.1), c^H is a $p \times p$ matrix and hence $c^H X(t)$ is a $p \times 1$ matrix; that is, an $\mathbb{R}^p$ -valued vector. Similarly, in (9.2.1), c^E is a $q \times q$ matrix, $c^E t$ is an $\mathbb{R}^q$ -valued vector and $X(c^E t)$ is an $\mathbb{R}^p$ -valued vector.

Figure 9.1 illustrates the mapping $S \ni x \mapsto c^A x$ in two dimensions for several choices of A and c , and for the ellipse $S = \{(x_1, x_2)' \in \mathbb{R}^2 : x_1^2 + 4x_2^2 = 1\}$ (the smallest ellipse in the three plots).

It is useful to discuss some basic but key properties of matrix exponentials. For a diagonal matrix $A = \text{diag}(a_1, \dots, a_m) \in \mathbb{C}^{m \times m}$, we have

$$c^A = \text{diag}(c^{a_1}, \dots, c^{a_m}).$$

In particular, if A is the identity matrix I , then c^A is the diagonal matrix cI . If

$$A = W A_0 W^{-1}$$

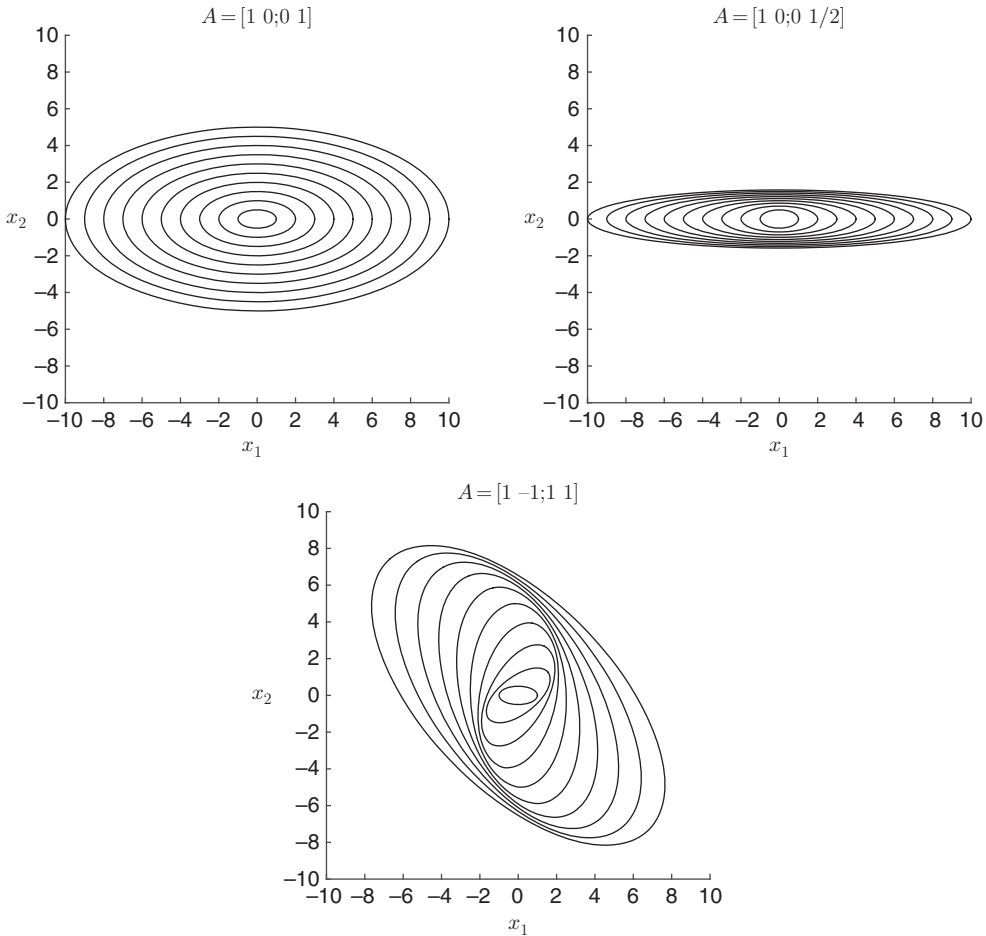


Figure 9.1 The mapping $S \ni x \mapsto c^A x$ in two dimensions for $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (left), $A = \begin{bmatrix} 1 & 0 \\ 0 & 1/2 \end{bmatrix}$ (right) and $A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$ (bottom), $c = 1, 2, \dots, 10$, and for the ellipse $S = \{(x_1, x_2)' \in \mathbb{R}^2 : x_1^2 + 4x_2^2 = 1\}$ (the smallest ellipse in the three plots).

for some invertible matrix W (with the inverse W^{-1}), then

$$\begin{aligned}
 c^A &= \sum_{k=0}^{\infty} \frac{(\log c)^k A^k}{k!} = \sum_{k=0}^{\infty} \frac{(\log c)^k (W A_0 W^{-1})^k}{k!} \\
 &= \sum_{k=0}^{\infty} \frac{(\log c)^k (W A_0 W^{-1})(W A_0 W^{-1}) \dots (W A_0 W^{-1})}{k!} \\
 &= \sum_{k=0}^{\infty} \frac{(\log c)^k W A_0^k W^{-1}}{k!} = W c^{A_0} W^{-1}.
 \end{aligned} \tag{9.2.4}$$

In particular, if $A_0 = \text{diag}(a_{0,1}, \dots, a_{0,m}) \in \mathbb{C}^{m \times m}$ and thus A is diagonalizable over $\mathbb{C}$, then

$$c^A = W \operatorname{diag}(c^{a_{0,1}}, \dots, c^{a_{0,m}}) W^{-1}.$$

It is *not* true in general that

$$c^{A+B} = c^A c^B = c^B c^A. \quad (9.2.5)$$

These relations hold, however, when A and B commute; that is, $AB = BA$.

To shed light on the operator self-similarity relation (9.2.1), suppose that E and H are diagonalizable over $\mathbb{R}$ with

$$E = W_E E_0 W_E^{-1}, \quad H = W_H H_0 W_H^{-1}, \quad (9.2.6)$$

where $W_E \in \mathbb{R}^{q \times q}$, $W_H \in \mathbb{R}^{p \times p}$, $E_0 = \operatorname{diag}(e_{0,1}, \dots, e_{0,q}) \in \mathbb{R}^{q \times q}$ and $H_0 = \operatorname{diag}(h_{0,1}, \dots, h_{0,p}) \in \mathbb{R}^{p \times p}$. One special case of (9.2.6) is when, for example, E is symmetric and W_E is orthogonal (that is, $W_E^{-1} = W_E'$). If (9.2.6) holds, by using (9.2.4), the relation (9.2.1) can be expressed as

$$\{X(W_E c^{E_0} W_E^{-1} t)\}_{t \in \mathbb{R}^q} \stackrel{d}{=} \{W_H c^{H_0} W_H^{-1} X(t)\}_{t \in \mathbb{R}^q}$$

or

$$\{W_H^{-1} X(W_E c^{E_0} W_E^{-1} t)\}_{t \in \mathbb{R}^q} \stackrel{d}{=} \{c^{H_0} W_H^{-1} X(t)\}_{t \in \mathbb{R}^q} \stackrel{d}{=} \{c^{H_0} W_H^{-1} X(W_E W_E^{-1} t)\}_{t \in \mathbb{R}^q}$$

or, after setting $s = W_E^{-1} t$,

$$\{W_H^{-1} X(W_E c^{E_0} s)\}_{s \in \mathbb{R}^q} \stackrel{d}{=} \{c^{H_0} W_H^{-1} X(W_E s)\}_{s \in \mathbb{R}^q}. \quad (9.2.7)$$

Thus, letting

$$Y(s) = W_H^{-1} X(W_E s), \quad s \in \mathbb{R}^q, \quad (9.2.8)$$

we get that

$$\{Y(c^{E_0} s)\}_{s \in \mathbb{R}^q} \stackrel{d}{=} \{c^{H_0} Y(s)\}_{s \in \mathbb{R}^q}. \quad (9.2.9)$$

In an expanded form, denoting $s = (s_1, \dots, s_q)'$ and $Y(s) = (Y_1(s), \dots, Y_p(s))'$, this can be expressed as

$$\left\{ \begin{pmatrix} Y_1 \begin{pmatrix} c^{e_{0,1}} s_1 \\ \vdots \\ c^{e_{0,q}} s_q \end{pmatrix} \\ \vdots \\ Y_p \begin{pmatrix} c^{e_{0,1}} s_1 \\ \vdots \\ c^{e_{0,q}} s_q \end{pmatrix} \end{pmatrix} \right\}_{s \in \mathbb{R}^q} \stackrel{d}{=} \left\{ \begin{pmatrix} c^{h_{0,1}} Y_1 \begin{pmatrix} s_1 \\ \vdots \\ s_q \end{pmatrix} \\ \vdots \\ c^{h_{0,p}} Y_p \begin{pmatrix} s_1 \\ \vdots \\ s_q \end{pmatrix} \end{pmatrix} \right\}_{s \in \mathbb{R}^q}. \quad (9.2.10)$$

- For vector processes (that is, $q = 1$), the relation (9.2.10) can be written as

$$\left\{ \begin{pmatrix} Y_1(c^{e_0} s) \\ \vdots \\ Y_p(c^{e_0} s) \end{pmatrix} \right\}_{s \in \mathbb{R}} \stackrel{d}{=} \left\{ \begin{pmatrix} c^{h_{0,1}} Y_1(s) \\ \vdots \\ c^{h_{0,p}} Y_p(s) \end{pmatrix} \right\}_{s \in \mathbb{R}}$$

or, supposing $e_0 \neq 0$,

$$\left\{ \begin{pmatrix} Y_1(cs) \\ \vdots \\ Y_p(cs) \end{pmatrix} \right\}_{s \in \mathbb{R}} \stackrel{d}{=} \left\{ \begin{pmatrix} c^{h_{0,1}/e_0} Y_1(s) \\ \vdots \\ c^{h_{0,p}/e_0} Y_p(s) \end{pmatrix} \right\}_{s \in \mathbb{R}}. \quad (9.2.11)$$

- For random fields (that is, $p = 1$), the relation (9.2.10) can be written as

$$\left\{ Y \begin{pmatrix} c^{e_{0,1}} s_1 \\ \vdots \\ c^{e_{0,q}} s_q \end{pmatrix} \right\}_{s \in \mathbb{R}^q} \stackrel{d}{=} \left\{ c^{h_0} Y \begin{pmatrix} s_1 \\ \vdots \\ s_q \end{pmatrix} \right\}_{s \in \mathbb{R}^q}$$

or, supposing $h_0 \neq 0$,

$$\left\{ Y \begin{pmatrix} c^{e_{0,1}/h_0} s_1 \\ \vdots \\ c^{e_{0,q}/h_0} s_q \end{pmatrix} \right\}_{s \in \mathbb{R}^q} \stackrel{d}{=} \left\{ c Y \begin{pmatrix} s_1 \\ \vdots \\ s_q \end{pmatrix} \right\}_{s \in \mathbb{R}^q}. \quad (9.2.12)$$

Relations (9.2.10)–(9.2.12) resemble the one-dimensional self-similarity relation (2.5.1). For example, (9.2.11) shows that each one-dimensional component process $Y_j(s)$ is self-similar in the sense of (2.5.1) with the scalar self-similarity parameter $H_j = h_{0,j}/e_0$. The self-similarity parameter may be different for different values of j . In (9.2.12), the self-similarity of the field Y holds even if it scales differently across the one-dimensional parameters s_j with self-similarity parameters $H_j = h_0/e_{0,j}$, $j = 1, \dots, q$.

Finally, note that (9.2.8) corresponds to changing the coordinate systems in $\mathbb{R}^q$ and $\mathbb{R}^p$. Thus, after this change of coordinate systems, the process X becomes the process Y , and scales entry-wise.

9.3 Vector Operator Fractional Brownian Motions

We consider here extensions of FBM to the vector setting $p \geq 1$, $q = 1$. FBM was introduced in Definition 2.6.2 as a self-similar Gaussian process with stationary increments and $0 < H \leq 1$. We adopt an analogous definition in the vector setting, replacing self-similarity by operator self-similarity.

Definition 9.3.1 A vector-valued ($\mathbb{R}^p$ -valued) process $\{X(t)\}_{t \in \mathbb{R}}$ is called *vector operator fractional Brownian motion* (vector OFBM) if it is Gaussian, has stationary increments and is operator self-similar.³

Stationarity of increments in Definition 9.3.1 means as in (2.5.2) that, for any $h \in \mathbb{R}$,

$$\{X(t+h) - X(h)\}_{t \in \mathbb{R}} \stackrel{d}{=} \{X(t) - X(0)\}_{t \in \mathbb{R}}.$$

For the definition of *operator self-similarity*, we shall take: for any $c > 0$,

$$\{X(ct)\}_{t \in \mathbb{R}} \stackrel{d}{=} \{c^H X(t)\}_{t \in \mathbb{R}}, \quad (9.3.1)$$

³ Vector OFBM is often referred to simply as OFBM. We use the term *vector* OFBM to emphasize its difference from the operator fractional Brownian fields introduced below in Section 9.5.

that is, the relationship (9.2.1) with $E = 1$ and $H \in \mathbb{R}^{p \times p}$. We shall continue denoting a vector OFBM by $\{B_H(t)\}_{t \in \mathbb{R}}$. The properties of the matrix H will be described below.

Assumption We assume that a vector OFBM is *proper* in the sense that, for each $t \neq 0$, the distribution of $B_H(t)$ is not contained in a proper subspace of $\mathbb{R}^p$, and that it is continuous in distribution at all t .

In what follows, we will present integral representations, consider several special subclasses and discuss basic identifiability questions for vector OFBMs.

9.3.1 Integral Representations

The following result provides spectral domain representations of vector OFBMs. Compare it with the spectral representation when $p = 1$, given in Proposition 2.6.11.

Theorem 9.3.2 Let $\{B_H(t)\}_{t \in \mathbb{R}}$ be a vector OFBM with exponent $H \in \mathbb{R}^{p \times p}$. Suppose that the eigenvalues h_k of H satisfy

$$0 < \Re(h_k) < 1, \quad k = 1, \dots, p. \quad (9.3.2)$$

Then, $\{B_H(t)\}_{t \in \mathbb{R}}$ admits the integral representation

$$\{B_H(t)\}_{t \in \mathbb{R}} \stackrel{d}{=} \left\{ \int_{\mathbb{R}} \frac{e^{itx} - 1}{ix} (x_+^{-D} A + x_-^{-D} \bar{A}) \tilde{B}(dx) \right\}_{t \in \mathbb{R}} \quad (9.3.3)$$

for some $A \in \mathbb{C}^{p \times p}$. Here, the bar denotes component-wise complex conjugate,

$$D = H - \frac{1}{2}I \in \mathbb{R}^{p \times p} \quad (9.3.4)$$

and $\tilde{B}(dx)$ is a vector random measure consisting of independent components $\tilde{B}_k(dx)$, $k = 1, \dots, p$, which are Hermitian Gaussian random measures with identical control measures $\mathbb{E}|\tilde{B}_k(dx)|^2 = dx$.

Proof For notational simplicity, set $X = B_H$. Since X has stationary increments, we have

$$X(t) - X(s) = \int_{\mathbb{R}} \frac{e^{itx} - e^{isx}}{ix} \tilde{Y}(dx), \quad (9.3.5)$$

$\tilde{Y}(dx)$ is an orthogonal-increment⁴ random measure with values in $\mathbb{C}^p$. The relation (9.3.5) can be proved following the approach for the one-dimensional case found in Doob [325], p. 550, under the assumption that $\mathbb{E}|X(t+h) - X(t)|^2 \rightarrow 0$ as $h \rightarrow 0$, i.e., X is $L^2(\Omega)$ -continuous at every t (see also Yaglom [1021], p. 409, and Yaglom [1020], Theorem 7). The latter assumption is satisfied in our context because of the following. Property 2.1 in Maejima and Mason [667] states that, for an operator self-similar process Z with exponent H , if $\inf\{\Re(h_k) : k = 1, \dots, p\} > 0$, then $Z(0) = 0$ a.s. Thus, in view of (9.3.2), $X(0) = 0$ a.s. So, by stationarity of the increments,

$$X(t+h) - X(t) \stackrel{d}{=} X(h) \xrightarrow{d} X(0) = 0, \quad h \rightarrow 0, \quad (9.3.6)$$

⁴ A random measure $\tilde{Y}$ has orthogonal increments if $\mathbb{E}\tilde{Y}(E_1)\tilde{Y}(E_2)^* = 0$ whenever $E_1 \cap E_2 = \emptyset$. Recall that $*$ denotes the Hermitian transpose.

and hence in $L^2(\Omega)$ since X is Gaussian. The last convergence in (9.3.6) either follows from the assumption of the continuity in distribution made following Definition 9.3.1 or can be proved directly under the assumption (9.3.2) as in Property 2.1 in Maejima and Mason [667]. This proves (9.3.5). In particular,

$$X(t) = \int_{\mathbb{R}} \frac{e^{itx} - 1}{ix} \tilde{Y}(dx). \quad (9.3.7)$$

Now let

$$F_X(dx) = \mathbb{E} \tilde{Y}(dx) \tilde{Y}(dx)^* \quad (9.3.8)$$

be the vector spectral distribution of $\tilde{Y}(dx)$ and note that it is a $p \times p$ matrix.⁵

The rest of the proof is in three steps:

1. show the existence of a spectral density function $f_X(x) = F_X(dx)/dx$,
2. decorrelate the measure $\tilde{Y}(dx)$ component-wise by finding a filter based upon the spectral density function,
3. develop the form of the filter.

Step 1: Since X is operator self-similar with exponent H ,

$$\{X(ct)\} \stackrel{d}{=} \left\{ c^H \int_{\mathbb{R}} \frac{e^{itx} - 1}{ix} \tilde{Y}(dx) \right\}, \quad (9.3.9)$$

for $c > 0$. On the other hand, through a change of variables $x = c^{-1}v$ in (9.3.7),

$$X(ct) = \int_{\mathbb{R}} \frac{e^{itv} - 1}{iv} c \tilde{Y}(c^{-1}dv). \quad (9.3.10)$$

The relations (9.3.9) and (9.3.10) provide two spectral representations for the process $\{X(ct)\}_{t \in \mathbb{R}}$. As a consequence of the uniqueness of the spectral distribution function of the stationary process $\{X(t) - X(t-1)\}_{t \in \mathbb{R}}$ and of the fact that $|(e^{ix} - 1)/(ix)|^2 > 0$, $x \in \mathbb{R} \setminus \{2\pi k, k \in \mathbb{Z}\}$, we obtain, in view of (9.3.8), that

$$c^H F_X(dx) c^{H*} = c^2 F_X(c^{-1}dx), \quad c > 0, \quad (9.3.11)$$

which can be written, by a change of variables, $c^H F_X(cdx) c^{H*} = c F_X(dx) c$ or $F_X(cdx) = c^{I-H} F_X(dx) c^{(I-H)*}$. Thus, for $c > 0$,

$$\int_{(0,1]} F_X(cdx) = F_X(0, c] = c^{I-H} F_X(0, 1] c^{(I-H)*}, \quad (9.3.12)$$

$$\int_{(-1,0]} F_X(cdx) = F_X(-c, 0] = c^{I-H} F_X(-1, 0] c^{(I-H)*}. \quad (9.3.13)$$

One can write $I - H$ in its Jordan decomposition form as $I - H = PJP^{-1}$ $P \in \mathbb{C}^{p \times p}$ is invertible and $J = \text{diag}(J_{1-h_{i_1}}, \dots, J_{1-h_{i_s}})$ consists of Jordan blocks J_λ , defined in (9.1.2), and $i_1, \dots, i_s$ is a set of indices. Then, $c^{I-H} = P c^J P^{-1} = P \text{diag}(c^{J_{1-h_{i_1}}}, \dots, c^{J_{1-h_{i_s}}}) P^{-1}$. By the explicit formula for c^{J_λ} in Exercise 9.1, each individual entry $F_X(0, c]_{jk}$, $j, k =$

⁵ Note also that we use the notation F_X and not F_Y .

$1, \dots, p$, in the expression on the right-hand side of (9.3.12) is a linear combination (with complex weights) of terms of the form

$$\frac{(\log c)^m}{m!} c^{1-\bar{h}_r} \frac{(\log c)^n}{n!} c^{1-h_s}, \quad r, s = 1, \dots, p, \quad m, n = 0, \dots, p-1. \quad (9.3.14)$$

By (9.3.2), the eigenvalues h_k of H satisfy $0 < \Re(h_k) < 1$. Since (9.3.14) are differentiable in c over $(0, \infty)$, so are $F_X(0, c]_{jk}$. Hence, $F_X(c)$ is differentiable in c over $(0, \infty)$ as well, since $F_X(0, c]_{jk} = F_X(c)_{jk} - F_X(0)_{jk}$. The differentiability of F_X on $(-\infty, 0)$ follows from (9.3.13) and an analogous argument.

The above proves that F_X is differentiable and hence absolutely continuous on $\mathbb{R} \setminus \{0\}$. The function F_X will also be absolutely continuous on $\mathbb{R}$ if it is continuous at zero. Note that

$$F_X(-c, c] = c^{I-H} F_X(-1, 1] c^{(I-H)^*} \rightarrow 0,$$

as $c \rightarrow 0^+$. The limit holds because $\|c^{I-H}\| \rightarrow 0$, as $c \rightarrow 0^+$, $\|\cdot\|$ is the matrix norm defined in (9.1.3), which in turn follows from Proposition 2.1, (ii), in Maejima and Mason [667] under the assumption that $\Re(h_k) < 1$, $k = 1, \dots, p$.

Step 2: Denote the spectral density of X by f_X . Since $|(1 - e^{-ix})/(ix)|^2 f_X(x)$ is the spectral density of the stationary process $\{X(t) - X(t-1)\}_{t \in \mathbb{R}}$, $f_X(x)$ is a positive semidefinite Hermitian-symmetric matrix dx -a.e. (Hannan [450], Theorem 1, p. 34). Let

$$\widehat{a}(x) = f_X(x)^{1/2} \in \mathbb{C}^{p \times p} \quad (9.3.15)$$

be the (unique) positive semidefinite square root of $f_X(x)$ (see Section 9.1.1). Let also $\widetilde{B}(dx)$ be a complex-valued vector measure as in the statement of the theorem. Then, X can also be represented as

$$\{X(t)\} \stackrel{d}{=} \left\{ \int_{\mathbb{R}} \frac{e^{itx} - 1}{ix} \widehat{a}(x) \widetilde{B}(dx) \right\} \quad (9.3.16)$$

because

$$\mathbb{E}(\widehat{a}(x) \widetilde{B}(dx) \widetilde{B}(dx)^* \widehat{a}(x)^*) = \widehat{a}(x) \widehat{a}(x)^* dx = f_X(x) dx = F_X(dx),$$

and the processes on both sides of (9.3.16) are Gaussian and real-valued.

Step 3: It follows from the relation (9.3.11) in Step 1 that for every $c > 0$, $F_X(dx) = c^{I-H} F_X(c^{-1} dx) c^{I-H}$. By (9.3.15), $F_X(c^{-1} dx) = f_X(c^{-1} x) c^{-1} dx = \widehat{a}(c^{-1} x) \widehat{a}(c^{-1} x)^* c^{-1} dx$ and by (9.3.4), we have $(I - H) - (I/2) = (I/2) - H = -D$. Hence, for every $c > 0$,

$$\widehat{a}(x) \widehat{a}(x)^* = c^{-D} \widehat{a}\left(\frac{x}{c}\right) \widehat{a}\left(\frac{x}{c}\right)^* c^{-D^*} \quad dx\text{-a.e.} \quad (9.3.17)$$

The relation (9.3.17) also holds $dx dc$ -a.e. This is a consequence of the following Fubini-based argument which is often used. If A denotes the set of x, c for which the two functions in (9.3.17) are equal, we have $\int_{\mathbb{R}} 1_A(x, c) dx = 0$ for every c by (9.3.17) and hence by Fubini's theorem, $\int_{\mathbb{R}} \int_{\mathbb{R}} 1_A(x, c) dx dc = 0$; that is, $A = \mathbb{R}^2$ $dx dc$ -a.e. or the two functions are also equal $dx dc$ -a.e.

Consider $x > 0$. A change of variables c to x/v leads to

$$\widehat{a}(x) \widehat{a}(x)^* = x^{-D} v^D \widehat{a}(v) \widehat{a}(v)^* v^{D^*} x^{-D^*} \quad dx dv\text{-a.e.}$$

Thus, one can choose $v_+ > 0$ such that

$$\widehat{a}(x)\widehat{a}(x)^* = x^{-D}v_+^D\widehat{a}(v_+)\widehat{a}(v_+)^*v_+^{D*}x^{-D*} \quad dx\text{-a.e.} \quad (9.3.18)$$

This means, in particular, that if we set, for $dx\text{-a.e. } x > 0$,

$$\widehat{\alpha}_+(x) = x^{-D}v_+^D\widehat{a}(v_+) = x_+^{-D}v_+^D\widehat{a}(v_+),$$

then $\widehat{\alpha}_+(x)\widehat{\alpha}_+(x)^* = \widehat{a}(x)\widehat{a}(x)^* = f_X(x)$ for $x > 0$.

We now suppose $x < 0$. Again by considering the stationary process $\{X(t) - X(t-1)\}_{t \in \mathbb{R}}$, and by applying Theorem 3 in Hannan [450], p. 41, which states that for continuity points $x_2 > x_1$ of F_X ,

$$F_X(x_2) - F_X(x_1) = \lim_{T \rightarrow \infty} \frac{1}{2\pi} \int_{-T}^T (\mathbb{E}X(t)X(0)') \frac{e^{ix_2t} - e^{ix_1t}}{it} dt,$$

we conclude that $\overline{F_X(x_2)} - \overline{F_X(x_1)} = F_X(-x_1) - F_X(-x_2)$ and hence that $\int_{x_1}^{x_2} \overline{f_X(x)} dx = \int_{-x_2}^{-x_1} f_X(x) dx = \int_{x_1}^{x_2} f_X(-x) dx$. Since F_X has at most a countable number of discontinuity points (Exercise 9.3), the last equality holds for all $x_2 > x_1$ and hence $f_X(-x) = \overline{f_X(x)}$ a.e. dx . Thus,

$$\widehat{a}(-x)\widehat{a}(-x)^* = f_X(-x) = \overline{f_X(x)} = x^{-D}v_+^D\overline{\widehat{a}(v_+)\widehat{a}(v_+)^*}v_+^{D*}x^{-D*} \quad dx\text{-a.e.}$$

Hence, for $x < 0$, we can set

$$\widehat{\alpha}_-(x) = (-x)_+^{-D}v_+^D\overline{\widehat{a}(v_+)} = x_-^{-D}v_+^D\overline{\widehat{a}(v_+)}$$

and, for $x \in \mathbb{R}$, we have

$$\widehat{\alpha}(x) = x_+^{-D}v_+^D\widehat{a}(v_+) + x_-^{-D}v_+^D\overline{\widehat{a}(v_+)} \quad dx\text{-a.e.},$$

and

$$\widehat{\alpha}(x)\widehat{\alpha}(x)^* = f_X(x) \quad dx\text{-a.e.}$$

Therefore, we can use $\widehat{\alpha}$ in place of $\widehat{a}$ in the spectral representation (9.3.16) of X . In view (9.3.16), setting $A = v_+^D\widehat{a}(v_+)$, which is a $p \times p$ matrix, establishes the relation (9.3.3). $\square$

Remark 9.3.3 As a consequence of Corollary 2.1 in Maejima and Mason [667], the eigenvalues h_k of the exponent H of a vector OFBM must satisfy $\Re(h_k) \leq 1$, $k = 1, \dots, p$. However, the extension of the definition of vector OFBM to the case of H with at least one eigenvalue h_k satisfying $\Re(h_k) = 1$ can be delicate (see Didier and Pipiras [307], Remark 3.2).

Theorem 9.3.2 shows that vector OFBM is characterized by a (potentially non-unique; see Section 9.3.4) operator self-similarity exponent H and a matrix A . For the sake of simplicity, we will continue to use the notation B_H for vector OFBM instead of a more correct notation $B_{H,A}$.

We shall next provide integral representations of vector OFBMs in the time domain, which is done in Theorem 9.3.8 below. The key technical step in the proof is the calculation of the (entry-wise) Fourier transform of the kernels

$$(t-u)_\pm^D - (-u)_\pm^D = \exp(\log(t-u)_\pm D) - \exp(\log(-u)_\pm D), \quad (9.3.19)$$

which are the multivariate analogues of the corresponding univariate FBM time domain kernels.

It is natural and convenient to carry out this step in the framework of the so-called “primary matrix functions.” The latter allows one to define naturally matrix analogues $f(D)$, $D \in \mathbb{R}^{p \times p}$, of univariate functions $f(d)$, $d \in \mathbb{R}$, and to say when two such matrix-valued functions are equal based on their univariate counterparts. We shall need to define, in particular, $\Gamma(D + I)$ and $\sin(D\pi/2)$ where D is a $p \times p$ matrix.

We shall now recall the definition of primary matrix functions (more details and properties can be found in Horn and Johnson [482], Sections 6.1 and 6.2; see also Higham [475]). Let $\Lambda \in \mathbb{C}^{p \times p}$ and consider its Jordan decomposition $\Lambda = PJP^{-1}$, $J = \text{diag}(J_{\lambda_1}, \dots, J_{\lambda_N})$ is in Jordan canonical form with the Jordan blocks $J_{\lambda_1}, \dots, J_{\lambda_N}$, defined in (9.1.2) and of respective sizes $r_k \times r_k$, $k = 1, \dots, N$, on the diagonal, and $\lambda_1, \dots, \lambda_N$ are the eigenvalues of Λ .

Let $U \subseteq \mathbb{C}$ be an open set. Given a function $h : U \rightarrow \mathbb{C}$ and some $\Lambda \in \mathbb{C}^{p \times p}$ as above, consider the conditions:

(M1) $\lambda_k \in U$, $k = 1, \dots, N$;

(M2) if $r_k > 1$ (with r_k referring to the size of J_{λ_k}), then $h(z)$ is analytic in a vicinity $U_k \ni \lambda_k$, $U_k \subseteq U$.

Let $\mathcal{M}_h = \{\Lambda \in \mathbb{C}^{p \times p} : \text{conditions (M1) and (M2) hold at the eigenvalues } \lambda_1, \dots, \lambda_N \text{ of } \Lambda\}$.

We now define the *primary matrix function* $h(\Lambda)$ associated with the scalar-valued function $h(z)$.

Definition 9.3.4 The *primary matrix function* $h : \mathcal{M}_h \rightarrow \mathbb{C}^{p \times p}$ is defined as

$$h(\Lambda) = Ph(J)P^{-1} = P \begin{pmatrix} h(J_{\lambda_1}) & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & h(J_{\lambda_N}) \end{pmatrix} P^{-1},$$

where $h(J_{\lambda_k})$ is the following $r_k \times r_k$ matrix:

$$h(J_{\lambda_k}) = \begin{pmatrix} h(\lambda_k) & 0 & \dots & 0 \\ h'(\lambda_k) & h(\lambda_k) & \dots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ \frac{h^{(r_k-1)}(\lambda_k)}{(r_k-1)!} & \dots & h'(\lambda_k) & h(\lambda_k) \end{pmatrix}.$$

Thus one can express $h(\Lambda) = h(PJP^{-1})$ as $h(\Lambda) = Ph(J)P^{-1}$. Note also that the matrix $h(\Lambda)$ is invertible if $h(\lambda_k) \neq 0$, $k = 1, \dots, N$. The following example illustrates the notion of a primary matrix function.

Example 9.3.5 Consider the Jordan block

$$J = \begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix}$$

and let $h(x) = x^2$, so that $h'(x) = 2x$. Then

$$h(J) = \begin{pmatrix} h(2) & 0 \\ h'(2) & h(2) \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 4 & 4 \end{pmatrix},$$

which is also equal to J^2 .

Example 9.3.6 (Primary matrix functions) (i) The function $h(z) = \sin(z)$ is analytic for $z \in \mathbb{C} =: U$. According to Definition 9.3.4, the primary matrix function $h(\Lambda) = \sin(\Lambda)$ can be defined for any matrix Λ as

$$\sin(\Lambda) = P \sin(J) P^{-1} = P \begin{pmatrix} \sin(J_{\lambda_1}) & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sin(J_{\lambda_N}) \end{pmatrix} P^{-1},$$

where $\Lambda = P J P^{-1}$ is the Jordan decomposition of Λ , and

$$\sin(J_{\lambda_k}) = \begin{pmatrix} \sin(\lambda_k) & 0 & \dots & 0 \\ \cos(\lambda_k) & \sin(\lambda_k) & \dots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ \frac{\sin^{(r_k-1)}(\lambda_k)}{(r_k-1)!} & \dots & \cos(\lambda_k) & \sin(\lambda_k) \end{pmatrix}.$$

(ii) When $h(z) = e^z$, one can similarly define the primary matrix function $h(\Lambda) = e^\Lambda$ for any matrix Λ . It should be noted that the definition of the matrix exponential based on a series is equivalent to that based on primary matrix functions. See Exercise 9.2.

The functions $(t-u)_\pm^D$, $\Gamma(D+I)$, $|x|^{-D}$, $e^{\mp i \operatorname{sign}(x) D \pi/2}$ appearing in Proposition 9.3.7 below, and the functions $\sin(D\pi/2)$, $\cos(D\pi/2)$ appearing in Theorem 9.3.8 below are all defined as primary matrix functions.

The proof of the following technical result can be found in Didier and Pipiras [307], Proposition 3.1.

Proposition 9.3.7 Under (9.3.2) and the condition (9.3.21) in Theorem 9.3.8 below,

$$\int_{\mathbb{R}} e^{ixu} \left((t-u)_\pm^D - (-u)_\pm^D \right) du = \frac{e^{itx} - 1}{ix} |x|^{-D} \Gamma(D+I) e^{\mp i \operatorname{sign}(x) D \pi/2}. \quad (9.3.20)$$

Note that, when $p = 1$, the relation (9.3.20) agrees with (2.6.4) and (6.1.42) (together with (6.1.24) and (6.1.35)). But we emphasize again that the functions $(t-u)_\pm^D$, $\Gamma(D+I)$, $|x|^{-D}$, $e^{\mp i \operatorname{sign}(x) D \pi/2}$ appearing in (9.3.20) are all defined as primary matrix functions for general p .

Next, we construct time domain representations for vector OFBMs.

Theorem 9.3.8 Let $\{B_H(t)\}_{t \in \mathbb{R}}$ be a vector OFBM with operator self-similarity exponent H having the spectral representation (9.3.3) with $A = A_1 + i A_2$, $A_1, A_2 \in \mathbb{R}^{p \times p}$.

(i) Suppose that $H \in \mathbb{R}^{p \times p}$ has eigenvalues satisfying (9.3.2) and

$$\operatorname{Re}(h_k) \neq \frac{1}{2}, \quad k = 1, \dots, p. \quad (9.3.21)$$

Then, there are matrices $M_+, M_- \in \mathbb{R}^{p \times p}$ such that

$$\{B_H(t)\}_{t \in \mathbb{R}} \stackrel{d}{=} \left\{ \int_{\mathbb{R}} \left(((t-u)_+^D - (-u)_+^D) M_+ + ((t-u)_-^D - (-u)_-^D) M_- \right) B(du) \right\}_{t \in \mathbb{R}}, \quad (9.3.22)$$

where $B(du)$ is a vector-valued random measure consisting of independent Gaussian random measures $B_k(du)$ with control measures $\mathbb{E} B_k(du) B_k(du)^* = du$. Moreover, the matrices M_+, M_- can be taken as

$$M_{\pm} = \sqrt{\frac{\pi}{2}} \left(\left(\sin \frac{D\pi}{2} \right)^{-1} \Gamma(D+I)^{-1} A_1 \pm \left(\cos \frac{D\pi}{2} \right)^{-1} \Gamma(D+I)^{-1} A_2 \right). \quad (9.3.23)$$

(ii) Suppose that $H = (1/2)I$. Then, there exist matrices $M, N \in \mathbb{R}^{p \times p}$ such that

$$\{B_H(t)\}_{t \in \mathbb{R}} \stackrel{d}{=} \left\{ \int_{\mathbb{R}} \left((\operatorname{sign}(t-u) - \operatorname{sign}(-u)) M + \log \left(\frac{|t-u|}{|u|} \right) N \right) B(du) \right\}_{t \in \mathbb{R}}, \quad (9.3.24)$$

where $B(du)$ is as in (9.3.22). Moreover, the matrices M, N can be taken as

$$M = \sqrt{\frac{\pi}{2}} A_1, \quad N = -\sqrt{\frac{2}{\pi}} A_2. \quad (9.3.25)$$

Proof (i) Denote the process on the right-hand side of (9.3.22) by X_H . By using the Jordan decomposition of D , one can show that X_H is well-defined (Exercise 9.5). To prove the theorem, it suffices to show that there are matrices $M_{\pm}$ such that the covariance structure of X_H matches that of the vector OFBM B_H given by its spectral representation (9.3.3) with $A = A_1 + iA_2$. By using Plancherel's identity (A.1.10) and (9.3.20), note first that

$$\begin{aligned} \mathbb{E} X_H(t) X_H(s)^* &= \frac{1}{2\pi} \int_{\mathbb{R}} \frac{(e^{itx} - 1)(e^{-isx} - 1)}{|x|^2} \\ &\quad \times \left(|x|^{-D} \Gamma(D+I) [e^{-i\operatorname{sign}(x)D\pi/2} M_+ + e^{i\operatorname{sign}(x)D\pi/2} M_-] \right) \\ &\quad \cdot \left([M_+^* e^{i\operatorname{sign}(x)D^*\pi/2} + M_-^* e^{-i\operatorname{sign}(x)D^*\pi/2}] \Gamma(D+I)^* |x|^{-D^*} \right) dx, \end{aligned}$$

where $*$ is just the transpose in the case of $X_H(t) \in \mathbb{R}^p$ and $M_+, M_-, D, \Gamma(D+I) \in \mathbb{R}^{p \times p}$. Meanwhile, for B_H , we have by (9.3.3),

$$\mathbb{E} B_H(t) B_H(s)^* = \int_{\mathbb{R}} \frac{(e^{itx} - 1)(e^{-isx} - 1)}{|x|^2} (x_+^{-D} A A^* x_+^{-D^*} + x_-^{-D} \overline{A A^*} x_-^{-D^*}) dx. \quad (9.3.26)$$

Thus, by using the relation $e^{i\Theta} = \cos(\Theta) + i \sin(\Theta)$, $\Theta \in \mathbb{R}^{p \times p}$, it is sufficient to find $M_{\pm} \in \mathbb{R}^{p \times p}$ such that

$$\begin{aligned} A A^* &= \frac{1}{2\pi} \Gamma(D+I) (e^{-iD\pi/2} M_+ + e^{iD\pi/2} M_-) \\ &\quad \cdot (M_+^* e^{iD^*\pi/2} + M_-^* e^{-iD^*\pi/2}) \Gamma(D+I)^* \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2\pi} \Gamma(D+I) \left(\left[\cos \frac{D\pi}{2} - i \sin \frac{D\pi}{2} \right] M_+ + \left[\cos \frac{D\pi}{2} + i \sin \frac{D\pi}{2} \right] M_- \right) \\
&\quad \cdot \left(M_+^* \left[\cos \frac{D^*\pi}{2} + i \sin \frac{D^*\pi}{2} \right] + M_-^* \left[\cos \frac{D^*\pi}{2} - i \sin \frac{D^*\pi}{2} \right] \right) \Gamma(D+I)^* \\
&= \frac{1}{2\pi} \Gamma(D+I) \left(\sin \left(\frac{D\pi}{2} \right) (M_+ - M_-) (M_+^* - M_-^*) \sin \left(\frac{D^*\pi}{2} \right) \right. \\
&\quad + \cos \left(\frac{D\pi}{2} \right) (M_+ + M_-) (M_+^* + M_-^*) \cos \left(\frac{D^*\pi}{2} \right) \\
&\quad + i \left(\cos \left(\frac{D\pi}{2} \right) (M_+ + M_-) (M_+^* - M_-^*) \sin \left(\frac{D^*\pi}{2} \right) \right. \\
&\quad \left. \left. - \sin \left(\frac{D\pi}{2} \right) (M_+ - M_-) (M_+^* + M_-^*) \cos \left(\frac{D^*\pi}{2} \right) \right) \right) \Gamma(D+I)^*. \quad (9.3.27)
\end{aligned}$$

On the other hand,

$$AA^* = (A_1 + iA_2)(A_1^* - iA_2^*) = (A_1A_1^* + A_2A_2^*) + i(A_2A_1^* - A_1A_2^*). \quad (9.3.28)$$

We now compare (9.3.28) and (9.3.27). A natural way to proceed is to view M_+ and M_- as solutions to the system

$$A_1 = \frac{1}{\sqrt{2\pi}} \Gamma(D+I) \sin \left(\frac{D\pi}{2} \right) (M_+ - M_-), \quad A_2 = \frac{1}{\sqrt{2\pi}} \Gamma(D+I) \cos \left(\frac{D\pi}{2} \right) (M_+ + M_-). \quad (9.3.29)$$

By assumption (9.3.21), $\sin(\frac{D\pi}{2})$, $\cos(\frac{D\pi}{2})$ and $\Gamma(D+I)$ are invertible (see the note following Definition 9.3.4). Then,

$$\begin{aligned}
M_+ - M_- &= \sqrt{2\pi} \left(\sin \frac{D\pi}{2} \right)^{-1} \Gamma(D+I)^{-1} A_1, \\
M_+ + M_- &= \sqrt{2\pi} \left(\cos \frac{D\pi}{2} \right)^{-1} \Gamma(D+I)^{-1} A_2
\end{aligned}$$

and hence we obtain the solution given by (9.3.23).

(ii) In this case, $H = (1/2)I$ and one can readily compute the inverse Fourier transform of the integrand in (9.3.3), that is (up to $(2\pi)^{-1}$),

$$\begin{aligned}
&\int_{\mathbb{R}} e^{-iux} \frac{e^{itx} - 1}{ix} \left(1_{\{x>0\}} A + 1_{\{x<0\}} \bar{A} \right) dx \\
&= \int_{\mathbb{R}} \frac{\cos((t-u)x) - \cos(ux) + i(\sin((t-u)x) + \sin(ux))}{ix} \left(1_{\{x>0\}} A + 1_{\{x<0\}} \bar{A} \right) dx. \quad (9.3.30)
\end{aligned}$$

One can show (Exercise 9.6) that this becomes

$$-2 \log \left(\frac{|t-u|}{|u|} \right) A_2 + (\text{sign}(t-u) - \text{sign}(-u)) \pi A_1. \quad (9.3.31)$$

Then, by considering second moments and using Plancherel's identity, representation (9.3.24) holds with $M = (2\pi)^{-1/2} \pi A_1$ and $N = (2\pi)^{-1/2} (-2) A_2$. It is well-defined because the integrand comes from the inverse Fourier transform of a square-integrable function and hence is also square-integrable. $\square$

Remark 9.3.9 When (9.3.21) does not hold and $H \neq (1/2)I$, the general form of time domain representations can be quite intricate. See Didier and Pipiras [307], Example 3.1.

Vector OFBMs provide a large class of processes indexed by matrices H and A . We examine next several interesting subclasses, the first subclass consisting of time reversible vector OFBMs, and the second class consisting of vector OFBMs with a diagonal self-similarity matrix H .

9.3.2 Time Reversible Vector OFBMs

Consider a vector OFBM B_H ($\mathbb{R}^p$ -valued) whose exponent H has eigenvalues h_k with positive real parts. By using operator self-similarity and stationarity of increments, one can argue as in the one-dimensional case (see Proposition 2.5.6, (e)) that

$$\begin{aligned} \mathbb{E}B_H(t)B_H(s)^* + \mathbb{E}B_H(s)B_H(t)^* \\ &= \mathbb{E}B_H(t)B_H(t)^* + \mathbb{E}B_H(s)B_H(s)^* - \mathbb{E}(B_H(t) - B_H(s))(B_H(t) - B_H(s))^* \\ &= |t|^H \Gamma(1, 1) |t|^{H^*} + |s|^H \Gamma(1, 1) |s|^{H^*} - |t - s|^H \Gamma(1, 1) |t - s|^{H^*}, \quad (9.3.32) \\ \Gamma(1, 1) &= \mathbb{E}B_H(1)B_H(1)^* \end{aligned}$$

and we continue using $*$ instead of the prime for the transpose of real-valued matrices as in the previous section. In contrast to the one-dimensional case, it is *not* generally true that

$$\mathbb{E}B_H(t)B_H(s)^* = \mathbb{E}B_H(s)B_H(t)^*, \quad (9.3.33)$$

and hence the vector OFBM is not characterized by H and a matrix $\Gamma(1, 1)$. As will be seen below, the condition (9.3.33) corresponds to time reversibility. We shall provide next conditions for vector OFBM to be time reversible.

Recall (cf. Section 9.1.2) that a process X is said to be *time reversible* if

$$\{X(t)\}_{t \in \mathbb{R}} \stackrel{d}{=} \{X(-t)\}_{t \in \mathbb{R}}. \quad (9.3.34)$$

When X is a zero-mean multivariate Gaussian stationary process, (9.3.34) is equivalent to

$$\mathbb{E}X(t)X(s)^* = \mathbb{E}X(-t)X(-s)^*, \quad s, t \in \mathbb{R},$$

which by adding $t + s$ to each argument on the right-hand side, is in turn equivalent to

$$\mathbb{E}X(t)X(s)^* = \mathbb{E}X(s)X(t)^*, \quad s, t \in \mathbb{R}.$$

The next proposition provides necessary and sufficient conditions for time reversibility in the case of Gaussian processes with stationary increments. It is stated without proof, since the latter is elementary.

Proposition 9.3.10 *Let X be an $\mathbb{R}^p$ -valued Gaussian process with stationary increments and spectral representation*

$$\{X(t)\}_{t \in \mathbb{R}} \stackrel{d}{=} \left\{ \int_{\mathbb{R}} \frac{e^{itx} - 1}{ix} \tilde{Y}(dx) \right\}_{t \in \mathbb{R}},$$

where $\tilde{Y}(dx)$ is an orthogonal-increment random measure taking values in $\mathbb{C}^n$. The following statements are equivalent:

- (i) X is time reversible;
- (ii) $\mathbb{E}\tilde{Y}(dx)\tilde{Y}(dx)^* = \mathbb{E}\tilde{Y}(-dx)\tilde{Y}(-dx)^*$;
- (iii) $\mathbb{E}X(t)X(s)^* = \mathbb{E}X(s)X(t)^*$, $s, t \in \mathbb{R}$.

The following result on time reversibility of vector OFBMs is a direct consequence of Proposition 9.3.10.

Theorem 9.3.11 *Let $\{B_H(t)\}_{t \in \mathbb{R}}$ be a vector OFBM with exponent H and the spectral representation (9.3.3). Let $A = A_1 + iA_2$, $A_1, A_2 \in \mathbb{R}^{p \times p}$. Then, B_H is time reversible if and only if*

$$AA^* = \overline{AA^*} \quad \text{or, equivalently,} \quad A_2A_1^* = A_1A_2^*. \quad (9.3.35)$$

Proof By (9.3.3) and Proposition 9.3.10, (ii), time reversibility is equivalent to

$$\begin{aligned} & \mathbb{E}\left((x_+^{-D}A + x_-^{-D}\overline{A})\tilde{B}(dx)\tilde{B}(dx)^*(A^*x_+^{-D*} + \overline{A^*}x_-^{-D*})\right) \\ &= \mathbb{E}\left((x_-^{-D}A + x_+^{-D}\overline{A})\tilde{B}(-dx)\tilde{B}(-dx)^*(A^*x_-^{-D*} + \overline{A^*}x_+^{-D*})\right) \end{aligned}$$

or

$$x_+^{-D}AA^*x_+^{-D*} + x_-^{-D}\overline{AA^*}x_-^{-D*} = x_-^{-D}AA^*x_-^{-D*} + x_+^{-D}\overline{AA^*}x_+^{-D*} \quad dx\text{-a.e.}$$

or, for $x > 0$,

$$x^{-D}AA^*x^{-D*} = x^{-D}\overline{AA^*}x^{-D*} \quad dx\text{-a.e.}$$

Since x^D is invertible for $x > 0$ (with the inverse x^{-D}), this is equivalent to the first relation in (9.3.35).

Let us now prove the statement in (9.3.35). $AA^* = \overline{AA^*}$ implies that AA^* is real-valued. But $AA^* = (A_1 + iA_2)(A_1^* - iA_2^*) = A_1A_1^* + A_2A_2^* + i(A_2A_1^* - A_1A_2^*)$ so that $A_2A_1^* = A_1A_2^*$ as desired. $\square$

Corollary 9.3.12 *The time reversible $\mathbb{R}^p$ -valued $\{B_H(t)\}_{t \in \mathbb{R}}$ in Theorem 9.3.11 has covariance matrix-valued function*

$$\mathbb{E}B_H(t)B_H(s)^* = \int_{\mathbb{R}} \frac{(e^{itx} - 1)(e^{-isx} - 1)}{|x|^2} |x|^{-D}AA^*|x|^{-D*} dx, \quad s, t \in \mathbb{R}. \quad (9.3.36)$$

Proof The result follows from (9.3.35) and (9.3.26). $\square$

Corollary 9.3.13 *Let $\{B_H(t)\}_{t \in \mathbb{R}}$ be an $\mathbb{R}^p$ -valued vector OFBM with the time domain representation given by (9.3.22), and exponent H satisfying (9.3.2) and (9.3.21). Then, B_H is time reversible if and only if*

$$\begin{aligned} & \cos\left(\frac{D\pi}{2}\right)(M_+ + M_-)(M_+^* - M_-^*) \sin\left(\frac{D^*\pi}{2}\right) \\ &= \sin\left(\frac{D\pi}{2}\right)(M_+ - M_-)(M_+^* + M_-^*) \cos\left(\frac{D^*\pi}{2}\right). \end{aligned} \quad (9.3.37)$$

Proof As in the proof of Theorem 9.3.8, under (9.3.21) the matrices $\sin(D\pi/2)$, $\cos(D\pi/2)$ and $\Gamma(D + I)$ are invertible, and thus by using (9.3.29) to define $A = A_1 + iA_2$, one can equivalently reexpress the condition $A_2A_1^* = A_1A_2^*$ in (9.3.35) as (9.3.37). $\square$

A consequence of Theorem 9.3.11 is that non-time reversible vector OFBMs can only emerge in the multivariate context, since in the univariate context condition (9.3.35) is always satisfied. Another elementary consequence of Proposition 9.3.10 is the following result, which partially justifies the interest in time reversibility in the case of vector OFBMs.

Proposition 9.3.14 *Let $\{B_H(t)\}_{t \in \mathbb{R}}$ be a vector OFBM with H satisfying (9.3.2). If $\{B_H(t)\}_{t \in \mathbb{R}}$ is time reversible, then its covariance structure is given by the function: for $s, t \in \mathbb{R}$,*

$$\mathbb{E}B_H(t)B_H(s)^* = \frac{1}{2} \left(|t|^H \Gamma(1, 1) |t|^{H^*} + |s|^H \Gamma(1, 1) |s|^{H^*} - |t-s|^H \Gamma(1, 1) |t-s|^{H^*} \right), \quad (9.3.38)$$

where $\Gamma(1, 1) = \mathbb{E}B_H(1)B_H(1)^*$. Conversely, a vector OFBM with covariance function (9.3.38) is time reversible.

Proof This follows from (9.3.32) and Proposition 9.3.10, (iii). $\square$

Remark 9.3.15 Relation (9.3.38) holds with $\Gamma(1, 1)$ being equal to $\mathbb{E}B_H(1)B_H(1)^*$. There is a priori no reason to believe that, for a fixed exponent $H \in \mathbb{R}^{p \times p}$, every positive definite matrix $\Gamma(1, 1) \in \mathbb{R}^{p \times p}$ leads to a valid covariance function (9.3.38) for time reversible vector OFBMs (see Exercise 9.7). This issue is a problem, for instance, in the context of simulation methods that require knowledge of the covariance function. For time reversible vector OFBMs with diagonalizable H , one natural way to parameterize $\Gamma(1, 1)$ is through the formula (9.3.36). Moreover, as shown in Helgason, Pipiras, and Abry [465], Section 3.3.4, the expression (9.3.36) with $s = t = 1$ can be evaluated explicitly in terms of the parameters D and A .

9.3.3 Vector Fractional Brownian Motions

Another class of vector OFBMs of interest is when the operator self-similarity matrix H is diagonal.

Definition 9.3.16 An $\mathbb{R}^p$ -valued vector OFBM with a diagonal operator self-similarity matrix $H = \text{diag}(h_1, \dots, h_p) \in \mathbb{R}^{p \times p}$ is called *vector fractional Brownian motion* (VFBM).

A VFBM $X = B_H$ with $H = \text{diag}(h_1, \dots, h_p)$ thus satisfies the relation: for $c > 0$,

$$\{X(ct)\}_{t \in \mathbb{R}} = \left\{ \begin{pmatrix} X_1(ct) \\ \vdots \\ X_p(ct) \end{pmatrix} \right\}_{t \in \mathbb{R}} \stackrel{d}{=} \left\{ \begin{pmatrix} c^{h_1} X_1(t) \\ \vdots \\ c^{h_p} X_p(t) \end{pmatrix} \right\}_{t \in \mathbb{R}} = \{\text{diag}(c^{h_1}, \dots, c^{h_p}) X(t)\}_{t \in \mathbb{R}}. \quad (9.3.39)$$

Processes X satisfying (9.3.39) will be called *vector self-similar* with vector self-similarity matrix $H = \text{diag}(h_1, \dots, h_p)$. Note that vector self-similar processes are naturally expected as limits of suitably normalized vectors. The following result is a simple extension of Lamperti's theorem (Theorem 2.8.5).

Theorem 9.3.17 (i) *If a vector-valued stochastic process Y satisfies*

$$f(\xi)^{-1} \left(Y(\xi t) + g(\xi) \right) \xrightarrow{fdd} X(t), \quad t \geq 0, \text{ as } \xi \rightarrow \infty, \quad (9.3.40)$$

for some proper vector-valued process $X(t)$, deterministic function $g(\xi) = (g_1(\xi), \dots, g_p(\xi))' \in \mathbb{R}^p$ and $f(\xi) = \text{diag}(f_1(\xi), \dots, f_p(\xi))$ with some positive deterministic functions $f_k(\xi) \rightarrow \infty$, as $\xi \rightarrow \infty$, then there is $H = \text{diag}(h_1, \dots, h_p)$ with $h_k \geq 0$ such that

$$f(\xi) = \xi^H L(\xi), \quad g(\xi) = \xi^H L(\xi) w(\xi), \quad (9.3.41)$$

where $\xi^H = \text{diag}(\xi^{h_1}, \dots, \xi^{h_p})$, $L(\xi) = \text{diag}(L_1(\xi), \dots, L_p(\xi))$ with slowly varying functions $L_k(\xi)$ at infinity and $w(\xi) = (w_1(\xi), \dots, w_p(\xi))'$ is a function which converges to a constant $w = (w_1, \dots, w_p)'$ as $\xi \rightarrow \infty$. Moreover, the process $\{X(t) - w\}_{t \geq 0}$ is vector self-similar with the exponent H .

(ii) *If $\{X(t) - w\}_{t \geq 0}$ is vector self-similar with exponent H , then there are functions $f(\xi)$, $g(\xi)$ and a process Y such that the above convergence (9.3.40) holds.*

Proof (i) The relations in (9.3.41) follow from the one-dimensional Lamperti's theorem (Theorem 2.8.5) by considering the individual components of the relation (9.3.40). To see that $\{X(t) - w\}_{t \geq 0}$ is vector self-similar with the exponent H , suppose without loss of generality that $w = 0$. Observe that, for $c > 0$,

$$f(\xi)^{-1} \left(Y(\xi ct) + g(\xi) \right) \xrightarrow{fdd} X(ct).$$

On the other hand, by (9.3.41),

$$\begin{aligned} f(\xi)^{-1} \left(Y(\xi ct) + g(\xi) \right) &= (f(\xi)^{-1} f(c\xi)) f(c\xi)^{-1} \left(Y(c\xi t) + g(c\xi) \right) + f(\xi)^{-1} (g(\xi) - g(c\xi)) \\ &= c^H (L(\xi)^{-1} L(c\xi)) f(c\xi)^{-1} \left(Y(c\xi t) + g(c\xi) \right) + w(\xi) - c^H L(\xi)^{-1} L(c\xi) w(c\xi) \xrightarrow{fdd} c^H X(t), \end{aligned}$$

as $\xi \rightarrow \infty$, since $L(\xi)^{-1} L(c\xi) \rightarrow 1$, $w(\xi) \rightarrow 0 (= w)$ and $f(c\xi)^{-1} (Y(c\xi t) + g(c\xi)) \xrightarrow{fdd} X(t)$. This implies that $X(ct)$ and $c^H X(t)$ have the same finite-dimensional distributions; that is, X is vector self-similar.

(ii) The proof is analogous to that of Theorem 2.8.5, (ii). □

VFBMs underly many vector OFBMs through the change of coordinate systems as explained in (9.2.6)–(9.2.11). Being vector self-similar, VFBMs are also naturally expected as limits of component-wise normalized vector processes. Moreover, VFBMs can be readily used in applications as there is a way to estimate their parameters (Amblard and Coeurjolly [25]).

Note that each component $B_{H,k}$ of VFBM is a one-dimensional FBM, and its covariance function is

$$\mathbb{E}B_{H,k}(t)B_{H,k}(s) = \frac{\mathbb{E}B_{H,k}(1)^2}{2} \left\{ |t|^{2H_k} + |s|^{2H_k} - |t-s|^{2H_k} \right\}, \quad s, t \in \mathbb{R}, \quad (9.3.42)$$

This is not the case for general vector OFBMs. Based on the following auxiliary result, we will provide an analogous relation for the cross-covariance function $\mathbb{E}B_{H,j}(t)B_{H,k}(s)$, $j \neq k$.

Lemma 9.3.18 *Let $H \in (0, 1)$ and $s, t \in \mathbb{R}$. Then,*

$$\begin{aligned} \int_{\mathbb{R}} \frac{(e^{itx} - 1)(e^{-isx} - 1)}{|x|^2} x_{\pm}^{-2H+1} dx &= C_1(H) \left(|t|^{2H} + |s|^{2H} - |t-s|^{2H} \right) \\ &\pm i \left\{ \begin{aligned} &C_2(H) \left(\text{sign}(t-s)|t-s|^{2H} - \text{sign}(t)|t|^{2H} + \text{sign}(s)|s|^{2H} \right), & \text{if } H \neq \frac{1}{2} \\ &-(t-s) \log |t-s| + t \log |t| - s \log |s|, & \text{if } H = \frac{1}{2} \end{aligned} \right\}, \end{aligned} \quad (9.3.43)$$

where

$$C_1(H) = \begin{cases} \frac{\Gamma(2-2H) \cos(H\pi)}{2H(1-2H)}, & \text{if } H \neq \frac{1}{2} \\ \frac{\pi}{2}, & \text{if } H = \frac{1}{2} \end{cases}, \quad C_2(H) = \frac{\Gamma(2-2H) \sin(H\pi)}{2H(1-2H)}. \quad (9.3.44)$$

Proof Note that the relation (9.3.43) with x_- follows from (9.3.43) with x_+ after the change of variables $x \rightarrow (-x)$ and interchanging s and t . It is therefore enough to consider the integral in (9.3.43) with x_+ ; that is,

$$\begin{aligned} &\int_0^\infty (e^{itx} - 1)(e^{-isx} - 1)x^{-2H-1} dx \\ &= \int_0^\infty (e^{i(t-s)x} - e^{itx} - e^{-isx} + 1)x^{-2H-1} dx = I_1 + iI_2, \end{aligned}$$

where

$$\begin{aligned} I_1 &= \int_0^\infty (\cos((t-s)x) - \cos(tx) - \cos(sx) + 1)x^{-2H-1} dx, \\ I_2 &= \int_0^\infty (\sin((t-s)x) - \sin(tx) + \sin(sx))x^{-2H-1} dx. \end{aligned}$$

To compute I_1 , write

$$\begin{aligned} I_1 &= \int_0^\infty (\cos(|t-s|x) - \cos(|t|x) - \cos(|s|x) + 1)x^{-2H-1} dx \\ &= \int_0^\infty (\cos(|t-s|x) - 1)x^{-2H-1} dx - \int_0^\infty (\cos(|t|x) - 1)x^{-2H-1} dx \end{aligned}$$

$$\begin{aligned}
& - \int_0^\infty (\cos(|s|x) - 1)x^{-2H-1}dx \\
&= \int_0^\infty (1 - \cos y)y^{-2H-1}dy \left(|t|^{2H} + |s|^{2H} - |t-s|^{2H} \right),
\end{aligned}$$

after the change of variables $|z|x = y$ in the integrals $\int_0^\infty (\cos(|z|x) - 1)x^{-2H-1}dx$, $z = t - s, t, s$. Observe now that, after integration by parts,

$$\int_0^\infty (1 - \cos y)y^{-2H-1}dy = \frac{1}{2H} \int_0^\infty \sin(y) y^{-2H}dy = C_1(H),$$

by using Formula 3.761.4 when $H \neq 1/2$, and Formula 3.782.2 when $H = 1/2$, in Gradshteyn and Ryzhik [421], pp. 436 and 447 (see also (2.6.10)).

To compute I_2 , we consider the cases $H < 1/2$, $H > 1/2$ and $H = 1/2$ separately.

- When $H < 1/2$, I_2 can be written as

$$\begin{aligned}
I_2 &= \int_0^\infty \sin((t-s)x)x^{-2H-1}dx - \int_0^\infty \sin(tx)x^{-2H-1}dx + \int_0^\infty \sin(sx)x^{-2H-1}dx \\
&= \text{sign}(t-s) \int_0^\infty \sin(|t-s|x)x^{-2H-1}dx - \text{sign}(t) \int_0^\infty \sin(|t|x)x^{-2H-1}dx \\
&\quad + \text{sign}(s) \int_0^\infty \sin(|s|x)x^{-2H-1}dx \\
&= \int_0^\infty \sin(y) y^{-2H-1}dy \left(\text{sign}(t-s)|t-s|^{2H} - \text{sign}(t)|t|^{2H} + \text{sign}(s)|s|^{2H} \right),
\end{aligned}$$

where $\int_0^\infty \sin(y) y^{-2H-1}dy = C_2(H)$ by Formula 3.761.4 in Gradshteyn and Ryzhik [421], p. 436.

- When $H > 1/2$, we have

$$\begin{aligned}
I_2 &= \int_0^\infty (\sin((t-s)x) - (t-s)x)x^{-2H-1}dx - \int_0^\infty (\sin(tx) - tx)x^{-2H-1}dx \\
&\quad + \int_0^\infty (\sin(sx) - sx)x^{-2H-1}dx \\
&= \int_0^\infty (\sin y - y)y^{-2H-1}dy \left(\text{sign}(t-s)|t-s|^{2H} - \text{sign}(t)|t|^{2H} + \text{sign}(s)|s|^{2H} \right)
\end{aligned}$$

and after integration by parts,

$$\begin{aligned}
\int_0^\infty (\sin y - y)y^{-2H-1}dy &= \frac{1}{2H} \int_0^\infty (\cos y - 1)y^{-2H}dy \\
&= \frac{1}{2H(1-2H)} \int_0^\infty \sin(y) y^{1-2H}dy = C_2(H)
\end{aligned}$$

by Formula 3.761.4 in Gradshteyn and Ryzhik [421], p. 436.

- Finally, when $H = 1/2$, the above arguments do not apply since the integral $\int_0^\infty \sin(y) y^{-2}dy$ is not defined at $y = 0$, and the integral $\int_0^\infty (\sin y - y)y^{-2}dy$ is not defined at $y = +\infty$. In this case, we write I_2 as

$$\begin{aligned}
I_2 &= \int_0^\infty (\sin((t-s)x) - (t-s)x1_{\{|t-s|x \leq 1\}})x^{-2}dx - \int_0^\infty (\sin(tx) - tx1_{\{|t|x \leq 1\}})x^{-2}dx \\
&\quad + \int_0^\infty (\sin(sx) - sx1_{\{|s|x \leq 1\}})x^{-2}dx + \int_0^\infty ((t-s)1_{\{|t-s|x \leq 1\}} - t1_{\{|t|x \leq 1\}} \\
&\quad + s1_{\{|s|x \leq 1\}})x^{-1}dx \\
&= \int_0^\infty (\sin(y) - y1_{\{|y| \leq 1\}})y^{-2}dy \left(\text{sign}(t-s)|t-s| - \text{sign}(t)|t| + \text{sign}(s)|s| \right) \\
&\quad + \int_0^\infty ((t-s)1_{\{a \leq x \leq |t-s|^{-1}\}} - t1_{\{a \leq x \leq |t|^{-1}\}} \\
&\quad + s1_{\{a \leq x \leq |s|^{-1}\}})x^{-1}dx + \int_0^a ((t-s) - t + s)x^{-1}dx =: I_{2,1} + I_{2,2} + I_{2,3},
\end{aligned}$$

where $a = \min\{|s|^{-1}, |t|^{-1}, |t-s|^{-1}\}$. Note that $I_{2,1} = 0$ since $\text{sign}(t-s)|t-s| - \text{sign}(t)|t| + \text{sign}(s)|s| = (t-s) - t + s = 0$. Also $I_{2,3} = 0$. Hence,

$$\begin{aligned}
I_2 &= I_{2,2} = (t-s) \int_a^{|t-s|^{-1}} \frac{dx}{x} - t \int_a^{|t|^{-1}} \frac{dx}{x} + s \int_a^{|s|^{-1}} \frac{dx}{x} \\
&= -(t-s) \log |t-s| + t \log |t| - s \log |s|.
\end{aligned}$$

This establishes (9.3.43) with x_+ . $\square$

The following proposition provides the cross-covariances $\mathbb{E}B_{H,j}(t)B_{H,k}(s)$ between the components of a VFBM.

Proposition 9.3.19 *Let $\{B_H(t)\}_{t \in \mathbb{R}}$ be a VFBM with exponent $H = \text{diag}(h_1, \dots, h_p)$ and the spectral representation (9.3.3) with matrix A . Then, for $j \neq k$, $s, t \in \mathbb{R}$,*

$$\mathbb{E}B_{H,j}(t)B_{H,k}(s) = \frac{\sigma_j \sigma_k}{2} (w_{jk}(t)|t|^{h_j+h_k} + w_{jk}(-s)|s|^{h_j+h_k} - w_{jk}(t-s)|t-s|^{h_j+h_k}), \quad (9.3.45)$$

where $\sigma_j^2 = \mathbb{E}B_{H,j}(1)^2$ and

$$w_{jk}(u) = \begin{cases} \rho_{jk} - \eta_{jk} \text{sign}(u), & \text{if } h_j + h_k \neq 1, \\ \rho_{jk} - \eta_{jk} \text{sign}(u) \log |u|, & \text{if } h_j + h_k = 1 \end{cases} \quad (9.3.46)$$

with $\rho_{jk} \in [-1, 1]$ and $\eta_{jk} \in \mathbb{R}$. Moreover, letting $C_1(\cdot)$ and $C_2(\cdot)$ be given in (9.3.44) and denoting $AA^* = C = (c_{jk})_{j,k=1,\dots,p}$, we have

$$\sigma_j \sigma_k \rho_{jk} = 4C_1\left(\frac{h_j + h_k}{2}\right) \Re(c_{jk}), \quad \sigma_j \sigma_k \eta_{jk} = 4C_2\left(\frac{h_j + h_k}{2}\right) \Im(c_{jk}) \quad (9.3.47)$$

if $h_j + h_k \neq 1$, and

$$\sigma_j \sigma_k \rho_{jk} = 4C_1\left(\frac{1}{2}\right) \Re(c_{jk}), \quad \sigma_j \sigma_k \eta_{jk} = 4\Im(c_{jk}) \quad (9.3.48)$$

if $h_j + h_k = 1$.

Proof As in (9.3.26),

$$\mathbb{E}B_H(t)B_H(s)^* = \int_{\mathbb{R}} \frac{(e^{itx} - 1)(e^{-isx} - 1)}{|x|^2} (x_+^{-D}AA^*x_+^{-D*} + x_-^{-D}\overline{AA^*}x_-^{-D*})dx.$$

Since $d_j = h_j - 1/2$,

$$x_{\pm}^{-D}AA^*x_{\pm}^{-D*} = (c_{jk}x_{\pm}^{-(h_j+h_k)+1})_{j,k=1,\dots,p}. \quad (9.3.49)$$

It follows that

$$\mathbb{E}B_{H,j}(t)B_{H,k}(s) = \int_{\mathbb{R}} \frac{(e^{itx} - 1)(e^{-isx} - 1)}{|x|^2} (c_{jk}x_+^{-(h_j+h_k)+1} + \bar{c}_{jk}x_-^{-(h_j+h_k)+1})dx.$$

If $h_j + h_k \neq 1$, Lemma 9.3.18 yields

$$\begin{aligned} \mathbb{E}B_{H,j}(t)B_{H,k}(s) &= C_1\left(\frac{h_j + h_k}{2}\right)(c_{jk} + \bar{c}_{jk})\left(|t|^{h_j+h_k} + |s|^{h_j+h_k} - |t-s|^{h_j+h_k}\right) \\ &\quad + iC_2\left(\frac{h_j + h_k}{2}\right)(c_{jk} - \bar{c}_{jk})\left(\text{sign}(t-s)|t-s|^{h_j+h_k} \right. \\ &\quad \left. - \text{sign}(t)|t|^{h_j+h_k} + \text{sign}(s)|s|^{h_j+h_k}\right) \\ &= (\alpha_{jk} - \beta_{jk}\text{sign}(t))|t|^{h_j+h_k} + (\alpha_{jk} + \beta_{jk}\text{sign}(s))|s|^{h_j+h_k} \\ &\quad - (\alpha_{jk} - \beta_{jk}\text{sign}(t-s))|t-s|^{h_j+h_k}, \end{aligned}$$

where

$$\alpha_{jk} = 2C_1\left(\frac{h_j + h_k}{2}\right)\Re(c_{jk}), \quad \beta_{jk} = 2C_2\left(\frac{h_j + h_k}{2}\right)\Im(c_{jk}).$$

Note that $\mathbb{E}B_{H,j}(1)B_{H,k}(1) = 2\alpha_{jk}$. Hence, we can write

$$\alpha_{jk} = \frac{\sigma_j\sigma_k\rho_{jk}}{2}.$$

Since the covariance is $\mathbb{E}B_{H,j}(1)B_{H,k}(1) = 2\alpha_{jk}$, ρ_{jk} is the correlation between $B_{H,j}(1)$ and $B_{H,k}(1)$, and hence $\rho_{jk} \in [-1, 1]$. Writing

$$\beta_{jk} = \frac{\sigma_j\sigma_k\eta_{jk}}{2}$$

with $\eta_{jk} \in \mathbb{R}$, we obtain (9.3.45) and (9.3.47) when $h_j + h_k \neq 1$. The proof in the case of $h_j + h_k = 1$ is similar using Lemma 9.3.18. $\square$

Remark 9.3.20 Lavancier, Philippe, and Surgailis [597] used a different method and stronger assumptions on covariance structure to obtain the expression (9.3.45) directly from the definition of VFBMs.

The following result provides conditions for time reversibility of VFBMs in terms of the parameters in the cross-covariance function (9.3.45).

Proposition 9.3.21 *Let B_H be a VFBM with the cross-covariance structure (9.3.45). Then, B_H is time reversible if and only if $\eta_{jk} = 0$, $j, k = 1, \dots, p$, $j \neq k$, in (9.3.46).*

Proof Proposition 9.3.14 implies that $w_{jk}(u)$ in (9.3.46) should not depend on u . $\square$

The next result provides conditions for (9.3.42) and (9.3.45) to be a valid covariance structure: that is, the covariance structure of a VFBM.

Proposition 9.3.22 *The covariance structure given by (9.3.42) and (9.3.45) is that of a VFBM if and only if the matrix $Q = (q_{jk})_{j,k=1,\dots,p}$ is positive semidefinite, where*

$$q_{jk} = \frac{\rho_{jk}}{C_1(\frac{h_j+h_k}{2})} + i \frac{\eta_{jk}}{C_2(\frac{h_j+h_k}{2})}, \quad \text{if } h_j + h_k \neq 1, \quad (9.3.50)$$

$$q_{jk} = \frac{\rho_{jk}}{C_1(\frac{1}{2})} + i \eta_{jk}, \quad \text{if } h_j + h_k = 1. \quad (9.3.51)$$

If $j = k$, it is assumed in (9.3.50) and (9.3.51) that $\rho_{jj} = 1$ and $\eta_{jj} = 0$.

Proof Suppose first that B_H is a VFBM and let $C = (c_{jk})_{j,k=1,\dots,p} = AA^*$ as in (9.3.49). Hence the matrix C is positive semidefinite. It follows from (9.3.47) and (9.3.48) that

$$c_{jk} = \Re(c_{jk}) + i \Im(c_{jk}) = \frac{\sigma_j \sigma_k}{4} q_{jk}.$$

Since $C = (\frac{\sigma_j \sigma_k}{4} q_{jk})_{j,k=1,\dots,p}$ is positive semidefinite, so is the matrix $Q = (q_{jk})_{j,k=1,\dots,p}$.

Conversely, if Q is positive semidefinite, so is the matrix $C = (\frac{\sigma_j \sigma_k}{4} q_{jk})_{j,k=1,\dots,p}$. Then, there is A such that $C = AA^*$ (see Section 9.1.1). The proof of Proposition 9.3.19 shows that the covariance structure (9.3.42) and (9.3.45) is that of a VFBM given by the spectral representation (9.3.3) with matrices H and A . $\square$

When $p = 2$, the condition in Proposition 9.3.22 can be written in an alternative and more revealing form. See Exercise 9.8.

9.3.4 Identifiability Questions

One interesting fact about vector OFBMs and, more generally, vector operator self-similar processes is that an operator self-similarity matrix exponent H does not have to be unique. This fact has fundamental connections to algebra and matrix analysis. Our goal in this section is to give some idea of these connections and to present some fundamental results. We begin with a simple example illustrating the non-uniqueness of operator self-similarity exponents.

Example 9.3.23 (*Single-parameter vector OFBM*) Suppose $B_H = \{B_H(t)\}_{t \in \mathbb{R}}$ consists of p independent copies of a univariate FBM with parameter $h \in (0, 1)$. Note that B_H is a vector OFBM with an operator self-similarity exponent $H = hI$. It will be called a *single-parameter vector OFBM*. The matrix $H = hI$ is not the only operator self-similarity exponent of a single-parameter vector OFBM. Take $L \in so(p)$, where $so(p)$ stands for the space of skew-symmetric matrices (that is, matrices $M \in \mathbb{R}^{p \times p}$ such that $M^* = -M$). Then, for any $c > 0$, $c^L = e^{L \log(c)} \in O(p)$, where $O(p)$ is the space of orthogonal matrices (that is, matrices $M \in \mathbb{R}^{p \times p}$ such that $M^*M = I$). In particular, for any $c > 0$,

$$\{c^L B_H(t)\}_{t \in \mathbb{R}} \stackrel{d}{=} \{B_H(t)\}_{t \in \mathbb{R}},$$

since the processes on both sides are Gaussian and have the same covariance structure. To see that the covariance structure is the same, note that $\mathbb{E}(c^L B_H(t))(c^L B_H(s))^* = c^L \mathbb{E} B_H(t) B_H(s)^* (c^L)^* = \Gamma_h(t, s) c^L c^{L*} = \Gamma_h(t, s) c^{L-L} = \Gamma_h(t, s) c^0 = \Gamma_h(t, s) I$, $\Gamma_h(t, s) = \mathbb{E} B_h(t) B_h(s)$ is the covariance of a univariate FBM with parameter $h \in (0, 1)$. Skew-symmetry was used in the step $c^L c^{L*} = c^{L-L}$. This shows that, for any $c > 0$,

$$\{B_H(ct)\}_{t \in \mathbb{R}} \stackrel{d}{=} \{c^H B_H(t)\}_{t \in \mathbb{R}} \stackrel{d}{=} \{c^H c^L B_H(t)\}_{t \in \mathbb{R}} = \{c^{H+L} B_H(t)\}_{t \in \mathbb{R}},$$

in the last step we used the fact that $H = hI$ and L commute. Thus, there is lack of uniqueness in the exponent since for any $L \in so(p)$, $H + L$ is another operator self-similarity exponent of a single-parameter vector OFBM.

It turns out (Hudson and Mason [499]) that, for an operator self-similar process, the class of its operator self-similarity exponents can be characterized in terms of the so-called symmetry group.

Definition 9.3.24 For a proper $\mathbb{R}^p$ -valued process $\{X(t)\}_{t \in \mathbb{R}}$, its *symmetry group* is defined as

$$G_X = \left\{ C \in GL(p) : \{X(t)\}_{t \in \mathbb{R}} \stackrel{d}{=} \{CX(t)\}_{t \in \mathbb{R}} \right\}, \quad (9.3.52)$$

where $GL(p)$ is the multiplicative group of invertible matrices in $\mathbb{R}^{p \times p}$.

Note that, if X is a zero mean, Gaussian process, then

$$G_X = \left\{ C \in GL(p) : \mathbb{E} X(t) X(s)^* = C \mathbb{E} X(s) X(t)^* C^*, \quad \text{for all } s, t \in \mathbb{R} \right\}. \quad (9.3.53)$$

To state the main result, we need some notation. For an operator self-similar process, let $\mathcal{E}(X)$ denote the collection of all its operator self-similarity exponents. It will be characterized using

$$T(G_X) = \left\{ C : C = \lim_{n \rightarrow \infty} \frac{C_n - I}{d_n}, \quad \text{for some } \{C_n\} \subset G_X, 0 < d_n \rightarrow 0 \right\}, \quad (9.3.54)$$

the so-called *tangent space* of the symmetry group G_X .

Theorem 9.3.25 Let $X = \{X(t)\}_{t \in \mathbb{R}}$ be a proper operator self-similar process. Then, with the above notation,

$$\mathcal{E}(X) = H + T(G_X), \quad (9.3.55)$$

where H is any exponent of the process X .

For a proof of Theorem 9.3.25, see Hudson and Mason [499], Theorem 2.

Example 9.3.26 (Single-parameter vector OFBM) Let B_H be the single-parameter vector OFBM in Example 9.3.23. Its covariance function is $\mathbb{E} B_H(t) B_H(s) = \Gamma_h(t, s) I$, $\Gamma_h(t, s)$ is the covariance function of a univariate FBM with parameter h . In view of (9.3.53), the factors $\Gamma_h(t, s)$ cancel out, so that the symmetry group G_{B_H} turns out to be

$$G_{B_H} = \{C \in GL(p) : I = CC^*\}.$$

Consequently, $G_{B_H} = O(p)$, the space of orthogonal matrices. Moreover, it is known (e.g., Jurek and Mason [542], Example 1.5.7) that the tangent space of orthogonal matrices is the space of skew-symmetric matrices; that is, $T(O(p)) = so(p)$, and hence, by Theorem 9.3.25,

$$\mathcal{E}(B_H) = H + so(p),$$

which is in agreement with Example 9.3.23.

Symmetry groups, their structure and implications on the classes of exponents for vector OFBMs were studied in greater depth by Didier and Pipiras [308]. The results are formulated in terms of the matrix parameters H , $D = H - (1/2)I$ and A in the spectral domain representation (9.3.3) of vector OFBM. The following assumption was made: suppose $\Re(AA^*)$ has full rank, and hence can be factored as

$$\Re(AA^*) = S_R \Lambda_R^2 S_R^* =: W^2, \quad (9.3.56)$$

with an orthogonal matrix $S_R \in \mathbb{R}^{p \times p}$, a diagonal $\Lambda_R \in \mathbb{R}^{p \times p}$ and a positive definite W (having inverse W^{-1}).

Denote the symmetry group of a vector OFBM B_H by G_H . It can be shown (Didier and Pipiras [308], Theorem 3.1) that

$$G_H = W \left(\bigcap_{x>0} G(\Pi_x) \cap G(\Pi_I) \right) W^{-1}, \quad (9.3.57)$$

where W is defined in (9.3.56), $\Pi_I = W^{-1} \Im(AA^*) W^{-1}$,

$$\Pi_x = W^{-1} x^{-D} \Re(AA^*) x^{-D^*} W^{-1} = x^{-M} x^{-M^*}$$

with $M = W^{-1} D W$, and

$$G(\Pi) = \{O \in O(p) : O\Pi = \Pi O\} \quad (9.3.58)$$

is the so-called *centralizer* of a matrix Π in the group $O(p)$ of orthogonal matrices.⁶ A more refined analysis of the symmetry groups of vector OFBMs can thus be reduced to the study of centralizers and matrix commutativity. The latter is a well-known and well-studied problem in matrix analysis (e.g., Gantmacher [381]; Section 2.3 in Didier and Pipiras [308]).

Example 9.3.27 (Single-parameter vector OFBM) Let B_H be the single-parameter vector OFBM in Examples 9.3.23 and 9.3.26. It can be represented as (9.3.3) with $A = cI$ for a constant c . The relation (9.3.56) then holds with $W = cI$. The matrices Π_I and Π_x in (9.3.57) become $\Pi_I = 0$ (zero matrix) since $\Im(AA^*) = 0$ and $\Pi_x = x^{-2h+1}I$. Their centralizers are $G(\Pi_I) = G(\Pi_x) = O(p)$ (for the last equality since x^{-2h+1} is a scalar and I commutes with orthogonal matrices), leading to $G_H = O(p)$ as expected by using (9.3.57).

⁶ Note that in (9.3.57), G_H , $G(\Pi_x)$ and $G(\Pi_I)$ are groups and thus collections of matrices, and that W is a matrix. For a collection G of matrices, the notation WGW^{-1} used on the right-hand side of (9.3.57) stands for the collections of matrices $\{WMW^{-1} : M \in G\}$.

By using the results on centralizers and matrix commutativity from matrix analysis, Didier and Pipiras [308] derived a number of results concerning the symmetry group G_H . Necessary and sufficient conditions were provided for G_H to be of maximal type: that is, to be conjugate to $O(p)$ in the sense that $G_H = WO(p)W^{-1}$ (Theorem 4.1 in Didier and Pipiras [308]). Sufficient conditions for G_H to be of minimal type (that is, $G_H = \{I, -I\}$) were given (Proposition 4.1 in Didier and Pipiras [308]). It was shown that the minimal type is the most “common” in a suitable sense.⁷ A complete description of all possible symmetry groups in dimensions $p = 2$ and $p = 3$ was also provided, along with implications for the corresponding classes of exponents (Section 5 in Didier and Pipiras [308]).

Remark 9.3.28 Though operator self-similarity exponents are generally not unique as discussed above, several quantities related to them turn out to be unique. An important example is the real spectrum (that is, the real parts of the eigenvalues) of the exponents, whose uniqueness is established in e.g., Meerschaert and Scheffler [710], Corollary 11.1.7. The real spectrum determines the growth behavior of the operator self-similar process in any given radial direction; see Meerschaert and Scheffler [710], Section 11.2.

9.4 Vector Long-Range Dependence

We have focused so far on the vector operator FBM. We now turn to time series.

We are interested here in long-range dependence for vector-valued time series. The basics of stationary vector time series are recalled in Section 9.1.2. Definitions and basic properties of vector long-range dependence are given in Section 9.4.1. Several classes of models are considered in Sections 9.4.2 and 9.4.3. We consider vector FARIMA(0, D , 0) in Section 9.4.2 and vector FGN in Section 9.4.3. Section 9.4.4 concerns the so-called *fractional cointegration*. As will be seen below, a number of new and interesting issues arise when one departs from the univariate case.

9.4.1 Definitions and Basic Properties

We begin with time and spectral domain conditions which extend conditions II and IV in Section 2.1 to the vector setting. In contrast to the conditions in Section 2.1, however, we do not consider general slowly varying functions, only those behaving as a constant asymptotically. On one hand, these slowly varying functions appear below in a (positive semidefinite) matrix form, and have, apparently, not been considered in the literature. On the other hand, the conditions we provide are most suitable for statistical inference. The suffix “-v” in the conditions refers to “vector”, in order to distinguish them from the conditions in Section 2.1. There will also be a collection of parameters which we gather in a diagonal matrix

$$D = \text{diag}(d_1, \dots, d_p)$$

and suppose that $d_j \in (0, 1/2)$, $j = 1, \dots, p$.

⁷ Minimal, or any finite symmetry groups, lead necessarily to unique exponents in view of the relation (9.3.55).

Condition II-v The autocovariance matrix function of the time series $X = \{X_n\}_{n \in \mathbb{Z}}$ satisfies: for $n \geq 1$,

$$\gamma_X(n) = n^{D-(1/2)I} L_2(n) n^{D-(1/2)I}, \quad (9.4.1)$$

where L_2 is an $\mathbb{R}^{p \times p}$ -valued function satisfying

$$L_2(u) \sim R, \quad \text{as } u \rightarrow +\infty, \quad (9.4.2)$$

for some $R = (r_{jk})_{j,k=1,\dots,p} \in \mathbb{R}^{p \times p}$. Equivalently, the relation (9.4.1) can be written component-wise as: for $j, k = 1, \dots, p$,

$$\gamma_{X,jk}(n) = L_{2,jk}(n) n^{(d_j+d_k)-1} \sim r_{jk} n^{(d_j+d_k)-1}, \quad \text{as } n \rightarrow +\infty. \quad (9.4.3)$$

Condition IV-v The time series $X = \{X_n\}_{n \in \mathbb{Z}}$ has a spectral density matrix satisfying: for $\lambda \in (0, \pi]$,

$$f_X(\lambda) = \lambda^{-D} L_4(\lambda) \lambda^{-D*}, \quad (9.4.4)$$

where $L_4(\lambda) \in \mathbb{C}^{p \times p}$ is Hermitian symmetric and positive semidefinite, satisfying

$$L_4(\lambda) \sim G, \quad \text{as } \lambda \rightarrow 0, \quad (9.4.5)$$

where $G = (g_{jk})_{j,k=1,\dots,p} \in \mathbb{C}^{p \times p}$ is also Hermitian symmetric and positive semidefinite. Equivalently, the relations (9.4.4) and (9.4.5) can be written component-wise as: for $j, k = 1, \dots, p$,

$$f_{X,jk}(\lambda) = L_{4,jk}(\lambda) \lambda^{-(d_j+d_k)} \sim g_{jk} \lambda^{-(d_j+d_k)} =: \alpha_{jk} e^{i\phi_{jk}} \lambda^{-(d_j+d_k)}, \quad \text{as } \lambda \rightarrow 0, \quad (9.4.6)$$

where $\alpha_{jk} \in \mathbb{R}$ and $\phi_{jk} \in (-\pi, \pi]$. See (9.4.7) and (9.4.8) below for two common specifications of α_{jk}, ϕ_{jk} .

Definition 9.4.1 A vector second-order stationary time series $X = \{X_n\}_{n \in \mathbb{Z}}$ is called *long-range dependent* (LRD, in short) if one of the non-equivalent conditions II-v and IV-v above holds with $d_j \in (0, 1/2)$, $j = 1, \dots, p$.

The extension of condition I of Section 2.1 is discussed in Exercise 9.9. A number of comments regarding Definition 9.4.1 and conditions II-v and IV-v are in place.

First, note that the individual component series $\{X_{j,n}\}_{n \in \mathbb{Z}}$, $j = 1, \dots, p$, of a vector LRD series are LRD with parameters d_j , $j = 1, \dots, p$. (Note that $\phi_{jj} = 0$ in condition IV-v since the matrix G is Hermitian symmetric and hence has real-valued entries on the diagonal.) Another possibility would be to require that at least one of the individual series $\{X_{j,n}\}_{n \in \mathbb{Z}}$ be LRD. This could be achieved assuming in condition IV-v that $d_j \in [0, 1/2)$, $j = 1, \dots, p$, and that at least one $d_j > 0$.

Second, note that the structure of (9.4.4) is such that $f_X(\lambda)$ is Hermitian symmetric and positive semidefinite. D^* appearing in (9.4.4) can be replaced by D since it is diagonal and consists of real-valued entries. We write D^* to make the semidefiniteness of $f_X(\lambda)$ more evident. The entries ϕ_{jk} in (9.4.6) are referred to as *phase parameters* of vector LRD series (around zero frequency) and α_{jk} as *amplitude parameters*. The role of phase parameters will become more apparent in Proposition 9.4.2 below where conditions II-v and IV-v are

compared. Recall from (9.1.13)–(9.1.14) that there are two common but slightly different ways to represent the amplitude α_{jk} and the phase ϕ_{jk} , namely,

$$\alpha_{jk} = |g_{jk}|, \quad \phi_{jk} = \text{Arg}(g_{jk}) \quad (9.4.7)$$

and

$$\alpha_{jk} = \frac{\Re g_{jk}}{\cos \phi_{jk}}, \quad \phi_{jk} = \arctan \frac{\Im g_{jk}}{\Re g_{jk}}. \quad (9.4.8)$$

In the case (9.4.8), $\phi_{jk} \in (-\pi/2, \pi/2)$.

It should also be noted that phase parameters are unique to the vector LRD case. Indeed, since they appear in the cross spectral density (spectrum) $f_{X,jk}(\lambda)$, $j \neq k$, the (nonzero) phases do not arise in the univariate case. Likewise, taking

$$\sum_{n=-\infty}^{\infty} \|\gamma_X(n)\|_2 < \infty \quad (9.4.9)$$

for the definition of vector *short-range dependent* (SRD) series, where $\|A\|_2 = (\sum_{j,k} |a_{jk}|^2)^{1/2}$ is the Frobenius norm of $A = (a_{jk})$, we have

$$f_X(\lambda) = \frac{1}{2\pi} \sum_{n=-\infty}^{\infty} e^{-in\lambda} \gamma_X(n) \quad (9.4.10)$$

(cf. Lemma 2.2.22). In particular, since $\gamma_X(n) \in \mathbb{R}^{p \times p}$, $f_X(0) = (2\pi)^{-1} \sum_{n=-\infty}^{\infty} \gamma_X(n)$ is in $\mathbb{R}^{p \times p}$ as well, and

$$f_X(\lambda) \sim G (= f_X(0)), \quad \text{as } \lambda \rightarrow 0, \quad (9.4.11)$$

where $G \in \mathbb{R}^{p \times p}$. The relation (9.4.11) corresponds to (9.4.6) with $d_j = 0$, $j = 1, \dots, p$, and all phase parameters $\phi_{jk} = 0$.

Third, the squared coherence function $\mathcal{H}_{jk}^2(\lambda) = |f_{X,jk}(\lambda)|^2 / (f_{X,jj}(\lambda) f_{X,kk}(\lambda))$ satisfies $0 \leq \mathcal{H}_{jk}^2(\lambda) \leq 1$ by Cauchy–Schwarz (e.g., Brockwell and Davis [186], p. 436). As $\lambda \rightarrow 0$, this translates into

$$0 \leq \lim_{\lambda \rightarrow 0} \frac{|g_{jk}|^2 \lambda^{-2(d_j+d_k)}}{g_{jj} \lambda^{-2d_j} g_{kk} \lambda^{-2d_k}} = \frac{|g_{jk}|^2}{g_{jj} g_{kk}} \leq 1$$

and also explains why the choice of $\lambda^{-(d_j+d_k)}$ is natural for the cross-spectral density $f_{X,jk}(\lambda)$.

Fourth, note also that (9.4.6) is considered for $\lambda > 0$, $\lambda \rightarrow 0$. For real-valued λ , since $f_X(-\lambda) = \overline{f_X(\lambda)}$, (9.4.6) can be replaced by

$$f_{X,jk}(\lambda) = L_{4,jk}(\lambda) |\lambda|^{-(d_j+d_k)} \sim \alpha_{jk} e^{i \text{sign}(\lambda) \phi_{jk}} |\lambda|^{-(d_j+d_k)}, \quad \text{as } \lambda \rightarrow 0. \quad (9.4.12)$$

It follows that $L_{4,jk}(-\lambda) = \overline{L_{4,jk}(\lambda)}$; that is,

$$\Re L_{4,jk}(-\lambda) = \Re L_{4,jk}(\lambda) \quad \text{and} \quad \Im L_{4,jk}(-\lambda) = -\Im L_{4,jk}(\lambda) \quad (9.4.13)$$

if both positive and negative λ s are considered. (Note that since G is Hermitian symmetric, we have $\alpha_{jk} = \alpha_{kj} \in \mathbb{R}$.)

The following proposition relates conditions II-v and IV-v. Quasi-monotonic functions are defined in Definition 2.2.11.

Proposition 9.4.2 (i) Suppose that the component functions $L_{2,jk}$ are quasi-monotone slowly varying. Then, condition II-v implies condition IV-v with

$$g_{jk} = \frac{\Gamma(d_j + d_k)}{2\pi} \left\{ (r_{jk} + r_{kj}) \cos \left((d_j + d_k) \frac{\pi}{2} \right) - i(r_{jk} - r_{kj}) \sin \left((d_j + d_k) \frac{\pi}{2} \right) \right\} \quad (9.4.14)$$

in the relation (9.4.6). In the specification (9.4.8) of $g_{jk} = \alpha_{jk} e^{i\phi_{jk}}$, one has

$$\phi_{jk} = -\arctan \left\{ \frac{r_{jk} - r_{kj}}{r_{jk} + r_{kj}} \tan \left((d_j + d_k) \frac{\pi}{2} \right) \right\}, \quad (9.4.15)$$

$$\alpha_{jk} = \frac{\Gamma(d_j + d_k)(r_{jk} + r_{kj}) \cos((d_j + d_k) \frac{\pi}{2})}{2\pi \cos(\phi_{jk})}. \quad (9.4.16)$$

(ii) Suppose that the component functions $\Re L_{4,jk}$, $\Im L_{4,jk}$ are quasi-monotone slowly varying. Then, condition IV-v implies condition II-v with

$$r_{jk} = 2\Gamma(1 - (d_j + d_k)) \left\{ \Re g_{jk} \sin \left((d_j + d_k) \frac{\pi}{2} \right) - \Im g_{jk} \cos \left((d_j + d_k) \frac{\pi}{2} \right) \right\} \quad (9.4.17)$$

in the relation (9.4.3).

Proof (i) As in Proposition 2.2.14, we have

$$f_{X,jk}(\lambda) = \frac{1}{2\pi} \sum_{n=-\infty}^{\infty} e^{-in\lambda} \gamma_{X,jk}(n).$$

(This is immediate by the proposition for the real part of $f_{X,jk}$, and the relation for the imaginary part can be deduced similarly.) Then, by using $\gamma_{X,jk}(-n) = \gamma_{X,kj}(n)$ (see (9.1.5)),

$$\begin{aligned} f_{X,jk}(\lambda) &= \frac{1}{2\pi} \left\{ \sum_{n=-\infty}^{\infty} \cos(n\lambda) \gamma_{X,jk}(n) - i \sum_{n=-\infty}^{\infty} \sin(n\lambda) \gamma_{X,jk}(n) \right\} \\ &= \frac{1}{2\pi} \left\{ \gamma_{X,jk}(0) + \sum_{n=1}^{\infty} \cos(n\lambda) (\gamma_{X,jk}(n) + \gamma_{X,kj}(n)) \right\} \\ &\quad - \frac{i}{2\pi} \left\{ \sum_{n=1}^{\infty} \sin(n\lambda) (\gamma_{X,jk}(n) - \gamma_{X,kj}(n)) \right\} \\ &= \frac{1}{2\pi} \left\{ \gamma_{X,jk}(0) + \sum_{n=1}^{\infty} \cos(n\lambda) \frac{L_{2,jk}(n) + L_{2,kj}(n)}{n^{1-(d_j+d_k)}} \right\} \\ &\quad - \frac{i}{2\pi} \left\{ \sum_{n=1}^{\infty} \sin(n\lambda) \frac{L_{2,jk}(n) - L_{2,kj}(n)}{n^{1-(d_j+d_k)}} \right\}. \end{aligned}$$

It follows from Proposition A.2.1 in Appendix A.2 that

$$\begin{aligned} f_{X,jk}(\lambda) &\sim \frac{\Gamma(d_j + d_k)}{2\pi} \lambda^{-(d_j+d_k)} \left\{ (r_{jk} + r_{kj}) \cos \left((d_j + d_k) \frac{\pi}{2} \right) \right. \\ &\quad \left. - i(r_{jk} - r_{kj}) \sin \left((d_j + d_k) \frac{\pi}{2} \right) \right\}, \end{aligned}$$

showing (9.4.14). This also immediately yields (9.4.15)–(9.4.16) by using (9.4.8).

(ii) Note that

$$\begin{aligned}\gamma_{X,jk}(n) &= \int_{-\pi}^{\pi} e^{in\lambda} f_{X,jk}(\lambda) d\lambda = \int_{-\pi}^{\pi} e^{in\lambda} L_{4,jk}(\lambda) |\lambda|^{-(d_j+d_k)} d\lambda \\ &= \int_0^{\pi} e^{in\lambda} (\Re L_{4,jk}(\lambda) + i \Im L_{4,jk}(\lambda)) \lambda^{-(d_j+d_k)} d\lambda \\ &\quad + \int_0^{\pi} e^{-in\lambda} (\Re L_{4,jk}(-\lambda) + i \Im L_{4,jk}(-\lambda)) \lambda^{-(d_j+d_k)} d\lambda \\ &= 2 \int_0^{\pi} \cos(n\lambda) \Re L_{4,jk}(\lambda) \lambda^{-(d_j+d_k)} d\lambda - 2 \int_0^{\pi} \sin(n\lambda) \Im L_{4,jk}(\lambda) \lambda^{-(d_j+d_k)} d\lambda,\end{aligned}$$

where we used (9.4.13). By Proposition A.2.2 in Appendix A.2 and an analogous result where cosine is replaced by sine, we get that

$$\begin{aligned}\gamma_{X,jk}(n) &\sim 2\Gamma(1 - (d_j + d_k)) n^{(d_j+d_k)-1} \left\{ \Re g_{jk} \cos\left((1 - (d_j + d_k)) \frac{\pi}{2}\right) \right. \\ &\quad \left. - \Im g_{jk} \sin\left((1 - (d_j + d_k)) \frac{\pi}{2}\right) \right\},\end{aligned}$$

with g_{jk} defined in (9.4.6). This yields (9.4.3) with r_{jk} defined as in (9.4.17). $\square$

The relation (9.4.15) sheds light on the phase parameters ϕ_{jk} . Note that $\phi_{jk} = 0$ if and only if $r_{jk} = r_{kj}$. In view of (9.1.5), the last property corresponds to $\gamma_{X,jk}(n)$ being symmetric at large positive and large negative lags, that is, as $n \rightarrow \infty$ and $n \rightarrow -\infty$.

We provide next examples of two vector LRD series: the vector FARIMA and the vector FGN series.

9.4.2 Vector FARIMA(0, D, 0) Series

The univariate FARIMA was defined in Section 2.4.1 and included the backshift operator B , defined as $Ba_n = a_{n-1}$. Here, we will also use the forward-shift operator B^{-1} as well, defined as $B^{-1}a_n = a_{n+1}$.

Let $D = \text{diag}(d_1, \dots, d_p)$ with $d_j < 1/2$, $j = 1, \dots, p$. Then,

$$(I - B)^{-D} = \begin{pmatrix} (I - B)^{-d_1} & 0 & \dots & 0 \\ 0 & (I - B)^{-d_2} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & (I - B)^{-d_p} \end{pmatrix}$$

and $(I - B^{-1})^{-D}$ is defined similarly, with B^{-1} replacing B . Let $Q_+ = (q_{jk}^+)$, $Q_- = (q_{jk}^-) \in \mathbb{R}^{p \times p}$, and $\{Z_n\}_{n \in \mathbb{Z}}$ be an $\mathbb{R}^p$ -valued white noise series, satisfying $\mathbb{E}Z_n = 0$ and $\mathbb{E}Z_n Z_n' = I$.⁸ Define a *vector FARIMA(0, D, 0) series* as

$$X_n = (I - B)^{-D} Q_+ Z_n + (I - B^{-1})^{-D} Q_- Z_n. \quad (9.4.18)$$

⁸ Do not confuse the I in the operator $(I - B)^{-D}$ with the identity matrix I , appearing for example in $\mathbb{E}Z_n Z_n' = I$.

Note that Q_+ and Q_- introduce a dependence structure among the components of the noise vector Z . Since the second term in (9.4.18) includes B^{-1} , the series X_n is given by a two-sided linear representation (when Q_- is not identically zero). See also the discussion at the end of this section, and Exercise 9.9.

In the next result, we give the exact form of the autocovariance matrix function of the vector FARIMA(0, D , 0) series in (9.4.18).

Proposition 9.4.3 *The (j, k) component $\gamma_{X,jk}(n)$ of the autocovariance matrix function $\gamma_X(n)$ of the vector FARIMA(0, D , 0) series in (9.4.18) is given by*

$$\gamma_{X,jk}(n) = \frac{1}{2\pi} \left(b_{jk}^1 \gamma_{jk}^1(n) + b_{jk}^2 \gamma_{jk}^2(n) + b_{jk}^3 \gamma_{jk}^3(n) + b_{jk}^4 \gamma_{jk}^4(n) \right), \quad (9.4.19)$$

where

$$\begin{aligned} b_{jk}^1 &= \sum_{t=1}^p q_{jt}^- q_{jt}^- = (Q^- (Q^-)^*)_{jk}, & b_{jk}^3 &= \sum_{t=1}^p q_{jt}^+ q_{jt}^+ = (Q^+ (Q^+)^*)_{jk}, \\ b_{jk}^2 &= \sum_{t=1}^p q_{jt}^- q_{jt}^+ = (Q^- (Q^+)^*)_{jk}, & b_{jk}^4 &= \sum_{t=1}^p q_{jt}^+ q_{jt}^- = (Q^+ (Q^-)^*)_{jk}, \end{aligned} \quad (9.4.20)$$

and

$$\begin{aligned} \gamma_{jk}^1(n) &= \gamma_{kj}^3(n) = 2\Gamma(1-d_j-d_k) \sin(d_k\pi) \frac{\Gamma(n+d_k)}{\Gamma(n+1-d_j)}, \\ \gamma_{jk}^4(n) &= \gamma_{jk}^2(-n) = \begin{cases} 2\pi \frac{1}{\Gamma(d_j+d_k)} \frac{\Gamma(n+d_j+d_k)}{\Gamma(n+1)} & , \quad n = 0, 1, 2, \dots, \\ 0 & , \quad n = -1, -2, \dots \end{cases} \end{aligned} \quad (9.4.21)$$

Proof By using Theorem 11.8.3 in Brockwell and Davis [186], the vector FARIMA(0, D , 0) series in (9.4.18) has the spectral density matrix

$$f_X(\lambda) = \frac{1}{2\pi} G(\lambda) G(\lambda)^*, \quad (9.4.22)$$

where $G(\lambda) = (1 - e^{-i\lambda})^{-D} Q_+ + (1 - e^{i\lambda})^{-D} Q_-$. This can be expressed component-wise as

$$f_{X,jk}(\lambda) = \frac{1}{2\pi} g_j(\lambda) g_k(\lambda)^*, \quad (9.4.23)$$

where g_j is the j th row of G . Then, the (j, k) component of the autocovariance matrix is

$$\begin{aligned} \gamma_{X,jk}(n) &= \int_0^{2\pi} e^{in\lambda} f_{X,jk}(\lambda) d\lambda = \frac{1}{2\pi} \int_0^{2\pi} e^{in\lambda} g_j(\lambda) g_k(\lambda)^* d\lambda \\ &= \frac{1}{2\pi} \left(b_{jk}^1 \gamma_{jk}^1(n) + b_{jk}^2 \gamma_{jk}^2(n) + b_{jk}^3 \gamma_{jk}^3(n) + b_{jk}^4 \gamma_{jk}^4(n) \right), \end{aligned} \quad (9.4.24)$$

where $b_{jk}^1, b_{jk}^2, b_{jk}^3, b_{jk}^4$ are given in (9.4.20), and

$$\begin{aligned} \gamma_{jk}^1(n) &= \gamma_{kj}^3(n) = \int_0^{2\pi} e^{in\lambda} (1 - e^{i\lambda})^{-d_j} (1 - e^{-i\lambda})^{-d_k} d\lambda, \\ \gamma_{jk}^2(n) &= \int_0^{2\pi} e^{in\lambda} (1 - e^{i\lambda})^{-(d_j+d_k)} d\lambda, & \gamma_{jk}^4(n) &= \int_0^{2\pi} e^{in\lambda} (1 - e^{-i\lambda})^{-(d_j+d_k)} d\lambda \end{aligned}$$

(note that γ^1 is expressed with the indices j, k but γ^3 is expressed with the reverse indices k, j). Note that $\gamma_{jk}^4(n) = \gamma_{jk}^2(-n)$ as stated in (9.4.21), so that it is enough to consider γ_{jk}^2 , in addition to, for example, γ_{jk}^1 .

We shall now use the relation

$$\begin{aligned} 2 \sin\left(\frac{\lambda}{2}\right) e^{\pm i(\lambda - \pi)/2} &= 2 \sin\left(\frac{\lambda}{2}\right) \left(\cos\left(\frac{\lambda}{2} - \frac{\pi}{2}\right) \pm i \sin\left(\frac{\lambda}{2} - \frac{\pi}{2}\right) \right) \\ &= 2 \sin\left(\frac{\lambda}{2}\right) \left(\sin\left(\frac{\lambda}{2}\right) \mp i \cos\left(\frac{\lambda}{2}\right) \right) \\ &= 2 \sin^2\left(\frac{\lambda}{2}\right) \mp i 2 \sin\left(\frac{\lambda}{2}\right) \cos\left(\frac{\lambda}{2}\right) \\ &= (1 - \cos \lambda) \mp i \sin \lambda = 1 - e^{\pm i\lambda}. \end{aligned} \quad (9.4.25)$$

We have

$$\begin{aligned} \gamma_{jk}^1(n) &= \frac{e^{i(d_j - d_k)\pi/2}}{2^{d_j + d_k}} \int_0^{2\pi} e^{in\lambda} \sin^{-d_j - d_k} \left(\frac{\lambda}{2}\right) e^{i\lambda(d_k - d_j)/2} d\lambda \\ &= \frac{2e^{i(d_j - d_k)\pi/2}}{2^{d_j + d_k}} \int_0^\pi e^{i\omega(2n + d_k - d_j)} \sin^{-d_j - d_k}(\omega) d\omega. \end{aligned}$$

By using Formula 3.892.1 in Gradshteyn and Ryzhik [421], p. 485, we deduce that

$$\gamma_{jk}^1(n) = \frac{2e^{i(d_j - d_k)\pi/2}}{2^{d_j + d_k}} \frac{\pi e^{i\beta\pi/2}}{2^{v-1} \nu B\left(\frac{\nu + \beta + 1}{2}, \frac{\nu - \beta + 1}{2}\right)},$$

where $\beta = 2n + d_k - d_j$ and $\nu = 1 - d_k - d_j$. Then,

$$\gamma_{jk}^1(n) = 2\pi(-1)^n \frac{\Gamma(1 - d_j - d_k)}{\Gamma(1 - d_j + n)\Gamma(1 - d_k - n)}. \quad (9.4.26)$$

Similar calculations yield

$$\gamma_{jk}^2(n) = 2\pi(-1)^n \frac{\Gamma(1 - d_j - d_k)}{\Gamma(1 - n)\Gamma(1 + n - d_j - d_k)}. \quad (9.4.27)$$

The relations (9.4.21) can now be deduced from (9.4.26) and (9.4.27) by using the identities $\Gamma(z)\Gamma(1-z) = \pi/\sin(\pi z)$ and $\Gamma(z)\Gamma(1-z) = (-1)^n \Gamma(n+z)\Gamma(1-n-z)$, $0 < z < 1$. $\square$

By using the relation (2.4.3), observe that, as $n \rightarrow \infty$,

$$\gamma_{jk}^1(n) \sim 2\Gamma(1 - d_j - d_k) \sin(d_j\pi) \frac{\Gamma(n + d_j)}{\Gamma(n + 1 - d_k)} = \frac{2\pi \sin(d_j\pi)}{\Gamma(d_j + d_k) \sin((d_j + d_k)\pi)} n^{d_j + d_k - 1}, \quad (9.4.28)$$

$$\gamma_{jk}^4(n) \sim \frac{2\pi}{\Gamma(d_j + d_k)} n^{d_j + d_k - 1}. \quad (9.4.29)$$

These relations show that FARIMA(0, D , 0) series satisfies condition II-v in Section 9.4.1. The relation (9.4.22) shows that the series also satisfies condition IV-v.

We now comment on the presence of the second term in (9.4.18). Note that $Q_- \equiv 0$ (that is, no second term in (9.4.18)) implies by (9.4.23) and (9.4.25) that

$$f_{X,jk}(\lambda) = \frac{(Q + Q_+^*)_{jk}}{2\pi} (1 - e^{-i\lambda})^{-d_j} (1 - e^{i\lambda})^{-d_k} \sim \frac{(Q + Q_+^*)_{jk}}{2\pi} e^{-i(d_j - d_k)\frac{\pi}{2}} \lambda^{-(d_j + d_k)},$$

as $\lambda \rightarrow 0$. Thus, such FARIMA(0, D , 0) series necessarily have the *phase parameters*

$$\phi_{jk} = -(d_j - d_k) \frac{\pi}{2}. \quad (9.4.30)$$

The presence of Q_- in (9.4.18) allows one to obtain arbitrary phase parameters. To see this, note from (9.4.22) that

$$f_X(\lambda) \sim \lambda^{-D} W W^* \lambda^{-D^*}, \quad (9.4.31)$$

where

$$W = \frac{1}{2\pi} (e^{-iD\frac{\pi}{2}} Q_+ + e^{iD\frac{\pi}{2}} Q_-)$$

as $\lambda \rightarrow 0$, since $i = e^{i\pi/2}$. The right-hand side of (9.4.31) then captures the general behavior (9.4.4) in condition IV-v with a matrix $G = W W^*$ for some matrix W as above. But since G is positive semidefinite Hermitian symmetric, we have $G = Z Z^*$ for some $Z \in \mathbb{C}^{p \times p}$ (see Section 9.1.1). It remains to note that there are real-valued Q_+ , Q_- such that Z equals W . (The latter observation follows from the fact that any $z \in \mathbb{C}$ can be expressed as $z = e^{-id\frac{\pi}{2}} q_+ + e^{id\frac{\pi}{2}} q_-$ for some $q_+, q_- \in \mathbb{R}$, since z can be viewed as a vector in the complex plane, and $e^{-id\frac{\pi}{2}}$ and $e^{id\frac{\pi}{2}}$ are linearly independent vectors.)

Remark 9.4.4 Though the presence of Q_- in (9.4.18) allows for general phase as indicated above, this can be achieved by more than one choice of the parameters Q_+ , Q_- of vector FARIMA(0, D , 0) series. For example, in the argument given above, Q_+ , Q_- can be replaced by $Q_+ O$ and $Q_- O$ for an orthogonal matrix $O \in \mathbb{R}^{p \times p}$ satisfying $O O^* = I$. This identifiability issue is also apparent in the simplest bivariate case $p = 2$ where the specification (9.4.6) of bivariate LRD has six parameters (namely, $d_1, d_2, \alpha_{11}, \alpha_{12}, \alpha_{22}, \phi_{12}$), whereas the bivariate FARIMA model has eight parameters even after identification up to the products $Q_+ Q_+^*$ and $Q_- Q_-^*$ (namely, d_1, d_2 , the three parameters of $Q_+ Q_+^*$ and the three parameters of $Q_- Q_-^*$). An identifiable bivariate FARIMA(0, D , 0) model with six parameters and general phase is introduced in Kechagias and Pipiras [550], along with extensions incorporating AR and MA polynomials, estimation and application to a real data set.

Remark 9.4.5 When $Q_- \equiv 0$, FARIMA(0, D , 0) series have a one-sided linear representation

$$X_n = \sum_{m=0}^{\infty} \Psi_m \epsilon_{n-m}, \quad (9.4.32)$$

where Ψ_m are coefficients in $\mathbb{R}^{q \times q}$. The entries $\psi_{jk,m}$ of these coefficients follow a power-law behavior in the sense that $\psi_{jk,m} \sim C m^{d-1}$, as $m \rightarrow \infty$. In fact, as shown in Kechagias and Pipiras [551], Section 4.1, a one-sided time series with power-law coefficients necessarily has the phase parameters (9.4.30). One-sided linear representations can lead to general phase parameters when using a new class of coefficients, called *trigonometric power-law coefficients* (see Kechagias and Pipiras [551] for more information).

9.4.3 Vector FGN Series

The univariate FGN was defined in Section 2.8. We will focus here on the vector FGN. Let B_H be a vector FBM with a diagonal self-similarity matrix $H = \text{diag}(h_1, \dots, h_p) \in \mathbb{R}^{p \times p}$ (see Definition 9.3.16). In particular, we assume that B_H admits a spectral representation (9.3.3) with matrix A . Hence, its covariance is given in Proposition 9.3.19. In analogy to (2.8.3), we define a *vector fractional Gaussian noise* (FGN) series as

$$X_n = B_H(n) - B_H(n-1), \quad n \in \mathbb{Z}. \quad (9.4.33)$$

The series $\{X_n\}_{n \in \mathbb{Z}}$ is Gaussian, zero mean and stationary. For the autocovariance matrix function of $\{X_n\}_{n \in \mathbb{Z}}$, one can show by using the covariance of B_H in (9.3.45), that

$$\gamma_{X,jk}(n) = \frac{\sigma_j \sigma_k}{2} \left\{ w_{jk}(n+1)|n+1|^{h_j+h_k} + w_{jk}(n-1)|n-1|^{h_j+h_k} - 2w_{jk}(n)|n|^{h_j+h_k} \right\}, \quad (9.4.34)$$

where w_{jk} are defined in Proposition 9.3.19. By using (9.3.46), if $h_j + h_k \neq 1$ (and $n \neq 0$),

$$\gamma_{X,jk}(n) = \frac{\sigma_j \sigma_k}{2} (\rho_{jk} - \text{sign}(n)\eta_{jk}) \left\{ |n+1|^{h_j+h_k} + |n-1|^{h_j+h_k} - 2|n|^{h_j+h_k} \right\} \quad (9.4.35)$$

and arguing as in the proof of Proposition 2.8.1,

$$\gamma_{X,jk}(n) \sim \frac{\sigma_j \sigma_k}{2} (\rho_{jk} - \text{sign}(n)\eta_{jk})(h_j + h_k)(h_j + h_k - 1)|n|^{h_j+h_k-2}, \quad (9.4.36)$$

as $n \rightarrow \infty$. If $h_j + h_k = 1$ (and $|n| \geq 2$),

$$\begin{aligned} \gamma_{X,jk}(n) &= -\frac{\sigma_j \sigma_k \eta_{jk}}{2} \text{sign}(n) \left\{ |n+1| \log |n+1| + |n-1| \log |n-1| - 2|n| \log |n| \right\} \\ &\sim \frac{\sigma_j \sigma_k \eta_{jk}}{2} \text{sign}(n) |n|^{-1}, \end{aligned} \quad (9.4.37)$$

as $|n| \rightarrow \infty$. Deriving (9.4.34), (9.4.35)–(9.4.36) and (9.4.37) is left as an exercise (Exercise 9.4).

By using the spectral representation (9.3.3), one can show as in the proof of Proposition 2.8.1, that the spectral density matrix of $\{X_n\}_{n \in \mathbb{Z}}$ is given by

$$f_X(\lambda) = \frac{|1 - e^{-i\lambda}|^2}{|\lambda|^2} \sum_{n=-\infty}^{\infty} \left\{ (\lambda + 2\pi n)_+^{-D} C (\lambda + 2\pi n)_+^{-D} + (\lambda + 2\pi n)_-^{-D} \bar{C} (\lambda + 2\pi n)_-^{-D} \right\}, \quad (9.4.38)$$

where $C = (c_{jk}) = AA^*$ and $D = H - (1/2)I$. As $\lambda \rightarrow 0$, $f_X(\lambda) \sim \lambda^{-D} C \lambda^{-D}$ or, when written component-wise,

$$f_{X,jk}(\lambda) \sim c_{jk} \lambda^{-(d_j+d_k)} = c_{jk} \lambda^{-(h_j+h_k-1)}. \quad (9.4.39)$$

In view of (9.4.36) and (9.4.39), when $h_j \in (1/2, 1)$, $j = 1, \dots, p$, the vector FGN series satisfies conditions II-v and IV-v with $d_j = h_j - 1/2$. As an exercise, the reader can also verify that the expression (9.4.36) for $\gamma_X(n)$ and the expression (9.4.39) for $f_X(\lambda)$ are consistent with (9.4.3), (9.4.6) and (9.4.17) relating g_{jk} to r_{jk} .

The class of vector FGN series captures the general asymptotic structure of the spectral density matrix given in condition IV-v. Indeed, if condition IV-v holds with a positive

semidefinite Hermitian symmetric matrix G , then $G = AA^*$ for some $A \in \mathbb{C}^{p \times p}$ (see Section 9.1.1). But then the asymptotic relation (9.4.6) is (9.4.39) for vector FGN series when the underlying vector FBM has the matrix A in its spectral representation (9.3.3).

9.4.4 Fractional Cointegration

Fractional cointegration is another interesting phenomenon arising with vector LRD series (the other phenomenon being the existence of possibly nonzero phases around the origin, as discussed in Section 9.4.1). We introduce it here in the stationary context (see Section 9.8 for further notes). The following notation will be convenient. An LRD univariate series with parameter $d \in (0, 1/2)$ is denoted $I(d)$. An SRD univariate series is denoted $I(0)$. We adopt below condition IV stated in Section 2.1 and condition IV-v stated in Section 9.4 for the definitions of LRD in the univariate and multivariate contexts, respectively.

Example 9.4.6 (*Fractional cointegration for bivariate series*) Consider the $\mathbb{R}^2$ -valued series

$$X_n = \begin{pmatrix} X_{1,n} \\ X_{2,n} \end{pmatrix} = \begin{pmatrix} u_n^{(0)} \\ au_n^{(0)} + u_n^{(1)} \end{pmatrix}, \quad n \in \mathbb{Z}, \quad (9.4.40)$$

where $a \in \mathbb{R}$ ($a \neq 0$) is a constant, and $\{u_n^{(0)}\}$ and $\{u_n^{(1)}\}$ are uncorrelated $I(d_0)$ and $I(d_1)$ series, respectively, with $0 \leq d_1 < d_0 < 1/2$. Note that both component series $\{X_{1,n}\}$ and $\{X_{2,n}\}$ are $I(d_0)$. This is the case for $\{X_{1,n}\}$ by the assumption on $\{u_n^{(0)}\}$, and it holds for $\{X_{2,n}\}$ since $d_1 < d_0$. Though each component series $\{X_n\}$, $j = 1, 2$, is $I(d_0)$, the linear combination $X_{1,n} - (1/a)X_{2,n} = -(1/a)u_n^{(1)}$ is $I(d_1)$ with $d_1 < d_0$. This is precisely the idea of fractional cointegration: for a vector LRD series, each component series can be $I(d_0)$ but their linear combination can be $I(d_1)$ with $d_1 < d_0$; that is, can be less dependent (and possibly SRD when $d_1 = 0$).

Definition 9.4.7 Suppose $\{X_n\}_{n \in \mathbb{Z}}$ is an $\mathbb{R}^p$ -valued LRD series such that each component series $\{X_{j,n}\}$, $j = 1, \dots, p$, is $I(d_0)$ with $d_0 \in (0, 1/2)$. The series $\{X_n\}_{n \in \mathbb{Z}}$ is called *fractionally cointegrated* if there is $\beta \in \mathbb{R}^p$ such that $\beta'X_n$ is $I(d)$ with $0 \leq d < d_0$. A vector β is called a *cointegrating vector*.

If $\{X_n\}_{n \in \mathbb{Z}}$ is an $\mathbb{R}^p$ -valued LRD series as in Definition 9.4.7, and $\beta \in \mathbb{R}^p$, note that $\beta'X_n$ cannot be expected $I(d)$ with $d > d_0$. Indeed, if $\{X_n\}_{n \in \mathbb{Z}}$ satisfies condition IV-v with the spectral density matrix $f_X(\lambda)$, then $\{\beta'X_n\}_{n \in \mathbb{Z}}$ has the spectral density

$$\beta'f_X(\lambda)\beta \sim (\beta'G\beta)\lambda^{-2d_0}, \quad (9.4.41)$$

as $\lambda \rightarrow 0$, where G is defined in (9.4.5). Since G is positive semidefinite, $\beta'G\beta \geq 0$. Then, either $\beta'G\beta > 0$ in which case $\{\beta'X_n\}_{n \in \mathbb{Z}}$ is $I(d_0)$, or $\beta'G\beta = 0$ in which case $\{\beta'X_n\}_{n \in \mathbb{Z}}$ cannot be $I(d)$ with $d > d_0$. Note also that this argument shows that, for a cointegrating vector β ,

$$\beta'G\beta = 0. \quad (9.4.42)$$

Note that the converse does not have to be true; that is, if (9.4.42) holds, we can only deduce that $\{\beta' X_n\}_{n \in \mathbb{Z}}$ has the spectral density $\beta' f_X(\lambda) \beta$ satisfying $(\beta' f_X(\lambda) \beta) \lambda^{2d_0} \rightarrow 0$, whereas for a cointegrating vector β , we should have $\beta' f_X(\lambda) \beta \sim c \lambda^{-2d}$ with $d < d_0$.

There may be several, linearly independent cointegrating vectors. Their number has a special name.

Definition 9.4.8 The number of linearly independent cointegrating vectors, r , $1 \leq r \leq p - 1$, is called the *cointegrating rank*.

Under additional assumptions, one can arrive at a more refined fractional cointegration model as follows. Suppose that the matrix G in condition IV-v is real-valued. Suppose also that every vector β satisfying (9.4.42) is a cointegrating vector. Then,

$$r = p - \text{rank}(G), \quad (9.4.43)$$

where $\text{rank}(G)$ denotes the rank of a matrix G . In particular, there are linearly independent vectors $\beta_{0,1}, \dots, \beta_{0,p-r}, \eta_{0,1}, \dots, \eta_{0,r} \in \mathbb{R}^p$ such that

$$\begin{pmatrix} \beta'_{0,1} \\ \vdots \\ \beta'_{0,p-r} \end{pmatrix} X_n = \begin{pmatrix} u_{1,n}^{(0)} \\ \vdots \\ u_{p-r,n}^{(0)} \end{pmatrix} =: u_n^{(0)}, \quad (9.4.44)$$

where each $\{u_{j,n}^{(0)}\}_{n \in \mathbb{Z}}$, $j = 1, \dots, p - r$, is $I(d_0)$, and

$$\begin{pmatrix} \eta'_{0,1} \\ \vdots \\ \eta'_{0,r} \end{pmatrix} X_n = \begin{pmatrix} v_{1,n}^{(0)} \\ \vdots \\ v_{r,n}^{(0)} \end{pmatrix} =: v_n^{(0)}, \quad (9.4.45)$$

where for each $j = 1, \dots, r$, $\{v_{j,n}^{(0)}\}$ is $I(d_{0,j})$ with $d_{0,j} < d_0$. Moreover, the $\mathbb{R}^{p-r}$ -valued series $\{u_n^{(0)}\}$ is not fractionally cointegrated. We may also suppose without loss of generality that $d_{0,r} \leq \dots \leq d_{0,1}$.

Let now $p_1 \geq 1$ be such that $d_{0,p_1+1} < d_{0,p_1} = \dots = d_{0,1} =: d_1$; that is, the series $\{v_{1,n}^{(0)}\}, \dots, \{v_{p_1,n}^{(0)}\}$ are all $I(d_1)$. Let r_1 be the cointegrating rank of $v_n^{(0)}$ and suppose as above that $r_1 = p_1 - \text{rank}(G_1)$, where $\tilde{G}_1$ is the spectral density matrix of $v_n^{(0)}$. Then, as in (9.4.44) and (9.4.45), there are linearly independent vectors $\eta_{1,1}, \dots, \eta_{1,p_1-r_1}, \eta_{1,p_1-r_1+1}, \dots, \eta_{1,p_1}$ such that

$$\begin{pmatrix} \eta'_{1,1} \\ \vdots \\ \eta'_{1,p_1-r_1} \\ \eta'_{1,p_1-r_1+1} \\ \vdots \\ \eta'_{1,p_1} \end{pmatrix} v_n^{(0)} = \begin{pmatrix} u_{1,n}^{(1)} \\ \vdots \\ u_{p_1-r_1,n}^{(1)} \\ v_{1,n}^{(1)} \\ \vdots \\ v_{r_1,n}^{(1)} \end{pmatrix} =: \begin{pmatrix} u_n^{(1)} \\ v_n^{(1)} \end{pmatrix}, \quad (9.4.46)$$

where each component series $\{u_{j,n}^{(1)}\}$ of $\{u_n^{(1)}\}$ is $I(d_1)$, $\{u_n^{(1)}\}$ is not fractionally cointegrated, and a component series $\{v_{j,n}^{(1)}\}$ of $v_n^{(1)}$ is $I(d_{1,j})$ with $d_{1,j} < d_1$.

Let $a_0 = p - r$ and $a_1 = p_1 - r_1$. By combining (9.4.44), (9.4.45) and (9.4.46), there are $\beta_0 \in \mathbb{R}^{p \times a_0}$, $\beta_1 \in \mathbb{R}^{p \times a_1}$ and $\eta_2 \in \mathbb{R}^{p \times (p - a_0 - a_1)}$ with $(\beta_0, \beta_1, \eta_2)$ consisting of linearly independent vectors so that

$$\begin{pmatrix} \beta'_0 \\ \beta'_1 \\ \eta'_2 \end{pmatrix} X_n = \begin{pmatrix} u_n^{(0)} \\ u_n^{(1)} \\ v_n^{(1)} \end{pmatrix}, \quad (9.4.47)$$

where $\{u_n^{(0)}\}$, $\{u_n^{(1)}\}$ and $\{v_n^{(1)}\}$ are, respectively, $\mathbb{R}^{a_0}$ -, $\mathbb{R}^{a_1}$ - and $\mathbb{R}^{p - a_0 - a_1}$ -valued time series. Each component series of $\{u_n^{(0)}\}$ is $I(d_0)$. Each component series of $\{u_n^{(1)}\}$ is $I(d_1)$ with $d_1 < d_0$. Each component series of $\{v_n^{(1)}\}$ is $I(d)$ with $d < d_1$. Moreover, $\{u_n^{(0)}\}$ is not fractionally cointegrated, and neither is $\{u_n^{(1)}\}$.

By continuing the above procedure on $\{v_n^{(1)}\}$ and so on, and making suitable assumptions, we can conclude that there are $\beta_j \in \mathbb{R}^{p \times a_j}$, $j = 1, \dots, s$, $a_0 + \dots + a_s = p$, with $(\beta_j, j = 0, \dots, s)$ consisting of linearly independent vectors such that

$$\begin{pmatrix} \beta'_0 \\ \beta'_1 \\ \beta'_2 \\ \vdots \\ \beta'_s \end{pmatrix} X_n = \begin{pmatrix} u_n^{(0)} \\ u_n^{(1)} \\ \vdots \\ u_n^{(s)} \end{pmatrix} =: u_n, \quad (9.4.48)$$

where $\{u_n^{(j)}\}$, $j = 1, \dots, s$, are $\mathbb{R}^{a_j}$ -valued series which are not fractionally cointegrated and whose components series are $I(d_j)$ with

$$0 \leq d_s < d_{s-1} < \dots < d_1 < d_0. \quad (9.4.49)$$

Since $B = (\beta_0, \dots, \beta_s)$ consists of linearly independent vectors, the relation (9.4.48) can be inverted as $X_n = B^{-1}u_n =: Au_n$ and hence written as the following *fractional cointegration model*

$$X_n = Au_n = A_0u_n^{(0)} + A_1u_n^{(1)} + \dots + A_su_n^{(s)}, \quad (9.4.50)$$

where A_j , $j = 0, \dots, s$, are $\mathbb{R}^{p \times a_j}$ full rank matrices. Conversely, any series $\{X_n\}_{n \in \mathbb{Z}}$ given by (9.4.50) is fractionally cointegrated with the rank $r = p - a_0$.

There is a useful terminology of cointegrating subspaces related to the model (9.4.50). For any matrix A , let $\mathcal{M}(A)$ be the space spanned by the columns of A , and let $\mathcal{M}^\perp(A)$ denote the orthogonal complement of $\mathcal{M}(A)$. For the matrices $A_1, \dots, A_s$ in (9.4.50), let $\mathcal{B}_0 = \mathcal{M}(A_0)$, and $\mathcal{B}_k$, $k = 1, \dots, s$, be the subspaces such that

$$\mathcal{M}^\perp(A_0, \dots, A_{k-1}) = \mathcal{M}^\perp(A_0, \dots, A_k) \oplus \mathcal{B}_k \quad (9.4.51)$$

and

$$\mathcal{B}_k \perp \mathcal{M}^\perp(A_0, \dots, A_k). \quad (9.4.52)$$

One has $\mathcal{B}_j \perp \mathcal{B}_k$, $j \neq k$, $j, k = 0, \dots, s$, and

$$\mathbb{R}^p = \mathcal{B}_0 \oplus \mathcal{B}_1 \oplus \dots \oplus \mathcal{B}_s. \quad (9.4.53)$$

Note also that a nonzero vector $\beta \in \mathcal{B}_k$, $k = 1, \dots, s$, satisfies

$$\beta' A_l = 0, \quad l = 0, \dots, k-1, \quad \beta' A_k \neq 0. \quad (9.4.54)$$

The relation (9.4.54) implies that any nonzero vector $\beta \in \mathcal{B}_k$, $k = 1, \dots, s$, produces a fractionally cointegrated error series $\{\beta' X_n\}_{n \in \mathbb{Z}}$ which is $I(d_k)$. The space $\mathcal{B}_0$, on the other hand, consists of non-cointegrating vectors. This motivates the following definition.

Definition 9.4.9 The spaces $\mathcal{B}_k$, $k = 1, \dots, s$, defined in (9.4.51) and (9.4.52) above are called the *cointegrating spaces*.

The following simple example illustrates the terminology and notation introduced above.

Example 9.4.10 (*Fractional cointegration for bivariate series*) Let $\{X_n\}_{n \in \mathbb{Z}}$ be the series given in (9.4.40) of Example 9.4.6. It can be expressed as

$$X_n = \begin{pmatrix} u_n^{(0)} \\ au_n^{(0)} + u_n^{(1)} \end{pmatrix} = \begin{pmatrix} 1 \\ a \end{pmatrix} u_n^{(0)} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u_n^{(1)} = A_0 u_n^{(0)} + A_1 u_n^{(1)},$$

where $\{u_n^{(0)}\}$ and $\{u_n^{(1)}\}$, respectively, $\mathbb{R}^{a_0}$ - and $\mathbb{R}^{a_1}$ -valued time series with $a_0 = 1$ and $a_1 = 1$, thus univariate time series. It has the representation (9.4.50) with $A_0 = (1 \ a)'$, $A_1 = (0 \ 1)'$. Since $p = 2$, X_n has cointegrating rank $r = p - a_1 = 2 - 1 = 1$. One has $X_n = Au_n$ and thus $BX_n = u_n$ where $B = A^{-1} = (B_0, B_1) = \begin{pmatrix} 1 & 0 \\ -a & 1 \end{pmatrix}$ and thus $s = 1$. Finally, the space $\mathcal{B}_0 = \mathcal{M}(A_0) = \{c(1 \ a) : c \in \mathbb{R}\}$ and the single cointegrating space $\mathcal{B}_1 = \{c(-a \ 1) : c \in \mathbb{R}\}$, consisting of vectors orthogonal to those in $\mathcal{B}_0$, are both one-dimensional.

Note also that the cointegration model (9.4.50) is naturally related to vector operator self-similarity in (9.3.1). The following example expands on this point.

Example 9.4.11 (*Fractional cointegration and vector OFBM*) Let B_H be a vector OFBM with the diagonal self-similarity matrix

$$H = \text{diag}(\underbrace{h_0, \dots, h_0}_{a_0}, \dots, \underbrace{h_s, \dots, h_s}_{a_s}), \quad (9.4.55)$$

where $a_0 + \dots + a_{s-1} + a_s = p$ and

$$\frac{1}{2} \leq h_s < \dots < h_0 < 1. \quad (9.4.56)$$

Let also B_H be given by its spectral representations (9.3.3) with a positive definite matrix AA^* . Consider the process

$$X(t) = QB_H(t), \quad (9.4.57)$$

where $Q \in \mathbb{R}^{p \times p}$ is invertible. Note that the process X is zero mean, Gaussian and has stationary increments. Moreover, for $c > 0$,

$$\begin{aligned} \{X(ct)\}_{t \in \mathbb{R}} &= \{QB_H(ct)\}_{t \in \mathbb{R}} \stackrel{d}{=} \{Qc^H B_H(t)\}_{t \in \mathbb{R}} = \{Qc^H Q^{-1}X(t)\}_{t \in \mathbb{R}} \\ &= \{c^{QH} Q^{-1}X(t)\}_{t \in \mathbb{R}}, \end{aligned} \quad (9.4.58)$$

by using (9.2.4). Thus, X is vector OFBM with the (possibly non-diagonal) self-similarity matrix $QH Q^{-1}$. In view of (9.4.57), we can consider now the stationary vector FGN series

$$X_n = X(n) - X(n-1) = Q(B_H(n) - B_H(n-1)) =: Qu_n. \quad (9.4.59)$$

The series $\{u_n\}$ in (9.4.59) has the properties of that in (9.4.48) with $d_j = h_j - 1/2$, $j = 1, \dots, s$. The relation (9.4.59) corresponds to the fractional cointegration model (9.4.50) with A replaced by Q .

9.5 Operator Fractional Brownian Fields

We consider here extensions of FBM to the random fields setting $q \geq 1$, $p = 1$ (see Section 9.2). In this setting, recall that the operator self-similarity (9.2.1) reads, for any $c > 0$,

$$\{X(c^E t)\}_{t \in \mathbb{R}^q} \stackrel{d}{=} \{c^H X(t)\}_{t \in \mathbb{R}^q}, \quad (9.5.1)$$

where X is real-valued, $E \in \mathbb{R}^{q \times q}$ and $H \in \mathbb{R}$. Note that replacing c by $c^{1/H}$ (supposing $H \neq 0$) and denoting $E_H = E/H$, the relation (9.5.1) is equivalent to

$$\{X(c^{E_H} t)\}_{t \in \mathbb{R}^q} \stackrel{d}{=} \{cX(t)\}_{t \in \mathbb{R}^q}, \quad (9.5.2)$$

where $E_H \in \mathbb{R}^{q \times q}$. Though (9.5.2) is simpler and as general as (9.5.1), we shall take (9.5.1) for the definition of operator self-similarity in the random fields setting. One reason for this is that (9.5.1) is often considered in the literature with the constraint

$$\text{tr}(E) = q, \quad (9.5.3)$$

where $\text{tr}(E)$ denotes the trace of the matrix E . The idea behind (9.5.3) is that when $E = I$ and hence $\text{tr}(E) = q$, the two sides of the relation (9.5.1) become $X(ct)$ and $c^H X(t)$, in which case H can be viewed as the usual self-similarity parameter along all the q axes of $\mathbb{R}^q$. Considering (9.5.1) also provides a more general formulation in the sense that results for (9.5.2) can be deduced readily from those for (9.5.1) by setting $H = 1$. It should also be noted that operator self-similarity (9.5.3) is often called *operator scaling* in the spatial setting and, for example, operator self-similar Gaussian random fields are known as operator scaling Gaussian random fields.

Definition 9.5.1 A (real-valued) random field $\{X(t)\}_{t \in \mathbb{R}^q}$ is called an *operator fractional Brownian field* (OFBF) if it is Gaussian, has stationary increments and is operator self-similar in the sense of (9.5.1) with exponents $E \in \mathbb{R}^{q \times q}$ and $H > 0$.

The stationarity of increments in Definition 9.5.1 means, as in (2.5.2), that, for any $h \in \mathbb{R}^q$,

$$\{X(t+h) - X(h)\}_{t \in \mathbb{R}^q} \stackrel{d}{=} \{X(t) - X(0)\}_{t \in \mathbb{R}^q}.$$

We shall denote an OFBF by $\{B_{E,H}(t)\}_{t \in \mathbb{R}^q}$. Integral representations and basic properties of OFBFs are given in Section 9.5.2, and several special classes and examples are considered in Section 9.5.3. The next section provides some useful facts on the so-called homogeneous functions which will play an important role.

Remark 9.5.2 As with operator self-similar processes (see Section 9.3.4 and, in particular, the relation (9.3.25)), the operator self-similarity exponents E are generally not unique for the same random field X and fixed H . More specifically, as shown under mild assumptions in, e.g., Didier, Meerschaert, and Pipiras [309],

$$\mathcal{E}_H(X) = E + T(G_X^{\text{dom}}),$$

where $\mathcal{E}_H(X)$ is the set of all operator self-similarity exponents E satisfying (9.5.1),

$$G_X^{\text{dom}} = \left\{ A \in GL(q) : \{X(t)\}_{t \in \mathbb{R}} \stackrel{d}{=} \{X(At)\}_{t \in \mathbb{R}} \right\}$$

is the domain symmetry group, and $T(G_X^{\text{dom}})$ denotes the tangent space (defined in (9.3.54)).

9.5.1 M -Homogeneous Functions

We will need the notion of homogeneous functions defined next.

Definition 9.5.3 Let M be an $\mathbb{R}^{q \times q}$ matrix. A function $\phi : \mathbb{R}^q \rightarrow \mathbb{C}$ is called M -homogeneous if, for all $c > 0$ and $x \in \mathbb{R}^q \setminus \{0\}$,

$$\phi(c^M x) = c\phi(x). \quad (9.5.4)$$

When $q = 1$ and $M \neq 0$ is a real number, an M -homogeneous function can be written as a power function

$$\phi(x) = c_1 x_+^{1/M} + c_2 x_-^{1/M}$$

for constants $c_1 = \phi(1)$ and $c_2 = \phi(-1)$, which follows by taking $c = x$, $x = 1$ for $x > 0$ and $c = -x$, $x = -1$ for $x < 0$ in (9.5.4). Thus $\phi(bx) = b^{1/M} \phi(x)$ if $b > 0$ and $x \neq 0$. If $M = 1$, one would get the usual homogeneous function.

Homogeneous functions can thus be viewed as generalizations of power functions to higher dimensions. In higher dimensions, however, they cannot be expressed in general through elementary power functions in a simple and explicit way. Power functions were used in the various representations of fractional Brownian motion (see Proposition 2.6.5 and Remark 2.6.8, and Proposition 2.6.11). Homogeneous functions will appear similarly in the representations of operator fractional Brownian fields (see Theorems 9.5.9 and 9.5.11 below).

For example, the r th norm $\phi(x) = \|x\|_r = (|x_1|^r + \cdots + |x_q|^r)^{1/r}$, $r \geq 1$, is M -homogeneous with M equal to the $q \times q$ identity matrix I since $\phi(c^I x) = \|c^I x\|_r = c\|x\|_r = c\phi(x)$.

Lemma 9.5.4 Suppose that ϕ is M -homogeneous and let $a \in \mathbb{R}$, $a \neq 0$. Then $\psi(x) = (\phi(x))^a$ is (M/a) -homogeneous.

Proof Indeed, note that

$$\psi(c^{M/a}x) = [\phi(c^{M/a}x)]^a = [\phi((c^{1/a})^Mx)]^a = [c^{1/a}\phi(x)]^a = c\psi(x). \quad \square$$

We will also need a “polar coordinate” representation of $x \in \mathbb{R}^q$. Let M be an $\mathbb{R}^{q \times q}$ matrix and suppose that its eigenvalues have positive real parts. By Lemma 6.1.5 in Meerschaert and Scheffler [710] and a change of variables $s = e^{-t}$,

$$\|x\|_0 = \int_0^\infty \|e^{-Mt}x\| dt = \int_0^1 \|s^Mx\| s^{-1} ds \quad (9.5.5)$$

defines a norm on $\mathbb{R}^q$, where $\|\cdot\|$ is any norm on $\mathbb{R}^q$. Moreover, with the norm $\|\cdot\|_0$ and the unit sphere $S_0 = \{x \in \mathbb{R}^q : \|x\|_0 = 1\}$, the mapping $\Psi : (0, \infty) \times S_0 \rightarrow \mathbb{R}^q \setminus \{0\}$, $\Psi(r, \theta) = r^M\theta$, is a homeomorphism.⁹ Since Ψ is one-to-one, onto and has a continuous inverse, any $x \in \mathbb{R}^q \setminus \{0\}$ can be written uniquely as

$$x = (\tau_M(x))^M l_M(x), \quad (9.5.6)$$

for some *radial part* $\tau_M(x) > 0$ and some *direction* $l_M(x) \in S_0$ such that $\tau_M(x)$ and $l_M(x)$ are continuous in x . Since $\tau_M(x) \rightarrow 0$, as $x \rightarrow 0$, one can extend $\tau_M(x)$ to 0 continuously by setting $\tau_M(0) = 0$.

Note from the uniqueness of the vector representation (9.5.6) that $\tau_M(-x) = \tau_M(x)$ and $l_M(-x) = -l_M(x)$. Similarly, for $c > 0$, since $c^Mx = (c\tau_M(x))^M l_M(x)$, such uniqueness also yields

$$\tau_M(c^Mx) = c\tau_M(x), \quad l_M(c^Mx) = l_M(x). \quad (9.5.7)$$

In particular, the function $\tau_M(x)$ is M -homogeneous. Note also that, for $x = r^M\theta$ with $r > 0$ and $\theta \in S_0$,

$$\tau_M(\theta) = 1, \quad \tau_M(r^M\theta) = r, \quad l_M(r^M\theta) = \theta, \quad (9.5.8)$$

by the uniqueness of the polar coordinate representation.

Example 9.5.5 (*Polar coordinate representation associated with the identity matrix*) If $M = m_1 \cdot I$, where $m_1 > 0$ and I is the $q \times q$ identity matrix, then $\|x\|_0$ in (9.5.5) can be taken to be the Euclidean norm $\|x\|_2 = (|x_1|^2 + \cdots + |x_q|^2)^{1/2}$, and $S_0 = \{x \in \mathbb{R}^q : \|x\|_2 = 1\}$. Indeed, with $\|\cdot\| = m_1\|\cdot\|_2$ and $M = m_1 \cdot I$ in (9.5.5), note that $\|x\|_0 = \|x\|_2 m_1 \int_0^\infty e^{-m_1 t} dt = \|x\|_2$. Moreover,

$$\tau_M(x) = \|x\|_2^{1/m_1}, \quad l_M(x) = \left(\frac{x_1}{\|x\|_2}, \dots, \frac{x_q}{\|x\|_2} \right) = \frac{x}{\|x\|_2},$$

by the uniqueness of the polar coordinate representation.

Remark 9.5.6 The exponent M of a homogeneous function ϕ is not unique in general. By Theorem 5.2.13 in Meerschaert and Scheffler [710], the set of possible exponents has the form $M + T(S_\phi)$, where T denotes the tangent space as in (9.3.54) and $S_\phi = \{A : \phi(Ax) = \phi(x)\}$ is the so-called symmetry group of ϕ . For example, with $M = m_1 \cdot I$

⁹ $\Psi : (0, \infty) \times S_0 \rightarrow \mathbb{R}^q \setminus \{0\}$, $\Psi(r, \theta) = r^M\theta$, is a homeomorphism if Ψ is continuous, one-to-one, onto and has a continuous inverse.

and $\phi(x) = \|x\|_2^{1/m_1}$, $S_\phi = O(q)$, $T(S_\phi) = so(q)$ and the set of possible exponents is $M + so(q)$ (cf. Example 9.3.26). Thus, one can add to M any skewed symmetric matrix. For further discussion, see Remark 2.10 in Biermé, Meerschaert, and Scheffler [151].

Polar coordinate representations provide a convenient means of expressing M -homogeneous functions ϕ . Indeed, suppose $\phi(c^M x) = c\phi(x)$ and represent x through polar coordinates as $x = (\tau_M(x))^M l_M(x)$. Then, taking $c = \tau_M(x)^{-1}$ yields $\phi(\tau_M(x)^{-M} x) = \tau_M(x)^{-1} \phi(x)$ and hence

$$\phi(x) = \tau_M(x) \phi(\tau_M(x)^{-M} x) = \tau_M(x) \phi(l_M(x)), \quad (9.5.9)$$

since $x = (\tau_M(x))^M l_M(x)$. In particular, ϕ is determined by its values on S_0 through $\phi(l_M(x))$. Conversely, by using (9.5.7), the function

$$\tau_M(x) \phi_0(l_M(x)) \text{ is } M\text{-homogeneous}, \quad (9.5.10)$$

for any function ϕ_0 on S_0 .

The proof of the next technical lemma illustrates the use of the polar coordinate representation (9.5.6). The lemma is used in the proof of Theorem 9.5.11 below.

Lemma 9.5.7 *Let M be an $\mathbb{R}^{q \times q}$ matrix with eigenvalues having positive real parts. If the function $h : \mathbb{R}^q \rightarrow \mathbb{C}$ is such that, for all $c > 0$,*

$$h(c^M x) = ch(x) \quad \text{a.e. } dx, \quad (9.5.11)$$

then there is an M -homogeneous function ϕ such that $\phi(x) = h(x)$ a.e. dx .

Proof By the Fubini-based argument found following (9.3.17),

$$h(c^M x) = ch(x) \quad \text{a.e. } dcdx.$$

By using (9.5.6), we get

$$h(x) = c^{-1} h(c^M x) = c^{-1} h((c\tau_M(x))^M l_M(x)) \quad \text{a.e. } dcdx.$$

By making the change of variables (c, x) to $(c\tau_M(x)^{-1}, x)$, and using (9.5.6), we also have that

$$h(x) = c^{-1} \tau_M(x) h(c^M l_M(x)) \quad \text{a.e. } dcdx.$$

Now, there is c_0 such that the last relation holds a.e. dx ; that is,

$$h(x) = \tau_M(x) c_0^{-1} h(c_0^M l_M(x)) \quad \text{a.e. } dx$$

or

$$h(x) = \phi(x) \quad \text{a.e. } dx,$$

where $\phi(x) = \tau_M(x) \phi_0(l_M(x))$ with $\phi_0(\theta) = c_0^{-1} h(c_0^M \theta)$ for $\theta \in S_0$. As noted in (9.5.10), the function ϕ is M -homogeneous. $\square$

Example 9.5.8 (*Large class of homogeneous functions*) Let $e_1, \dots, e_q$ and $\theta_1, \dots, \theta_q$ be, respectively, the eigenvalues and eigenvectors of a matrix M^* : that is, $M^*\theta_j = e_j\theta_j$. Suppose that $e_1, \dots, e_q \in \mathbb{R} \setminus \{0\}$ and denote the inner product for $u, v \in \mathbb{R}^q$ by $\langle u, v \rangle = u^*v$. Then, for $C_1, \dots, C_q \geq 0$ and $\rho > 0$, the function

$$\phi(x) = \left(\sum_{j=1}^q C_j |\langle x, \theta_j \rangle|^{\rho/e_j} \right)^{1/\rho}$$

is M -homogeneous. This follows from observing that

$$(c^M x)^*\theta_j = x^*c^{M^*}\theta_j = x^* \sum_{k=0}^{\infty} \frac{(\log c)^k (M^*)^k}{k!} \theta_j = x^* \sum_{k=0}^{\infty} \frac{(\log c)^k e_j^k}{k!} \theta_j = c^{e_j} x^*\theta_j.$$

Consequently,

$$\phi(c^M x) = \left(\sum_{j=1}^q C_j |\langle c^M x, \theta_j \rangle|^{\rho/e_j} \right)^{1/\rho} = c \left(\sum_{j=1}^q C_j |\langle x, \theta_j \rangle|^{\rho/e_j} \right)^{1/\rho} = c\phi(x).$$

We shall use below several other auxiliary results related to the notions introduced above. The first result is a useful integration in polar coordinates formula. As above, let $M \in \mathbb{R}^{q \times q}$ with eigenvalues having positive real parts $0 < a_1 < \dots < a_q$. By Proposition 2.3 in Biermé et al. [151], for $f \in L^1(\mathbb{R}^q, dx)$,

$$\int_{\mathbb{R}^q} f(x) dx = \int_0^\infty \int_{S_0} f(r^M \theta) \sigma(d\theta) r^{\text{tr}(M)-1} dr, \quad (9.5.12)$$

for a unique finite Radon¹⁰ measure σ on S_0 , where $\text{tr}(M)$ denotes the trace of the matrix M .

We shall also use several bounds on a norm in terms of the radial part. Let $\|\cdot\|$ be any norm on $\mathbb{R}^q$ and let x be as in (9.5.6), namely $x = (\tau_M(x))^M l_M(x)$. We will first bound $\|x\|$ when $\|x\| \leq 1$, then when $\|x\| \geq 1$.

Suppose first $\|x\| \leq 1$, so that $r = \tau_M(x) \leq 1$ and consider any $\theta \in S_0$. Then, as shown in Lemma 2.1 in Biermé et al. [151] (and by using the fact that all norms in $\mathbb{R}^q$ are equivalent),

$$C_1 \tau_M(x)^{a_q+\delta} \leq \|x\| \leq C_2 \tau_M(x)^{a_1-\delta},$$

for arbitrarily small $\delta > 0$. This can be expressed in an expanded form as

$$C_1 r^{a_q+\delta} = C_1 \tau_M(x)^{a_q+\delta} \leq \|x\| = \|\tau_M(x)^M l_M(x)\| = \|r^M \theta\| \leq C_2 \tau_M(x)^{a_1-\delta} = C_2 r^{a_1-\delta} \quad (9.5.13)$$

for arbitrarily small $\delta > 0$. Suppose now $\|x\| \geq 1$, so that $r = \tau_M(x) \geq 1$ and again consider any $\theta \in S_0$. Then, as shown in Lemma 2.1 in Biermé et al. [151],

$$C_3 \tau_M(x)^{a_1-\delta} \leq \|x\| \leq C_4 \tau_M(x)^{a_q+\delta},$$

¹⁰ That is, a measure on the Borel sets of S_0 which is finite on compact sets of S_0 , and is inner regular in the sense that, for any Borel set A of S_0 and $\epsilon > 0$, there is a compact set K_ϵ such that $\sigma(A \setminus K_\epsilon) < \epsilon$.

for arbitrarily small $\delta > 0$. In an expanded form, this can also be written as

$$C_3 r^{a_1 - \delta} = C_3 \tau_M(x)^{a_1 - \delta} \leq \|x\| = \|\tau_M(x)^M l(x)\| = \|r^M \theta\| \leq C_4 \tau_M(x)^{a_q + \delta} = C_4 r^{a_q + \delta} \quad (9.5.14)$$

for arbitrarily small $\delta > 0$.

9.5.2 Integral Representations

We shall provide below integral representations of OFBFs in the spectral and time domains. The following result provides a spectral domain representation of OFBF.

Theorem 9.5.9 *Let $B_{E,H} = \{B_{E,H}(t)\}_{t \in \mathbb{R}^q}$ be an OFBF with $E \in \mathbb{R}^{q \times q}$ whose eigenvalues have positive real parts and $H > 0$. Assume also that $E \neq H \cdot I$. Then, $B_{E,H}$ has a spectral representation*

$$\{B_{E,H}(t)\}_{t \in \mathbb{R}^q} \stackrel{d}{=} \left\{ \int_{\mathbb{R}^q} (e^{i\langle x, t \rangle} - 1) \widehat{B}_F(dx) \right\}_{t \in \mathbb{R}^q}, \quad (9.5.15)$$

where $\widehat{B}_F$ is an Hermitian Gaussian random measure on $\mathbb{R}^q$ with the control measure $\mathbb{E}|\widehat{B}_F(dx)|^2 = F(dx)$ such that

$$F(c^{-E^*} dx) = c^{2H} F(dx) \quad (9.5.16)$$

or, equivalently, the corresponding distribution function F is $(-E^*/(2H))$ -homogeneous; that is,

$$F(c^{-E^*/(2H)} x) = c F(x). \quad (9.5.17)$$

If F is absolutely continuous with spectral density f , then $B_{E,H}$ can also be represented as

$$\{B_{E,H}(t)\}_{t \in \mathbb{R}^q} \stackrel{d}{=} \left\{ \int_{\mathbb{R}^q} (e^{i\langle x, t \rangle} - 1) g(x)^{-H - \text{tr}(E^*)/2} \widehat{B}(dx) \right\}_{t \in \mathbb{R}^q} \quad (9.5.18)$$

$$\stackrel{d}{=} \left\{ \int_{\mathbb{R}^q} (e^{i\langle x, t \rangle} - 1) \tau_{E^*}(x)^{-H - \text{tr}(E^*)/2} g_0(l_{E^*}(x)) \widehat{B}(dx) \right\}_{t \in \mathbb{R}^q}, \quad (9.5.19)$$

where $\widehat{B}$ is an Hermitian Gaussian random measure on $\mathbb{R}^q$ with the control measure dx , $\text{tr}(E^*)$ denotes the trace of the matrix E^* , g is a nonnegative, symmetric, E^* -homogeneous function and

$$g_0(y) = g(y)^{-H - \text{tr}(E^*)/2}. \quad (9.5.20)$$

Moreover, the function g_0 is $-E^*/(H + \text{tr}(E^*)/2)$ -homogeneous, and

$$g_0(y) = f(y)^{1/2} \quad \text{a.e. } dy. \quad (9.5.21)$$

Conversely, suppose that

$$0 < H < \min\{\Re(e_j) : j = 1, \dots, q\}, \quad (9.5.22)$$

where e_j are the eigenvalues of the matrix E . Suppose also that the function g_0 in (9.5.19) is bounded on S_0 . Then, the right-hand side of (9.5.19) is a well-defined OFBF with exponents E and H .

Proof We first prove (9.5.15). An OFBF $B_{E,H}$ has stationary increments and hence can be represented as

$$\{B_{E,H}(t)\}_{t \in \mathbb{R}^q} \stackrel{d}{=} \left\{ \int_{\mathbb{R}^q} (e^{i\langle x,t \rangle} - 1) \widehat{B}_F(dx) + \langle X, t \rangle + Y \right\}_{t \in \mathbb{R}^q}, \quad (9.5.23)$$

where $\widehat{B}_F(dx)$ is an Hermitian Gaussian random measure on $\mathbb{R}^q$ with a (symmetric) control measure $F(dx)$, X is a Gaussian $\mathbb{R}^q$ -vector with mean vector $\mu = \mathbb{E}X$ and covariance matrix $A = \mathbb{E}XX'$, which is independent of $\widehat{B}_F(dx)$, and Y is a random variable (Yaglom [1021], p. 436, Yaglom [1020]). Since $B_{E,H}(0) = B_{E,H}(c^E \cdot 0) \stackrel{d}{=} cB_{E,H}(0) \rightarrow 0$, as $c \rightarrow 0$, we have $B_{E,H}(0) = 0$ a.s. Hence, $Y = 0$ a.s. in the representation (9.5.23).

The measure F , the vector μ and the matrix A are known to be unique (Yaglom [1020]). We will exploit this uniqueness to derive the homogeneity of F by using the operator self-similarity of $B_{E,H}$. On one hand, note that

$$\begin{aligned} \{B_{E,H}(c^E t)\}_{t \in \mathbb{R}^q} &\stackrel{d}{=} \left\{ \int_{\mathbb{R}^q} (e^{i\langle x, c^E t \rangle} - 1) \widehat{B}_F(dx) + \langle X, c^E t \rangle \right\}_{t \in \mathbb{R}^q} \\ &= \left\{ \int_{\mathbb{R}^q} (e^{i\langle c^{E^*} x, t \rangle} - 1) \widehat{B}_F(dx) + \langle c^{E^*} X, t \rangle \right\}_{t \in \mathbb{R}^q} \\ &\stackrel{d}{=} \left\{ \int_{\mathbb{R}^q} (e^{i\langle y, t \rangle} - 1) \widehat{B}_{F_E}(dy) + \langle c^{E^*} X, t \rangle \right\}_{t \in \mathbb{R}^q}, \end{aligned} \quad (9.5.24)$$

by making the change of variables $c^{E^*}x = y$ ($x = c^{-E^*}y$), where the control measure F_E is defined as $F_E(dy) = F(c^{-E^*}dy)$. On the other hand, by the operator self-similarity, $\{B_{E,H}(c^E t)\}_{t \in \mathbb{R}^q}$ has the same distribution as $\{c^H B_{E,H}(t)\}_{t \in \mathbb{R}^q}$. The uniqueness then implies that

$$F(c^{-E^*}dy) = c^{2H}F(dy), \quad c^{E^*}X \stackrel{d}{=} c^H X. \quad (9.5.25)$$

Since $E \neq H \cdot I$ (or $E^* \neq H \cdot I$), the second relation in (9.5.25) holds only if $X = 0$ a.s. In terms of the spectral distribution function F , after setting $F(0) = 0$, the first relation in (9.5.25) becomes

$$F(c^{-E^*}x) = c^{2H}F(x) \quad (9.5.26)$$

and hence after changing c to $c^{1/(2H)}$ that $F(c^{-E^*/(2H)}x) = cF(x)$; that is, F is $(-E^*/(2H))$ -homogeneous, as stated in (9.5.17).

We now turn to the representations (9.5.18) and (9.5.19) of $B_{E,H}(t)$. Suppose now that F is absolutely continuous with a spectral density f . The relation (9.5.26) can then be written as

$$\int_0^{c^{-E^*}x} f(y)dy = c^{2H} \int_0^x f(y)dy$$

or, by making the change of variables $y = c^{-E^*}z$ in the first integral and using the fact that $\det(c^{E^*}) = c^{\text{tr}(E^*)}$, as

$$\int_0^x f(c^{-E^*}z) c^{-\text{tr}(E^*)} dz = c^{2H} \int_0^x f(y)dy.$$

By differentiating the last relation with respect to x , we have that, for any $c > 0$,

$$f(c^{-E^*}x)c^{-\text{tr}(E^*)} = c^{2H}f(x) \quad \text{a.e. } dx$$

or by changing c to $1/c$,

$$f(c^{E^*}x) = c^{-2H-\text{tr}(E^*)}f(x) \quad \text{a.e. } dx.$$

Equivalently,

$$f(c^{E^*}x)^{1/(-2H-\text{tr}(E^*))} = cf(x)^{1/(-2H-\text{tr}(E^*))} \quad \text{a.e. } dx$$

or

$$h(c^{E^*}x) = ch(x) \quad \text{a.e. } dx$$

by setting $f(x) = h(x)^{-2H-\text{tr}(E^*)}$. By applying Lemma 9.5.7, there is an E^* -homogeneous function g such that $g(x) = h(x)$ a.e. dx . Hence,

$$g(x) = f(x)^{1/(-2H-\text{tr}(E^*))} \quad \text{a.e. } dx$$

or

$$f(x)^{1/2} = g(x)^{-H-\text{tr}(E^*)/2} \quad \text{a.e. } dx \quad (9.5.27)$$

Combined with (9.5.23) and since $X = 0$ and $Y = 0$ as indicated above, this yields the representation (9.5.18). In turn, the representation (9.5.19) now follows from applying (9.5.9). The relation (9.5.21) is the same as (9.5.27), in view of (9.5.20).

Finally, we turn to the converse statement and show that the right-hand side of (9.5.19) is well-defined; that is,

$$\int_{\mathbb{R}^q} |e^{i\langle t, x \rangle} - 1|^2 \tau_{E^*}(x)^{-2H-\text{tr}(E^*)} |g_0(l_{E^*}(x))|^2 dx < \infty. \quad (9.5.28)$$

By using the formula (9.5.12), setting $x = r^{E^*}\theta$ and using (9.5.8), the last integral becomes

$$\int_0^\infty \int_{S_0} |e^{i\langle t, r^{E^*}\theta \rangle} - 1|^2 r^{-2H-\text{tr}(E^*)} |g_0(\theta)|^2 \sigma(d\theta) r^{\text{tr}(E^*)-1} dr \quad (9.5.29)$$

$$= \int_0^\infty \int_{S_0} |e^{i\langle t, r^{E^*}\theta \rangle} - 1|^2 r^{-2H-1} |g_0(\theta)|^2 \sigma(d\theta) dr, \quad (9.5.30)$$

where $\sigma(d\theta)$ is a Radon measure. By using the inequality $|e^{iz} - 1| \leq \min\{|z|, 2\} \leq 2 \min\{|z|, 1\}$ and (9.5.13), note that

$$\begin{aligned} |e^{i\langle t, r^{E^*}\theta \rangle} - 1|^2 &\leq 4 \min\{\langle t, r^{E^*}\theta \rangle^2, 1\} \leq 4 \min\{\|t\|^2 \|r^{E^*}\theta\|^2, 1\} \\ &\leq 4(1 + \|t\|^2) \min\{\|r^{E^*}\theta\|^2, 1\} \leq C(1 + \|t\|^2) \min\{r^{2(a_1-\delta)}, 1\}, \end{aligned}$$

for any $\delta \in (0, a_1 - H)$, where $a_1 = \min\{\Re(e_j) : j = 1, \dots, q\}$. Therefore, the integral in (9.5.30) can be bounded by

$$C(1 + \|t\|^2) \left(\sup_{\theta \in S_0} |g_0(\theta)|^2 \sigma(S_0) \right) \left(\int_0^\infty \min\{r^{2(a_1-\delta)}, 1\} r^{-2H-1} dr \right).$$

This is finite because g_0 is assumed bounded on S_0 , σ is finite, $H > 0$, so that $(-2H - 1) + 1 < 0$, and a_1 is assumed bigger than H , so that

$$2(a_1 - \delta) - 2H - 1 > -1. \quad \square$$

Remark 9.5.10 In contrast to vector OFBM (cf. Theorem 9.3.2), the control measure F in the representation (9.5.15) of OFBF is not necessarily absolutely continuous. For example, for $q = 2$, the measure

$$F_1(dx) = F_1(dx_1, dx_2) = |x_1|^{-\frac{2}{e_1}-1} dx_1 \delta_{x_1}(dx_2), \quad e_1 > 1,$$

is not absolutely continuous, and satisfies (9.5.16) with

$$E^* = E = \text{diag}(e_1, e_1) = e_1 \cdot I, \quad H = 1.$$

Take also any measure F_2 which is absolutely continuous and satisfies (9.5.16) with the same exponents E and H , and for which the corresponding OFBF is well-defined. Set $F = F_1 + F_2$, which is not absolutely continuous. The field given by (9.5.15) is then well-defined since

$$\int_{\mathbb{R}^2} |e^{i\langle x, t \rangle} - 1|^2 F_1(dx) = \int_{\mathbb{R}} |e^{ix_1(t_1+t_2)} - 1|^2 |x_1|^{-\frac{2}{e_1}-1} dx_1 < \infty,$$

supposing $e_1 > 1$. Hence, it is OFBF with the exponents E and H . (Note that the role of F_2 was just to ensure that the OFBF is proper. Indeed, without F_2 , the resulting “OFBF” is zero for $t_1 = -t_2$.)

The next result provides a time domain representation of OFBF. In contrast to Theorem 9.3.8 concerning the time domain representations of vector OFBMs, there are technical difficulties in going from the spectral domain to the time domain representations for OFBFs. We make additional assumptions on the function in the spectral representation and the exponents E, H .

Theorem 9.5.11 Let $B_{E,H} = \{B_{E,H}(t)\}_{t \in \mathbb{R}^q}$ be an OFBF with exponents E and H . Suppose that the assumptions of Theorem 9.5.9 hold and, in particular, that the control measure F is absolutely continuous. Suppose, in addition, that the function $g_0(y) = g(y)^{-H - \text{tr}(E^*)/2}$ in Theorem 9.5.9 is differentiable a.e. in the sense that

$$\frac{\partial^q}{\partial x_1 \dots \partial x_q} g_0(x) = \tilde{g}(x) \quad \text{a.e. } dx, \quad (9.5.31)$$

that $\tilde{g}$ is bounded on S_0 , that E is diagonalizable over $\mathbb{R}$ with real eigenvalues e_j , $j = 1, \dots, q$, and that

$$H + \text{tr}(E)/2 < q \min\{\Re(e_j) : j = 1, \dots, q\}. \quad (9.5.32)$$

Then, there is a function $h : \mathbb{R}^q \rightarrow \mathbb{R}$ satisfying

- (i) $h(t - \cdot) - h(-\cdot) \in L^2(\mathbb{R}^q)$, $t \in \mathbb{R}^q$,
- (ii) h is $E/(H - \text{tr}(E)/2)$ -homogeneous,

and such that

$$\{B_{E,H}(t)\}_{t \in \mathbb{R}^q} \stackrel{d}{=} \left\{ \int_{\mathbb{R}^q} (h(t-u) - h(-u)) B(du) \right\}_{t \in \mathbb{R}^q}, \quad (9.5.33)$$

where $B(du)$ is a Gaussian random measure on $\mathbb{R}^q$ with the Lebesgue control measure $\mathbb{E}|B(du)|^2 = du$.

Remark 9.5.12 If the function h in (9.5.33) is nonnegative, then one can write $h = \phi^{H-\text{tr}(E)/2}$ with a nonnegative E -homogeneous function ϕ (see Lemma 9.5.4), and the representation (9.5.33) as

$$\{B_{E,H}(t)\}_{t \in \mathbb{R}^q} \stackrel{d}{=} \left\{ \int_{\mathbb{R}^q} \left(\phi(t-u)^{H-\text{tr}(E)/2} - \phi(-u)^{H-\text{tr}(E)/2} \right) B(du) \right\}_{t \in \mathbb{R}^q}. \quad (9.5.34)$$

Proof Step 1: We shall first show that it is enough to consider the case of a diagonal E . Indeed, for a general diagonalizable $E = WE_0W^{-1}$ with $W \in \mathbb{R}^{q \times q}$ and diagonal $E_0 \in \mathbb{R}^{q \times q}$, the process $\tilde{B}_{E_0,H}(t) := B_{E,H}(Wt)$ is an OFBF with the diagonal exponent E_0 . Suppose that it can be represented as

$$\{\tilde{B}_{E_0,H}(t)\}_{t \in \mathbb{R}^q} \stackrel{d}{=} \left\{ \int_{\mathbb{R}^q} (h_0(t-u) - h_0(-u)) B(du) \right\}_{t \in \mathbb{R}^q},$$

where h_0 is $E_0/(H - \text{tr}(E_0)/2)$ -homogeneous. Then,

$$\begin{aligned} \{B_{E,H}(t)\}_{t \in \mathbb{R}^q} &\stackrel{d}{=} \left\{ \int_{\mathbb{R}^q} (h_0(W^{-1}t - u) - h_0(-u)) B(du) \right\}_{t \in \mathbb{R}^q} \\ &\stackrel{d}{=} \left\{ \int_{\mathbb{R}^q} (h_0(W^{-1}(t-v)) - h_0(-W^{-1}v)) |\det(W)|^{-1/2} B(dv) \right\}_{t \in \mathbb{R}^q} \\ &= \left\{ \int_{\mathbb{R}^q} (h(t-v) - h(-v)) B(dv) \right\}_{t \in \mathbb{R}^q} \end{aligned}$$

after the change of variables $u = W^{-1}v$ (so that $du = |\det(W)|^{-1}dv$), where

$$h(v) = |\det(W)|^{-1/2} h_0(W^{-1}v).$$

By using the homogeneity of h_0 and (9.5.4), we get

$$h_0(c^{E_0}W^{-1}v) = h_0\left((c^{H-\text{tr}(E_0)/2})^{\frac{E_0}{H-\text{tr}(E_0)/2}}W^{-1}v\right) = c^{H-\text{tr}(E_0)/2}h_0(W^{-1}v)$$

and since $\text{tr}(E_0) = \text{tr}(E)$, we have

$$\begin{aligned} h(c^E v) &= |\det(W)|^{-1/2} h_0(W^{-1}c^E v) = |\det(W)|^{-1/2} h_0(W^{-1}Wc^{E_0}W^{-1}v) \\ &= |\det(W)|^{-1/2} h_0(c^{E_0}W^{-1}v) = c^{H-\text{tr}(E_0)/2} |\det(W)|^{-1/2} h_0(W^{-1}v) = c^{H-\text{tr}(E)/2} h(v), \end{aligned}$$

that is, the function h is $E/(H - \text{tr}(E)/2)$ -homogeneous. This shows that the conclusion of the theorem holds for E .

Step 2: Thus, suppose without loss of generality that E is diagonal. Let

$$\widehat{f}_t(x) = (e^{i\langle x, t \rangle} - 1)g_0(x) \quad (\in L^2(\mathbb{R}^q))$$

be the kernel function appearing in the spectral representation (9.5.19) of OFBF $B_{E,H}$ (with a nonnegative, symmetric function g_0). Then, by using Plancherel's identity (see (A.1.10)), $B_{E,H}$ can also be represented as

$$\{B_{E,H}(t)\}_{t \in \mathbb{R}^q} \stackrel{d}{=} \left\{ \int_{\mathbb{R}^q} f_t(u) B(du) \right\}_{t \in \mathbb{R}^q},$$

where

$$\begin{aligned} f_t(u) &= \frac{1}{(2\pi)^q} \int_{\mathbb{R}^q} e^{-i\langle x, u \rangle} \widehat{f_t}(x) dx \quad (\in L^2(\mathbb{R}^q)) \\ &= \frac{1}{(2\pi)^q} \int_{\mathbb{R}^q} e^{-i\langle x, u \rangle} (e^{i\langle x, t \rangle} - 1) g_0(x) dx \\ &= \frac{1}{(2\pi)^q} \int_{\mathbb{R}^q} (\cos(\langle x, t - u \rangle) - \cos(\langle x, -u \rangle)) g_0(x) dx. \end{aligned} \quad (9.5.35)$$

We want to express next $f_t(u)$ as $h(t - u) - h(-u)$, where h is $E/(H - \text{tr}(E)/2)$ -homogeneous as stated in the theorem. Note that h cannot be defined directly as $h(v) = \int_{\mathbb{R}^q} \cos(\langle x, v \rangle) g_0(x) dx$ (see Remark 9.5.13 below).

Observe that, after the changes of variables for each orthant and the fact that g_0 is symmetric,

$$(2\pi)^q f_t(u) = \sum_{\sigma} \int_{[0, \infty)^q} (\cos(\langle x, (t - u)_{\sigma} \rangle) - \cos(\langle x, (-u)_{\sigma} \rangle)) g_0(x) dx =: \sum_{\sigma} f_{\sigma, t}(u), \quad (9.5.36)$$

where the sum is over all $\sigma = (\sigma_1, \dots, \sigma_q)$ with $\sigma_j \in \{-1, 1\}$, and $(v)_{\sigma} = (\sigma_1 v_1, \dots, \sigma_q v_q)$. It is thus enough to show that

$$f_{\sigma, t}(u) = h_{\sigma}(t - u) - h_{\sigma}(-u), \quad (9.5.37)$$

for some $E/(H - \text{tr}(E)/2)$ -homogeneous h_{σ} . For notational simplicity, set

$$b = \text{tr}(E) = \text{tr}(E^*).$$

Since in (9.5.20), g is E^* -homogeneous, we get that g_0 is $-E^*/(H + b/2)$ -homogeneous by Lemma 9.5.4. Denote by $[x, \infty)$ the Cartesian product $[x_1, \infty) \times \dots \times [x_q, \infty)$. Since g is assumed to be differentiable in the sense of (9.5.31), then by setting

$$g_0(x) = (-1)^q \int_{[x, \infty)} \widetilde{g}(y) dy, \quad (9.5.38)$$

we get that $\widetilde{g}$ is $-E^*/(H + 3b/2)$ -homogeneous (Exercise 9.11). Then,

$$\begin{aligned} f_{\sigma, t}(u) &= (-1)^q \int_{[0, \infty)^q} (\cos(\langle x, (t - u)_{\sigma} \rangle) - \cos(\langle x, (-u)_{\sigma} \rangle)) \left(\int_{[x, \infty)} \widetilde{g}(y) dy \right) dx \\ &= (-1)^q \int_{[0, \infty)^q} dy \widetilde{g}(y) \left(\int_{[0, y]} \cos(\langle x, (t - u)_{\sigma} \rangle) dx - \int_{[0, y]} \cos(\langle x, (-u)_{\sigma} \rangle) dx \right). \end{aligned} \quad (9.5.39)$$

Observe now that

$$\int_{[0, y]} \cos(\langle x, v \rangle) dx = \Re \int_{[0, y]} e^{i\langle x, v \rangle} dx = \frac{1}{v_1 \dots v_q} \Re \left(i^{-q} (e^{iy_1 v_1} - 1) \dots (e^{iy_q v_q} - 1) \right).$$

Then, by (9.5.39), the relation (9.5.37) holds with

$$\begin{aligned} h_\sigma(v) &= (-1)^q \int_{[0,\infty)^q} \frac{\tilde{g}(y)}{v_1 \dots v_q} \Re \left(i^{-q} (e^{iy_1 v_1} - 1) \dots (e^{iy_q v_q} - 1) \right) dy \\ &=: \int_{[0,\infty)^q} \frac{\tilde{g}(y)}{v_1 \dots v_q} k_{\sigma,v}(y) dy =: \frac{h_{0,\sigma}(v)}{v_1 \dots v_q}. \end{aligned} \quad (9.5.40)$$

Since E is diagonal, the function $h_\sigma(v)$ is $E^*/(H - b/2)$ -homogeneous. Indeed, by letting $E^* = \text{diag}(e_1^*, \dots, e_q^*)$ and using

$$k_{\sigma, c^{E^*/(H-b/2)}v}(y) = k_{\sigma,v}(c^{E^*/(H-b/2)}y),$$

note that

$$\begin{aligned} h_\sigma(c^{E^*/(H-b/2)}v) &= h_\sigma(c^{e_1^*/(H-b/2)}v_1, \dots, c^{e_q^*/(H-b/2)}v_q) \\ &= \int_{[0,\infty)^q} \frac{\tilde{g}(y)}{c^{b/(H-b/2)}v_1 \dots v_q} k_{\sigma, c^{E^*/(H-b/2)}v}(y) dy \\ &= \int_{[0,\infty)^q} \frac{\tilde{g}(y)}{c^{b/(H-b/2)}v_1 \dots v_q} k_{\sigma,v}(c^{E^*/(H-b/2)}y) dy \\ &= \int_{[0,\infty)^q} \frac{\tilde{g}(c^{-E^*/(H-b/2)}x)}{c^{b/(H-b/2)}v_1 \dots v_q} k_{\sigma,v}(x) c^{-b/(H-b/2)} dx \\ &= \int_{[0,\infty)^q} c^{(H+3b/2)/(H-b/2)} \frac{\tilde{g}(x)}{v_1 \dots v_q} k_{\sigma,v}(x) c^{-2b/(H-b/2)} dx = c h_\sigma(v), \end{aligned}$$

by using the $-E^*/(H + 3b/2)$ -homogeneity of $\tilde{g}$ and (9.5.40). It is then enough to show that the function $h_\sigma(v)$, or equivalently, the function $h_{0,\sigma}(v)$ in (9.5.40) is well-defined as a Lebesgue integral.

Note that, by the change of variables formula (9.5.12) with $f(y) = |\tilde{g}(y)| |k_{\sigma,v}(y)|$ and $M = -E^*/(H + 3b/2)$,

$$\begin{aligned} |h_{0,\sigma}(v)| &\leq \int_{[0,\infty)^q} |\tilde{g}(y)| |k_{\sigma,v}(y)| dy = \int_0^\infty \int_{S_0} |\tilde{g}(r^M \theta)| |k_{\sigma,v}(r^M \theta)| \sigma(d\theta) r^{\text{tr}(M)-1} dr \\ &= \int_0^\infty dr r^{-\frac{b}{H+3b/2}-1} r \left(\int_{S_0} |\tilde{g}(\theta)| |k_{\sigma,v}(r^{-\frac{E^*}{H+3b/2}} \theta)| \sigma(d\theta) \right), \end{aligned} \quad (9.5.41)$$

where we used the facts that $\tilde{g}(r^{-\frac{E^*}{H+3b/2}} \theta) = r \tilde{g}(\theta)$ by homogeneity and that

$$r^{\text{tr}(M)} = r^{-\text{tr}(E^*)/(H+3b/2)} = r^{-b/(H+3b/2)}.$$

Around $r = 0$, by bounding the integral over S_0 by a constant, the integrand behaves as $r^{-b/(H+3b/2)}$ and is integrable since

$$-\frac{b}{H+3b/2} + 1 = \frac{H+b/2}{H+3b/2} > 0$$

(by (9.5.22), $b = \text{tr}(E^*) = \text{tr}(E) > qH > 0$). For the behavior around $r = \infty$, observe that, for $y_1 \geq 0, \dots, y_q \geq 0$, $|k_{\sigma,v}(y)| \leq C y_1 \dots y_q \leq C(y_1 + \dots + y_q)^q \leq C' \|y\|^q$. Then, the integrand in (9.5.41) is bounded by

$$r^{-\frac{b}{H+3b/2}} \|r^{-\frac{E}{H+3b/2}}\|^q \leq Cr^{-\frac{b}{H+3b/2}} r^{-\frac{q(a_1-\delta)}{H+3b/2}}$$

for arbitrarily small $\delta > 0$, where we used the bound (9.5.13) with $a_1 = \min\{\Re(e_j) : j = 1, \dots, q\}$. The integrability around $r = \infty$ follows since

$$-\frac{b + qa_1}{H + 3b/2} + 1 = \frac{-qa_1 + H + b/2}{H + 3b/2} < 0,$$

in view of the assumption (9.5.32), which states that $H + b/2 < qa_1$. $\square$

Remark 9.5.13 It is illuminating to revisit the technical difficulties involved in the proof of Theorem 9.5.11 by considering the univariate context $q = 1$. In this case, one can pick the natural homogeneous specification $g_0(x) = c|x|^{-H-1/2}$ where $H = 1/E < 1$. Under such choice, the integral (9.5.35) can be recast as

$$c \int_{\mathbb{R}} (\cos(x(t-u)) - \cos(x(-u))) |x|^{-H-1/2} dx = \varphi(t-u) - \varphi(-u),$$

where

$$\varphi(v) = \begin{cases} c \int_{\mathbb{R}} \cos(xv) |x|^{-H-1/2} dx, & \text{if } H < 1/2, \\ c \int_{\mathbb{R}} (\cos(xv) - 1) |x|^{-H-1/2} dx, & \text{if } H > 1/2. \end{cases} \quad (9.5.42)$$

The expression (9.5.42) is interpreted as involving an improper Riemann integral when $H < 1/2$ and a Lebesgue integral when $H > 1/2$. Neither interpretation carries over to the spatial case $q \geq 2$. In view of (9.5.20), the integral (9.5.35) is

$$\int_{\mathbb{R}^q} (\cos(\langle x, t-u \rangle) - \cos(\langle x, -u \rangle)) g(x)^{-H-\text{tr}(E^*)/2} dx, \quad (9.5.43)$$

for an E^* -homogenous function g . On one hand, when $q \geq 2$, the homogeneous function g is no longer a simple power function as above, so that the integral

$$\int_{\mathbb{R}^q} \cos(\langle x, v \rangle) g(x)^{-H-\text{tr}(E^*)/2} dx$$

cannot be interpreted as an improper Riemann integral. On the other hand, one could subtract 1 from each $\cos$ in (9.5.43) and check when the integral $\int_{\mathbb{R}^q} (\cos(\langle x, v \rangle) - 1) g(x)^{-H-\text{tr}(E^*)/2} dx$ is well-defined in the Lebesgue sense; that is,

$$J = \int_{\mathbb{R}^q} |\cos(\langle x, v \rangle) - 1| g(x)^{-H-\text{tr}(E^*)/2} dx < \infty. \quad (9.5.44)$$

To do so, we use (9.5.12) and the E^* -homogeneity of g to get

$$J = \int_0^\infty \int_{S_0} |\cos(\langle t, r^{E^*} \theta \rangle) - 1| r^{-H-\text{tr}(E^*)/2} |g(\theta)^{-H-\text{tr}(E^*)/2}| \sigma(d\theta) r^{\text{tr}(E^*)-1} dr.$$

We now adapt the argument following (9.5.30), with the difference that the integrand is raised to the power 1 and not 2. This leads to the integrability condition at $r = \infty$ as $-H - \text{tr}(E^*)/2 + \text{tr}(E^*) - 1 < -1$; that is, $\text{tr}(E^*) < 2H$. However, the condition (9.5.22) for

the construction of a harmonizable representation requires $\min\{\Re(e_j) : j = 1, \dots, q\} > H$. But this implies that

$$\mathrm{tr}(E^*) = \sum_j e_j = \sum_j \Re(e_j) \geq q \min\{\Re(e_j) : j = 1, \dots, q\} > qH.$$

The conditions $\mathrm{tr}(E^*) < 2H$ and $\mathrm{tr}(E^*) > qH$ are not compatible for $q \geq 2$, and as a consequence, (9.5.44) can only play the role of the fractional filter $\varphi(v)$ in the univariate context, namely, when $q = 1$.

We also note that the condition (9.5.32) reduces to $H < 1/2$ when $q = 1$, and thus does not cover the case $H > 1/2$.

The next result provides conditions for (9.5.33) to be well-defined and hence be an OFBF with exponents E and H . The statement and proof follow Theorem 3.1 in Biermé et al. [151]. The following notion will be used.

Definition 9.5.14 Let $\beta > 0$ and $E \in \mathbb{R}^{q \times q}$. A continuous function $\phi : \mathbb{R}^q \rightarrow [0, \infty)$ is called (β, E) -admissible if $\phi(x) > 0$ for all $x \neq 0$ and, for any $0 < A < B$, there is a constant $C > 0$ such that, for any $A \leq \|y\| \leq B$,

$$\tau_E(x) \leq 1 \Rightarrow |\phi(x+y) - \phi(y)| \leq C\tau_E(x)^\beta, \quad (9.5.45)$$

where $\tau_E(x)$ is the radial part of E , as in (9.5.6), where the matrix is denoted M .

Example 9.5.15 (Admissible functions) The E -homogeneous function ϕ in Example 9.5.8 can be shown to be (β, E) -admissible for $\beta < \min\{e_1, \rho \frac{e_1}{e_q}\}$ if $e_1 \leq \rho$, and $\beta = \rho$ if $e_1 > \rho$ (Exercise 9.13).

Theorem 9.5.16 Let $\beta > 0$ and $E \in \mathbb{R}^{q \times q}$ with its eigenvalues having positive real parts. Let also $\phi : \mathbb{R}^q \rightarrow [0, \infty)$ be an E -homogeneous, (β, E) -admissible function with $\beta > H > 0$. Then, the random field

$$\int_{\mathbb{R}^q} \left(\phi(t-u)^{H-\mathrm{tr}(E)/2} - \phi(-u)^{H-\mathrm{tr}(E)/2} \right) B(du), \quad t \in \mathbb{R}^q, \quad (9.5.46)$$

is a well-defined OFBF with the exponents E and H , where $B(du)$ is a Gaussian random measure on $\mathbb{R}^q$ with the Lebesgue control measure $\mathbb{E}|B(du)|^2 = du$.

Proof It is enough to show that the random field (9.5.46) is well defined; that is, for fixed $t \in \mathbb{R}^q$,

$$\int_{\mathbb{R}^q} \left| \phi(t-u)^{H-\frac{b}{2}} - \phi(-u)^{H-\frac{b}{2}} \right|^2 du < \infty,$$

where $b = \mathrm{tr}(E)$. By the inequality $(a-b)^2 \leq 2a^2 + 2b^2$,

$$\begin{aligned} \left| \phi(t-u)^{H-\frac{b}{2}} - \phi(-u)^{H-\frac{b}{2}} \right|^2 &\leq 2(\phi(t-u)^{2H-b} + \phi(-u)^{2H-b}) \\ &\leq C(\tau_E(t-u)^{2H-b} + \tau_E(u)^{2H-b}), \end{aligned}$$

where we used the representation (9.5.9), the fact that τ is symmetric (see (9.5.7)) and the continuity of ϕ (in the definition of admissibility). For any $R > 0$, by using (9.5.12) and (9.5.8),

$$\begin{aligned} \int_{\tau_E(u) \leq R} \tau_E(u)^{2H-b} du &= \int_0^\infty \int_{S_0} 1_{\{\tau_E(r^E \theta) \leq R\}} \tau_E(r^E \theta)^{2H-b} \sigma(d\theta) r^{b-1} dr \\ &= \int_0^\infty \int_{S_0} 1_{\{r \tau_E(\theta) \leq R\}} r^{2H-b} \tau_E(\theta)^{2H-b} \sigma(d\theta) r^{b-1} dr = \sigma(S_0) \int_0^R r^{2H-1} dr < \infty. \end{aligned} \quad (9.5.47)$$

One can show (Lemma 2.2 in Biermé et al. [151]) that $\{u : \tau_E(t-u) \leq R\} \subset \{u : \tau_E(u) \leq K(R + \tau_E(t))\}$ for some constant K . Then, by the change of variables and arguing as above,

$$\begin{aligned} \int_{\tau_E(u) \leq R} \tau_E(t-u)^{2H-b} du &= \int_{\tau_E(t-u) \leq R} \tau_E(u)^{2H-b} du \\ &\leq \int_{\tau_E(u) \leq K(R + \tau_E(t))} \tau_E(u)^{2H-b} du < \infty. \end{aligned}$$

It remains to show that for some $R = R(t) > 0$, we have

$$\int_{\tau_E(u) > R} \left| \phi(t+u)^{H-\frac{b}{2}} - \phi(u)^{H-\frac{b}{2}} \right|^2 du < \infty.$$

Observe that for $\tau_E(u) > R$, we have by Definition 9.5.14 that $\phi(u) > 0$. Hence, we can write

$$\phi(t+u) = \phi\left(\phi(u)^E (\phi(u)^{-E} t + \phi(u)^{-E} u)\right) = \phi(u) \phi(\phi(u)^{-E} t + \phi(u)^{-E} u), \quad (9.5.48)$$

where in the last relation, we used the E -homogeneity of ϕ to write $\phi(c^E y) = c\phi(y)$, with $c = \phi(u)$ and $y = \phi(u)^{-E} (t+u)$. By using (9.5.9) and (9.5.6), observe also that $\|\phi(u)^{-E} u\| = \|\tau_E(u)^{-E} \phi(l_E(u))^{-E} \tau_E(u)^E l_E(u)\| = \|\phi(l_E(u))^{-E} l_E(u)\|$ and hence, since ϕ is continuous and positive by Definition 9.5.14, we have

$$0 < A \leq \|\phi(u)^{-E} u\| \leq B, \quad u \in \mathbb{R}^q. \quad (9.5.49)$$

Similarly, with $a_1 = \min\{\Re(e_j) : j = 1, \dots, q\}$ and eigenvalues e_j of E , we have by (9.5.13) that $\|\phi(u)^{-E} t\| = \|(\phi(u)\tau_E(t))^{-E} l_E(t)\| \leq C(\phi(u)\tau_E(t))^{-(a_1-\delta)}$ for arbitrarily small $\delta > 0$, as long as $(\phi(u)\tau_E(t))^{-1} \leq 1$. This yields

$$\|\phi(u)^{-E} t\| \leq C'(\tau_E(u))^{-(a_1-\delta)} \leq 1, \quad (9.5.50)$$

for $\tau_E(u) > R$ with large enough R . Since ϕ is (β, E) -admissible and $\phi(\phi(u)^{-E} u) = \phi(u)^{-1} \phi(u) = 1$, the relations (9.5.49) and (9.5.50) imply that there is $C > 0$ such that

$$\begin{aligned} |\phi(\phi(u)^{-E} t + \phi(u)^{-E} u) - 1| &= |\phi(\phi(u)^{-E} t + \phi(u)^{-E} u) - \phi(\phi(u)^{-E} u)| \\ &\leq C\tau_E(\phi(u)^{-E} t)^\beta = C\phi(u)^{-\beta} \tau_E(t)^\beta, \end{aligned} \quad (9.5.51)$$

where we also used the E -homogeneity of τ_E . Since ϕ is E -homogeneous, $\phi(u) = \phi(\tau_E(u)^E l_E(u)) = \tau_E(u) \phi(l_E(u))$, where $\phi(l_E(u))$ is bounded, and since $\beta > 0$, we can choose R large enough so that $C\phi(u)^{-\beta} \tau_E(t)^\beta < 1/2$ for all $\tau_E(u) > R$. By using (9.5.48),

$$\begin{aligned}
|\phi(t+u)^{H-\frac{b}{2}} - \phi(u)^{H-\frac{b}{2}}| &= \phi(u)^{H-\frac{b}{2}} \left| \phi(\phi(u)^{-E}t + \phi(u)^{-E}u)^{H-\frac{b}{2}} - 1^{H-\frac{b}{2}} \right| \\
&= \phi(u)^{1-\frac{b}{2}} \left| H - \frac{b}{2} \right| v_*^{H-\frac{b}{2}-1} \left| \phi(\phi(u)^{-E}t + \phi(u)^{-E}u) - 1 \right|
\end{aligned}$$

by applying the mean value theorem to the function $f(v) = v^{1-b/2}$, where v_* is a point between $v = 1$ and $v = \phi(\phi(u)^{-E}t + \phi(u)^{-E}u)$. Since $1/2 < v_* < 3/2$ by (9.5.51) and the observation following it for large enough R , we deduce that

$$\begin{aligned}
|\phi(t+u)^{H-\frac{b}{2}} - \phi(u)^{H-\frac{b}{2}}| &\leq C_0 \phi(u)^{H-\frac{b}{2}} \left| \phi(\phi(u)^{-E}t + \phi(u)^{-E}u) - 1 \right| \\
&\leq C_1 \phi(u)^{H-\frac{b}{2}-\beta} \tau_E(t)^\beta \leq C_2 \phi(u)^{H-\frac{b}{2}-\beta}
\end{aligned}$$

by (9.5.51), for all $\tau_E(u) > R$. Finally, by arguing in the same way as for (9.5.47) in the case $\tau_E(u) \leq R$ above, we conclude that

$$\begin{aligned}
\int_{\tau_E(u) > R} \left| \phi(t+u)^{H-\frac{b}{2}} - \phi(u)^{H-\frac{b}{2}} \right|^2 du &\leq C \int_{\tau_E(u) > R} \phi(u)^{2H-b-2\beta} du \\
&\leq C' \int_{\tau_E(u) > R} \tau_E(u)^{2H-b-2\beta} du \\
&= C'' \int_R^\infty r^{2H-b-2\beta} r^{b-1} dr = C'' \int_R^\infty r^{2H-2\beta-1} dr < \infty,
\end{aligned}$$

since $\beta > H$ by assumption. $\square$

9.5.3 Special Subclasses and Examples of OFBFs

It is instructive to describe first the covariance structure of an OFBF $B_{E,H}$. Setting $\sigma_\theta^2 = \mathbb{E}B_{E,H}(\theta)^2$, for $\theta \in S_0$, by the operator self-similarity (9.5.1) and the polar coordinate representation (9.5.6), we have that $\mathbb{E}B_{E,H}(t)^2 = \mathbb{E}B_{E,H}((\tau_E(t))^E l_E(t))^2 = \tau_E(t)^{2H} \mathbb{E}B_{E,H}(l_E(t))^2 = \tau_E(t)^{2H} \sigma_{l_E(t)}^2$. As in the univariate case, writing $2B_{E,H}(t)B_{E,H}(s) = B_{E,H}(t)^2 + B_{E,H}(s)^2 - (B_{E,H}(t) - B_{E,H}(s))^2$ and using the stationarity of increments, this yields the following result.

Proposition 9.5.17 *Let $B_{E,H} = \{B_{E,H}(t)\}_{t \in \mathbb{R}^q}$ be an OFBF. Then,*

$$\mathbb{E}B_{E,H}(t)B_{E,H}(s) = \frac{1}{2} \left\{ \tau_E(t)^{2H} \sigma_{l_E(t)}^2 + \tau_E(s)^{2H} \sigma_{l_E(s)}^2 - \tau_E(t-s)^{2H} \sigma_{l_E(t-s)}^2 \right\}, \quad s, t \in \mathbb{R}^q. \quad (9.5.52)$$

We next give several popular examples of OFBFs.

Example 9.5.18 (Isotropic FBFs) Take $E = E^* = e_1 \cdot I$, $H > 0$ with $e_1 > H$. By Example 9.5.5, one can take $\tau_{E^*}(x) = \|x\|_2^{1/e_1}$, $l_{E^*}(x) = x/\|x\|_2$. The spectral representation (9.5.19) of OFBF then becomes

$$\{B_{E,H}(t)\}_{t \in \mathbb{R}^q} \stackrel{d}{=} \left\{ \int_{\mathbb{R}^q} (e^{i\langle x, t \rangle} - 1) \|x\|_2^{-\frac{H}{e_1} - \frac{q}{2}} g_0\left(\frac{x}{\|x\|_2}\right) \widehat{B}(dx) \right\}_{t \in \mathbb{R}^q}. \quad (9.5.53)$$

The OFBF in (9.5.53) is well-defined by Theorem 9.5.9 (supposing g_0 is bounded) and is called an *isotropic fractional Brownian field* (isotropic FBF). Note that it satisfies

$$\{B_{E,H}(c^{e_1}t)\}_{t \in \mathbb{R}^q} \stackrel{d}{=} \{c^H B_{E,H}(t)\}_{t \in \mathbb{R}^q}.$$

Equivalently, $B_{E,H}(ct)$ and $B_{E,H}(c^{H_E}t)$ have the same finite-dimensional distributions, where $H_E = H/e_1 < 1$.

When $g_0(\theta) = c$, an isotropic FBF is referred to as simply an FBF.¹¹ It has the spectral representation

$$\{B_{E,H}(t)\}_{t \in \mathbb{R}^q} \stackrel{d}{=} \left\{ c_0 \int_{\mathbb{R}^q} (e^{i\langle x,t \rangle} - 1) \|x\|_2^{-\frac{H}{e_1} - \frac{q}{2}} \widehat{B}(dx) \right\}_{t \in \mathbb{R}^q} \quad (9.5.54)$$

and, by Proposition 9.5.17, the covariance structure

$$\mathbb{E} B_{E,H}(t) B_{E,H}(s) = \frac{\sigma^2}{2} \left\{ \|t\|_2^{2H/e_1} + \|s\|_2^{2H/e_1} - \|t-s\|_2^{2H/e_1} \right\}, \quad s, t \in \mathbb{R}^q.$$

An approximate sample realization of the field (9.5.54) with $q = 2$, $e_1 = 4/3$ and $H = 1$ (equivalently, $e_1 = 1$ and $H = 3/4$) is given in Figure 9.2, left plot. It is obtained by discretizing the integral (9.5.54) and using the two-dimensional FFT to make the computations numerically more efficient.

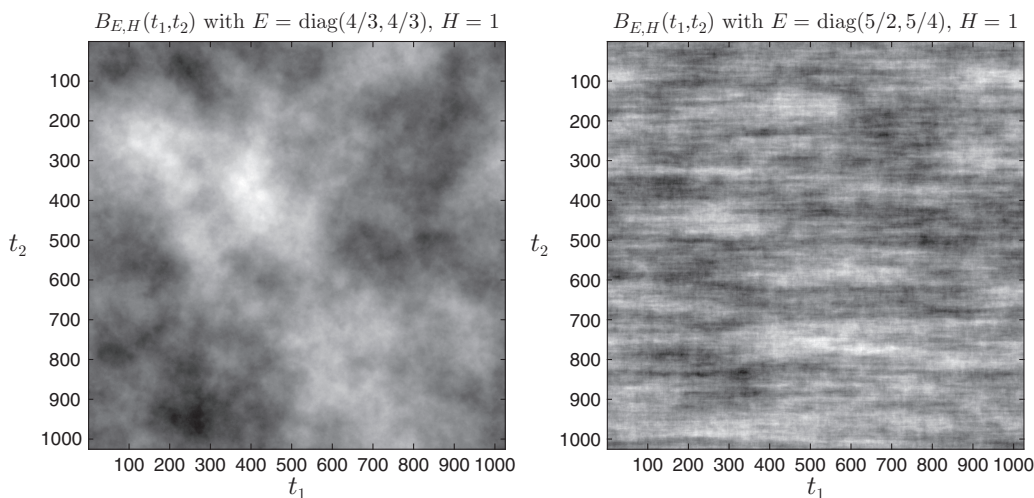


Figure 9.2 Sample realizations of OFBFs $B_{E,H}$. Left: The isotropic FBF (9.5.54) with $E = \text{diag}(4/3, 4/3)$ and $H = 1$ (equivalently, $E = I$ and $H = 3/4$). Right: The anisotropic FBF (9.5.55) with $E = \text{diag}(5/2, 5/4)$ and $H = 1$ (equivalently, $E = \text{diag}(4/3, 2/3)$ with $\text{tr}(E) = 2$ and $H = 8/15$). (The plots are in gray color map, where the white color represents the largest values and the black the smallest values.)

¹¹ Strictly speaking, only an FBF having the representation (9.5.54) could be called isotropic, since only for this field, the function g in its spectral representation depends solely on $\|x\|_2$. We call the fields having the representation (9.5.53) isotropic as well following Lavancier [595].

Example 9.5.19 (*Anisotropic FBFs*) OFBFs with $E = \text{diag}(e_1, \dots, e_q) \neq e \cdot I$, $e_j > H > 0$, and hence not having the integral representation (9.5.53) are generally referred to as *anisotropic fractional Brownian fields* (anisotropic FBFs). For example, the random field

$$\int_{\mathbb{R}^q} (e^{i\langle x, t \rangle} - 1) \left(\sum_{j=1}^q |x_j|^{\frac{1}{e_j}} \right)^{-H - \frac{\text{tr}(E)}{2}} \widehat{B}(dx), \quad t \in \mathbb{R}^q, \quad (9.5.55)$$

is an anisotropic FBF. To see that it is a well-defined OFBF with the exponents $E = \text{diag}(e_1, \dots, e_q)$ and H , use Theorem 9.5.9. The function g is E -homogeneous, since for $c > 0$, $g(c^E x) = \sum_{j=1}^q |c^{e_j} x_j|^{1/e_j} = c \sum_{j=1}^q |x_j|^{1/e_j} = cg(x)$. An approximate sample realization of the field (9.5.55) with $q = 2$ and $e_1 = 5/2$, $e_2 = 5/4$, $H = 1$ (equivalently, $e_1 = 4/3$, $e_2 = 2/3$, so that (9.5.3) holds, and $H = 8/15$), is given in Figure 9.2, right plot. Observe the anisotropic nature of the realization (cf. left plot).

The next example describes an interesting random field related to OFBFs.

Example 9.5.20 (*Fractional Brownian sheet*) Let $0 < H_1, \dots, H_q < 1$. Consider a stochastic process X given by the spectral representation

$$\{X(t)\}_{t \in \mathbb{R}^q} \stackrel{d}{=} \left\{ c \int_{\mathbb{R}^q} \prod_{j=1}^q \frac{e^{ix_j t_j} - 1}{|x_j|^{H_j+1/2}} \widehat{B}(dx) \right\}_{t \in \mathbb{R}^q}, \quad (9.5.56)$$

where $\widehat{B}(dx)$ is a Gaussian Hermitian random measure on $\mathbb{R}^q$ with the control measure dx . The function $(e^{ix_j t_j} - 1)/|x_j|^{H_j+1/2}$ is the kernel function appearing in the spectral representation (2.6.9) of FBM with parameter H_j . As for the univariate FBM, the process (9.5.56) is well-defined, and has the covariance structure

$$\mathbb{E}X(t)X(s) = c^2 \left(\prod_{j=1}^q c_2(H_j)^2 \right) \prod_{j=1}^q \frac{1}{2} \left\{ |t_j|^{2H_j} + |s_j|^{2H_j} - |t_j - s_j|^{2H_j} \right\}, \quad (9.5.57)$$

where $c_2(H)$ appears in (2.6.9). It is called a *fractional Brownian sheet* (FBS).

FBS satisfies the operator self-similarity relation (9.5.1) with $E = \text{diag}(1/H_1, \dots, 1/H_q)$ and $H = q$. Note, however, that FBS is not an OFBF as it does not have stationary increments. Even though FBS does not have stationary increments, it can still be used to construct discrete parameter stationary fields. For $j = 1, \dots, q$, and a process $Y(t) = Y(t_1, \dots, t_q)$, set

$$\Delta_j Y(t) = \Delta_j Y(t_1, \dots, t_q) = Y(t_1, \dots, t_j, \dots, t_q) - Y(t_1, \dots, t_j - 1, \dots, t_q).$$

For FBS X , consider

$$U_n = \Delta_1 \dots \Delta_q X(n), \quad n \in \mathbb{Z}^q.$$

One can check easily that $\{U_n\}_{n \in \mathbb{Z}^q}$ is stationary.

Finally, we also note that an OFBF analogue of FBS has the spectral representation

$$\left\{ c \int_{\mathbb{R}^q} (e^{i\langle x, t \rangle} - 1) \prod_{j=1}^q |x_j|^{-H_j-1/2} \widehat{B}(dx) \right\}_{t \in \mathbb{R}^q}. \quad (9.5.58)$$

One can check that this OFBF is well-defined when $0 < H_j < 1$ (Exercise 9.10).

9.6 Spatial Long-Range Dependence

A general notion of vector LRD was given in Section 9.4. In contrast, a general definition of LRD is not yet available in the spatial context. One reason for this may be explained through the analogy to vector OFBMs. In the vector setting, a homogeneous function $\phi : \mathbb{R} \rightarrow \mathbb{C}^p$ satisfies $\phi(cu) = c^M \phi(u)$ and may be expressed explicitly as $\phi(v) = v_+^M \phi_1 + v_-^M \phi_2$ with $\phi_1, \phi_2 \in \mathbb{C}^p$, leading to the representations (9.3.3) and (9.3.22), and then to Definition 9.4.1 of vector LRD in Section 9.4. In the spatial setting, there are no such explicit representations for a homogeneous function $\phi : \mathbb{R}^q \rightarrow \mathbb{C}$ satisfying $\phi(c^E x) = c \phi(x)$. Since a general definition of spatial LRD is not available (at least for practical purposes), we shall focus on several special cases, namely, the isotropic case and examples of anisotropic spatial LRD. Even the isotropic case is already mathematically quite rich and interesting! The basics of stationary spatial time series are recalled in Section 9.1.3.

9.6.1 Definitions and Basic Properties

The following are conditions for spatial LRD in the special case of isotropy. The suffix “-s” in the conditions refers to “spatial”, in order to distinguish them from the conditions in Sections 2.1 and 9.4. The suffix “-isot” in the conditions refers to “isotropic.” Let $d \in (0, 1/2)$ be a parameter.

Condition II-s-isot The autocovariance function of the random field $X = \{X_n\}_{n \in \mathbb{Z}^q}$ satisfies

$$\gamma_X(n) = L_2(\|n\|_2) \|n\|_2^{(2d-1)q} b_2\left(\frac{n}{\|n\|_2}\right), \quad n \in \mathbb{Z}^q, \quad (9.6.1)$$

where L_2 is a slowly varying function at infinity, $b_2 : S^{q-1} \rightarrow (0, \infty)$ is a function on the unit sphere S^{q-1} , and $\|\cdot\|_2$ is the usual Euclidean norm.

Condition IV-s-isot The random field $X = \{X_n\}_{n \in \mathbb{Z}^q}$ has a spectral density satisfying

$$f_X(\lambda) = L_4(\|\lambda\|_2) \|\lambda\|_2^{-2dq} b_4\left(\frac{\lambda}{\|\lambda\|_2}\right), \quad \lambda \in (0, \pi]^q, \quad (9.6.2)$$

where L_4 is a slowly varying function at zero and $b_4 : S^{q-1} \rightarrow (0, \infty)$ is a function on the unit sphere S^{q-1} .

Definition 9.6.1 A second-order stationary random field $X = \{X_n\}_{n \in \mathbb{Z}^q}$ is called *isotropic long-range dependent* (isotropic LRD, in short) if one of the non-equivalent conditions II-s-isot and IV-s-isot above holds with $d \in (0, 1/2)$. (For the terminology used, see also the footnote in Example 9.5.18.)

The relationship between conditions II-s-isot and IV-s-isot was studied by Wainger [993] (for a more recent account, see Leonenko and Olenko [610]). It involves the so-called *spherical harmonics*, which are special functions on the unit sphere S^{q-1} . See, for example, Andrews et al. [35], Atkinson and Han [55]. The functions are labeled and denoted as $Y_{l,m}(u')$ with

$$u' = \frac{u}{\|u\|_2} \in S^{q-1},$$

where $l = 0, 1, 2, \dots$ is the degree and $m = 1, \dots, n_{q,l}$ is the index referring to spherical harmonics of degree l . They form an orthonormal basis with respect to the inner product

$$\int_{S^{q-1}} Y_{l,m}(u') Y_{l',m'}(u') \sigma(du') = \delta_{ll'} \delta_{mm'},$$

where $\sigma(du')$ denotes the measure over the sphere S^{q-1} which is invariant under orthogonal transformations and satisfies $\int_{S^{q-1}} \sigma(du') = 2\pi^{q/2}/\Gamma(q/2)$ (see, for example, (1.19) in Atkinson and Han [55]). The number $n_{q,l}$ is known (e.g., (2.10) in Atkinson and Han [55]) but will not be used here in a general form. For example, for $l \geq 1$,

$$n_{2,l} = 2 \quad (q = 2), \quad n_{3,l} = 2l + 1 \quad (q = 3),$$

whereas $n_{q,0} = 1$ for all $q \geq 1$ with $Y_{0,1}(u') \equiv (\Gamma(q/2)/2\pi^{q/2})^{1/2}$.

When $q = 2$, one can take $Y_{0,1}(u') = 1/(2\pi)^{1/2}$ and, for $l \geq 1$,

$$Y_{l,1}(u') = \frac{\cos(l\theta)}{\sqrt{\pi}}, \quad Y_{l,2}(u') = \frac{\sin(l\theta)}{\sqrt{\pi}}, \quad (9.6.3)$$

where $u' = (\cos(\theta), \sin(\theta))$, $\theta \in [0, 2\pi)$, is the polar representation of $u' \in S^1$ (e.g., Atkinson and Han [55], bottom of p. 22). When $q = 3$, one can take $Y_{0,1}(u') \equiv (\Gamma(3/2)/2\pi^{3/2})^{1/2}$ and, for $l \geq 1$, the $2l + 1$ spherical harmonics as (e.g., Andrews et al. [35], p. 457, after correcting several typos)

$$A_0 P_l^0(\cos \theta), \quad A_m \cos(m\phi) P_l^m(\cos \theta), \quad A_m \sin(m\phi) P_l^m(\cos \theta), \quad m = 1, \dots, l, \quad (9.6.4)$$

where the normalizations are

$$A_0 = \sqrt{\frac{2l+1}{4\pi}}, \quad A_m = \sqrt{\frac{(l-m)!(2l+1)}{(l+m)!2\pi}},$$

$u' = (\cos(\theta) \sin(\phi), \sin(\theta) \sin(\phi), \cos(\phi))$, $\theta \in [0, 2\pi)$, $\phi \in [0, \pi)$, is the spherical representation of $u' \in S^2$, and $P_l^m(x)$ are the associated Legendre polynomials (e.g., (9.6.7) in Andrews et al. [35]). The functions in (9.6.4) are often also labeled as Y_l^0 , Y_l^m , Y_l^{-m} , $m = 1, \dots, l$.

Suppose that $b_2 \in C(S^{q-1})$ (with b_2 appearing in (9.6.1)) and consider its expansion in spherical harmonics

$$b_2(u') = \sum_{l,m} a(l,m) Y_{l,m}(u'), \quad u' \in S^{q-1}, \quad (9.6.5)$$

where the convergence is uniform in $C(S^{q-1})$ (e.g., Atkinson and Han [55]). If $b_2 \in C^\infty(S^{q-1})$, as assumed in Wainger [1993], then the coefficients $a(l,m)$ have an exponential decay in l .

Suppose that $b_2 \in C^\infty(S^{q-1})$ and that L_2 is infinitely differentiable on $[0, \infty)$ such that $h_0(t) := L_2(t)$ and $h_j(t) := t h'_{j-1}(t)$, $j = 1, 2, \dots$, belong to the Zygmund class.¹² Under

¹² A (positive) function f belongs to the Zygmund class if, for every $\delta > 0$, $t^\delta f(t)$ is ultimately increasing and $t^{-\delta} f(t)$ is ultimately decreasing. The Zygmund class is known to coincide with the so-called normalized slowly varying functions (Bingham et al. [157], p. 24).

these assumptions, Wainger [1993], Theorems 6 and 7, showed that condition II-s-isot implies condition IV-s-isot with

$$L_4(t) = L_2\left(\frac{1}{t}\right) \quad (9.6.6)$$

and

$$b_4(u') = 2^{(1-2d)q} \pi^{-q} \sum_{l,m} (-i)^l a(l, m) \frac{\Gamma(\frac{1}{2}(l - (2d - 2)q))}{\Gamma(\frac{1}{2}(l + (2d - 1)q))} Y_{l,m}(u'), \quad (9.6.7)$$

where the coefficients $a(l, m)$ appear in (9.6.5).

Example 9.6.2 (*Relation between functions b_2 and b_4*) If $b_2(u') \equiv c_2$ (constant), then $a(0, 1)$ is the only nonzero coefficient in (9.6.5) and the relation (9.6.7) reads

$$b_4(u') = 2^{(1-2d)q} \pi^{-q} \frac{\Gamma((1-d)q)}{\Gamma(\frac{q}{2}(2d-1))} c_2.$$

Example 9.6.3 (*Relation between functions b_2 and b_4*) Let $q = 3$,

$$u' = (\cos(\theta) \sin(\phi), \sin(\theta) \sin(\phi), \cos(\phi)), \quad \theta \in [0, 2\pi), \quad \phi \in [0, \pi),$$

be the spherical representation of $u' \in S^2$ and $b_2(u') \equiv c_2(4 + 3\cos^2\phi)$. By using the notation following (9.6.4), one has $Y_0^0(u') \equiv \frac{1}{2\sqrt{5}}$ and $Y_2^0(u') = \frac{1}{4}\sqrt{\frac{5}{\pi}}(3\cos^2\phi - 1)$, so that

$$b_2(u') = a_0 Y_0^0(u') + a_2 Y_2^0(u')$$

with $a_0 = c_2 2\sqrt{5} \cdot 5$, $a_2 = c_2 4\sqrt{\frac{\pi}{5}}$, and hence the relation (9.6.7) reads

$$\begin{aligned} b_4(u') &= 2^{(1-2d)3} \pi^{-3} \left(a_0 \frac{\Gamma(3-3d)}{\Gamma(3d-3/2)} Y_0^0(u') + a_2 \frac{\Gamma(4-3d)}{\Gamma(3d-1/2)} Y_2^0(u') \right) \\ &= 2^{(1-2d)3} \pi^{-3} \frac{\Gamma(3-3d)}{\Gamma(3d-3/2)} \left(a_0 Y_0^0(u') + a_2 \frac{1-d}{d-1/2} Y_2^0(u') \right), \end{aligned}$$

by using the identity $\Gamma(1+x) = x\Gamma(x)$.

Conditions II-s-isot and IV-s-isot lead to the definition of isotropic LRD fields. *Anisotropic LRD* generally refers to random fields with slowly decaying autocovariance functions or spectral densities with a pole at 0 which do not have the form (9.6.1) or (9.6.2), respectively. We provide next several conditions in the spectral domain which have been taken as special cases of anisotropic LRD. The suffix “-anisot” in the condition refers to “anisotropic.”

Condition IV-s-anisot The random field $X = \{X_n\}_{n \in \mathbb{Z}^q}$ has a spectral density satisfying one of the following conditions:

- For $d_1, \dots, d_q \in (0, 1/2)$ and $C > 0$

$$f(\lambda) \sim C |\lambda_1|^{-2d_1} \dots |\lambda_q|^{-2d_q}, \quad \text{as } \lambda \rightarrow 0. \quad (9.6.8)$$

- For $d \in (0, 1/2)$, $C > 0$ and $a \in \mathbb{R}^q$,

$$f(\lambda) \sim C|\langle a, \lambda \rangle|^{-2d}, \quad \text{as } \lambda \rightarrow 0. \quad (9.6.9)$$

- For an isotropic spectral density f_0 satisfying (9.6.2) and a nondegenerate matrix A ,

$$f(\lambda) = f_0(\|Aw\|_2). \quad (9.6.10)$$

- For a function $d : S^{q-1} \rightarrow (0, 1/2)$ and $C > 0$,

$$f(\lambda) \sim C\|\lambda\|_2^{-2d(\lambda')q}, \quad \text{as } \lambda \rightarrow 0, \quad (9.6.11)$$

where $\lambda' = \lambda/\|\lambda\|_2 \in S^{q-1}$.

- For $q = 2$, $\beta_1, \beta_2 > 0$, $\beta_1\beta_2 < \beta_1 + \beta_2$, $c > 0$ and $C > 0$,

$$f(\lambda_1, \lambda_2) \sim \frac{C}{\|\lambda_1\|^2 + \|\lambda_2\|^{2\beta_2/\beta_1} \|\lambda_1\|^{1/2}}, \quad \text{as } (\lambda_1, \lambda_2) \rightarrow 0. \quad (9.6.12)$$

The conditions on the various parameters above are to ensure that the spectral density f is integrable around $\lambda = 0$ (Exercise 9.12).

Random fields satisfying (9.6.8) were considered in Lavancier [595], Beran, Ghosh, and Schell [125], Boissy, Bhattacharyya, Li, and Richardson [168], Guo, Lim, and Meerschaert [440]. The condition (9.6.8) states that there is potentially a different LRD parameter d_j in each direction λ_j . The condition (9.6.9) was considered in Lavancier [595, 596] and represents the divergence of the spectral density along the direction $\langle a, \lambda \rangle = 0$. The condition (9.6.10) can be found in Leonenko and Olenko [610], and the condition (9.6.11) in Bonami and Estrade [173], Biermé and Richard [150].

Random fields satisfying (9.6.12) were considered in Puplinskaitė and Surgailis [835]. The latter work also established interesting scaling properties of aggregate sums of random fields satisfying (9.6.12) or (9.6.8). More specifically, given a random field $X = \{X_n\}_{n \in \mathbb{Z}^2}$, assume that for any $\gamma > 0$, there is a nontrivial random field $V_\gamma = \{V_\gamma(x)\}_{x \in \mathbb{R}^2}$ and a normalization $A_N(\gamma) \rightarrow \infty$ such that

$$A_N(\gamma)^{-1} \sum_{n \in K_N^\gamma(x)} X_n \xrightarrow{fdd} V_\gamma(x), \quad x \in \mathbb{R}^2, \quad N \rightarrow \infty, \quad (9.6.13)$$

where $K_N^\gamma(x) = K_N^\gamma(x_1, x_2) = \{n = (n_1, n_2) \in \mathbb{Z}^2 : 1 \leq n_1 \leq Nx_1, 1 \leq n_2 \leq N^\gamma x_2\}$. Puplinskaitė and Surgailis [835] say that X exhibits *scaling transition* if there is $\gamma_0 > 0$ such that

$$V_\gamma \stackrel{d}{=} V_+, \quad \gamma > \gamma_0, \quad V_\gamma \stackrel{d}{=} V_-, \quad \gamma < \gamma_0, \quad V_+ \not\stackrel{d}{=} aV_- \quad (\text{all } a > 0), \quad (9.6.14)$$

that is, the distributions of the fields V_γ do not depend on $\gamma > \gamma_0$ and $\gamma < \gamma_0$, and are different up to a multiplicative constant. Under additional suitable assumptions, Puplinskaitė and Surgailis [835] showed that the random fields satisfying (9.6.12) or (9.6.8) exhibit scaling transition.

9.6.2 Examples

We discuss here several examples of classes of spatial LRD series.

Example 9.6.4 (*Spatial LRD fields with explicit autocovariances*) In several instances (e.g., modeling in geostatistics, or simulations), it is desirable to have an explicit form for the autocovariance function of a second-order stationary random field. For example, for what functions $\gamma : [0, \infty) \rightarrow \mathbb{R}$, does an isotropic function $\gamma(\|u\|_2)$ become an autocovariance function? This and related questions have been well-studied in the case of general stationary (not necessarily LRD) random fields $\{X_u\}$ indexed by $u \in \mathbb{R}^q$. (Stationary random fields $\{Y_n\}$ with explicit autocovariances and indexed by $n \in \mathbb{Z}^q$ can then be obtained by setting $Y_n = X_{n\Delta}$ for some $\Delta > 0$.) The following is one classical result in this direction.

It is known (e.g., Schoenberg [890], Yaglom [1020]) that $\gamma(\|u\|_2)$ is the autocovariance function of a stationary random field $\{X_u\}_{u \in \mathbb{R}^q}$ if and only if there is a finite measure $G(d\lambda)$ on $(0, \infty)$ such that

$$\gamma(r) = \int_0^\infty Y_q(\lambda r) G(d\lambda), \quad (9.6.15)$$

where $Y_q(z)$ is the spherical Bessel function defined as

$$Y_1(z) = \cos z, \\ Y_q(z) = 2^{(q-2)/2} \Gamma\left(\frac{q}{2}\right) J_{(q-2)/2}(z) z^{(q-2)/2}, \quad z \geq 0, \quad q \geq 2,$$

and for $\nu > -1/2$,

$$J_\nu(z) = \sum_{m=0}^{\infty} (-1)^m \left(\frac{z}{2}\right)^{2m+\nu} \frac{1}{m\Gamma(m+\nu+1)}, \quad z > 0,$$

is the Bessel function of the first kind of order ν . A function $\gamma(\|u\|_2)$ is the autocovariance function of some $\{X_u\}_{u \in \mathbb{R}^q}$ for every q if and only if

$$\gamma(r) = \int_0^\infty e^{-ar^2} F(da), \quad (9.6.16)$$

for some measure $F(da)$ (that is, when $\gamma(r)$ is a normal scale mixture). A function $f(\|u\|_2^2)$ is the autocovariance function for every q if and only if f is completely monotone.¹³

The Cauchy family

$$\gamma(r) = \frac{1}{(1+r^\alpha)^{\beta/\alpha}}, \quad \alpha \in (0, 2], \quad \beta > 0, \quad (9.6.17)$$

is a well-known example of functions satisfying (9.6.16) (e.g., Schlather [888]). Setting $(2d-1)q = -\beta$, the corresponding random field is isotropic LRD when $\beta \in (0, q)$. When $\alpha = 2$, $\gamma(r) = f(r^2)$ with $f(r) = (1+r)^{-\beta/2}$, which can be checked to be completely monotone (Exercise 9.14).

Example 9.6.5 (*Anisotropic spatial FARIMA models*) Univariate FARIMA models considered in Section 2.4.2 can naturally be extended to the context of random fields in the following way. Consider the dimension $q = 2$ only (not to be confused with q in

¹³ Recall that a function $f : [0, \infty) \rightarrow \mathbb{R}$ is completely monotone if it satisfies $(-1)^n f^{(n)}(t) \geq 0$ for any n and $t > 0$, where $f^{(n)}$ denotes the n th derivative of f .

FARIMA(p, d, q) so that a random field is $X = \{X_{n_1, n_2}\}_{n_1, n_2 \in \mathbb{Z}}$. Consider the backshift operators B_1 and B_2 satisfying

$$B_1^k Y_{n_1, n_2} = Y_{n_1-k, n_2}, \quad B_2^k Y_{n_1, n_2} = Y_{n_1, n_2-k}, \quad k \in \mathbb{Z}, \quad (9.6.18)$$

that is, the backshift operators acting on the individual coordinates n_1 and n_2 . Denote $I = B_1^0 = B_2^0$.

An *anisotropic spatial FARIMA*((p_1, p_2), (d_1, d_2), (q_1, q_2)) random field $X = \{X_{n_1, n_2}\}_{n_1, n_2 \in \mathbb{Z}}$ is defined as

$$X_{n_1, n_2} = (I - B_1)^{-d_1} (I - B_2)^{-d_2} \Psi_1(B_1) \Psi_2(B_2) Z_{n_1, n_2}, \quad (9.6.19)$$

where $\{Z_{n_1, n_2}\}_{n_1, n_2 \in \mathbb{Z}}$ is a spatial white noise satisfying $\mathbb{E}Z_{n_1, n_2} = 0$ and $\mathbb{E}Z_{n_1, n_2} Z_{m_1, m_2} = \sigma_Z^2 \delta_{n_1, m_1} \delta_{n_2, m_2}$,

$$\Psi_1(B_1) = \varphi_1^{-1}(B_1) \theta_1(B_1), \quad \Psi_2(B_2) = \varphi_2^{-1}(B_2) \theta_2(B_2),$$

and, for $j = 1, 2$,

$$\varphi_j(z) = 1 - \varphi_{j,1}z - \cdots - \varphi_{j,p_j}z^{p_j}, \quad \theta_j(z) = 1 + \theta_{j,1}z + \cdots + \theta_{j,q_j}z^{q_j}.$$

For the convergence of the field (9.6.19), it is assumed that $d_1, d_2 < 1/2$ and that the polynomials $\varphi_1(z)$, $\varphi_2(z)$ do not have unit roots on the circle.

The spectral density of the field (9.6.19) is

$$f(\lambda_1, \lambda_2) = \frac{\sigma_Z^2}{4\pi^2} |1 - e^{-i\lambda_1}|^{-2d_1} |1 - e^{-i\lambda_2}|^{-2d_2} \left| \frac{\theta_1(e^{-i\lambda_1})}{\varphi_1(e^{-i\lambda_1})} \right|^2 \left| \frac{\theta_2(e^{-i\lambda_2})}{\varphi_2(e^{-i\lambda_2})} \right|^2. \quad (9.6.20)$$

When $d_1, d_2 \in (0, 1/2)$, it satisfies the condition (9.6.8) in the definition of anisotropic LRD. A different anisotropic FARIMA model is constructed in Example 9.6.7 below.

Example 9.6.6 (Isotropic spatial FARIMA models) The spatial FARIMA models considered in Example 9.6.5 are anisotropic as noted above. This is the result of the choice of the fractional integration operator $(I - B_1)^{-d_1} (I - B_2)^{-d_2}$ used in the definition (9.6.19). An isotropic FARIMA model can be obtained similarly by using a different fractional integration operator. More specifically, consider first the difference operator

$$\Delta(B_1, B_2) = \frac{1}{4} (I - B_1 + I - B_2 + I - B_1^{-1} + I - B_2^{-1}), \quad (9.6.21)$$

where $I = B_1^0 = B_2^0$, B_1, B_2 and their powers are defined in (9.6.18), so that

$$\begin{aligned} \Delta(B_1, B_2) Y_{n_1, n_2} &= \frac{1}{4} \sum_{|k_1|+|k_2|=1} (Y_{n_1, n_2} - Y_{n_1+k_1, n_2+k_2}) \\ &= \frac{1}{4} (Y_{n_1, n_2} - Y_{n_1-1, n_2} + Y_{n_1, n_2} - Y_{n_1, n_2-1} + Y_{n_1, n_2} - Y_{n_1+1, n_2} + Y_{n_1, n_2} - Y_{n_1, n_2+1}). \end{aligned}$$

An *isotropic spatial FARIMA*((p_1, p_2), d , (q_1, q_2)) random field $X = \{X_{n_1, n_2}\}_{n_1, n_2 \in \mathbb{Z}}$ can be defined as

$$X_{n_1, n_2} = \Delta(B_1, B_2)^{-d} \Psi_1(B_1) \Psi_2(B_2) Z_{n_1, n_2}, \quad (9.6.22)$$

where $d < 1/2$ and $\Psi_1(B_1)$, $\Psi_2(B_2)$, Z_{n_1, n_2} are as in (9.6.19). Since

$$|\Delta(e^{-i\lambda_1}, e^{-i\lambda_2})| = \frac{1}{2} |1 - \cos(\lambda_1) + 1 - \cos(\lambda_2)|,$$

an isotropic spatial FARIMA has spectral density

$$f(\lambda_1, \lambda_2) = \frac{\sigma_Z^2}{4\pi^2} 2^{2d} |1 - \cos(\lambda_1) + 1 - \cos(\lambda_2)|^{-2d} \left| \frac{\theta_1(e^{-i\lambda_1})}{\varphi_1(e^{-i\lambda_1})} \right|^2 \left| \frac{\theta_2(e^{-i\lambda_2})}{\varphi_2(e^{-i\lambda_2})} \right|^2. \quad (9.6.23)$$

When $d \in (0, 1/2)$, since $1 - \cos(\lambda) \sim \lambda^2/2$ as $\lambda \rightarrow 0$, it satisfies the condition (9.6.2) in the definition of isotropic LRD with the function b_4 being constant.

Example 9.6.7 (*Anisotropic spatial FARIMA models, cont'd*) Other anisotropic FARIMA models can be constructed as in Example 9.6.6 above by replacing the difference operator $\Delta(B_1, B_2)$ in (9.6.21) by its “anisotropic” counterpart. For example, one choice is to take

$$\Delta_\theta(B_1, B_2) = I - \theta B_1 - \frac{1 - \theta}{2} B_1(B_2 + B_2^{-1}) = I - B_1 + (1 - \theta) B_1 \left(I - \frac{B_2 + B_2^{-1}}{2} \right), \quad (9.6.24)$$

where $\theta \in (0, 1)$. Note that

$$\begin{aligned} |\Delta_\theta(e^{-i\lambda_1}, e^{-i\lambda_2})| &= |1 - e^{-i\lambda_1} + (1 - \theta)e^{-i\lambda_1}(1 - \cos \lambda_2)| \\ &= |e^{i\lambda_1} - 1 + (1 - \theta)(1 - \cos \lambda_2)| \\ &\sim |i\lambda_1 + (1 - \theta)\lambda_2^2/2| = (\lambda_1^2 + (1 - \theta)^2\lambda_2^4/4)^{1/2}, \end{aligned}$$

as $(\lambda_1, \lambda_2) \rightarrow 0$. Then, for example, the field

$$X_{n_1, n_2} = \Delta_\theta(B_1, B_2)^{-d} Z_{n_1, n_2}, \quad (9.6.25)$$

where $d < 3/2$ (see below) and Z_{n_1, n_2} is as in (9.6.19), has the spectral density

$$f(\lambda_1, \lambda_2) = \frac{\sigma_Z^2}{4\pi^2} |e^{i\lambda_1} - 1 + (1 - \theta)(1 - \cos \lambda_2)|^{-d} \sim \frac{\sigma_Z^2}{4\pi^2} (\lambda_1^2 + (1 - \theta)^2\lambda_2^4/4)^{-d}, \quad (9.6.26)$$

as $(\lambda_1, \lambda_2) \rightarrow 0$. When $d \in (0, 3/2)$, it satisfies the condition (9.6.12) in the definition of anisotropic LRD with $\beta_1 = 2d$, $\beta_2 = d$ (so that $\beta_1\beta_2 < \beta_1 + \beta_2$ when $d \in (0, 3/2)$), and can thus also be called an anisotropic FARIMA field.

9.7 Exercises

The symbols * and ** next to some exercises are explained in Section 2.12.

Exercise 9.1 Any matrix $A \in \mathbb{R}^{p \times p}$ has a Jordan decomposition $A = PJP^{-1}$, where $J = \text{diag}(J_{\lambda_1}, \dots, J_{\lambda_N})$ consists of Jordan blocks J_λ of size n_λ given by (9.1.2). Show that

$$z^{J_\lambda} = \begin{pmatrix} z^\lambda & 0 & 0 & \dots & 0 \\ (\log z)z^\lambda & z^\lambda & 0 & \dots & 0 \\ \frac{(\log z)^2}{2!}z^\lambda & (\log z)z^\lambda & z^\lambda & \dots & 0 \\ \vdots & \vdots & \ddots & \ddots & 0 \\ \frac{(\log z)^{n_\lambda-1}}{(n_\lambda-1)!}z^\lambda & \frac{(\log z)^{n_\lambda-2}}{(n_\lambda-2)!}z^\lambda & \dots & (\log z)z^\lambda & z^\lambda \end{pmatrix}. \quad (9.7.1)$$

Hint: Use $z^{J_\lambda} = e^{(\log z)J_\lambda} = \sum_{k=0}^{\infty} (\log z)^k J_\lambda^k / k!$ and evaluate J_λ^k .

Exercise 9.2 Show that the definitions of e^A , $A \in \mathbb{R}^{p \times p}$, in (9.2.3) and as a primary matrix function through Definition 9.3.4 are equivalent.

Exercise 9.3 Let F_X be the spectral distribution matrix function of a stationary vector-valued process $X = \{X(t)\}_{t \in \mathbb{R}}$. Show that F_X has at most a countable number of discontinuity points. (This result was used in the proof of Theorem 9.3.2.)

Exercise 9.4 Verify the relations (9.4.34), (9.4.35)–(9.4.36) and (9.4.37).

Exercise 9.5* Show that the process on the right-hand side of (9.3.22) is well-defined under the condition of Theorem 9.3.8, (i); that is, for any $t \in \mathbb{R}$,

$$\int_{\mathbb{R}} \left| (t-u)_{\pm}^D - (-u)_{\pm}^D \right|^2 du < \infty.$$

Hint: It is enough to show this for a Jordan block $D = J_\lambda$ in (9.1.2), in which case $v_{\pm}^D$ can be expressed as in (9.7.1).

Exercise 9.6 Show that (9.3.30) can be written as (9.3.31). *Hint:* Use the following identities from Gradshteyn and Ryzhik [421]: $\int_{\mathbb{R}} 1_{\{x < 0\}} \frac{\sin(ax)}{x} dx = \int_{\mathbb{R}} 1_{\{x > 0\}} \frac{\sin(ax)}{x} dx = \frac{\pi}{2} \text{sign}(a)$ (p. 423), $\int_{\mathbb{R}} 1_{\{x > 0\}} \frac{\cos(ax) - \cos(bx)}{x} dx = \log \frac{|b|}{|a|}$ (p. 447), $\int_0^\infty \log(x) \sin(ax) \frac{dx}{x} = -\frac{\pi}{2}(C + \log(a))$, $a > 0$ (p. 594), and $\int_0^\infty \log(x)(\cos(ax) - \cos(bx)) \frac{dx}{x} = \log\left(\frac{a}{b}\right) \left(C + \frac{1}{2} \log(ab)\right)$, $a, b > 0$ (p. 594), where C is Euler's constant.

Exercise 9.7 For $p = 2$, $\Gamma = I$ and the exponent

$$H = \begin{pmatrix} 1/2 & 0 \\ -1 & 1/2 \end{pmatrix},$$

show that there does not exist a time reversible vector OFBM B_H such that $\mathbb{E}B_H(1)B_H(1)^* = \Gamma$. See Remark 9.3.15 for a related discussion. *Hint:* From Corollary 9.3.12,

$$\mathbb{E}B_H(1)B_H(1)^* = \int_{\mathbb{R}} \left| \frac{e^{ix} - 1}{ix} \right|^2 |x|^{-D} AA^* |x|^{-D^*} dx,$$

where $D = H - (1/2)I$. Show that $\mathbb{E}B_H(1)B_H(1)^* = \Gamma = I$ is equivalent to

$$\begin{pmatrix} s_{11}r_0(d) & -s_{11}r_1(d) + s_{12}r_0(d) \\ -s_{11}r_1(d) + s_{12}r_0(d) & s_{11}r_2(d) - 2s_{12}r_1(d) + s_{22}r_0(d) \end{pmatrix} = I,$$

where $AA^* =: (s_{jk})_{j,k=1,2}$,

$$r_k(d) = 4 \int_0^\infty \frac{1 - \cos(x)}{x^2} (\log(x))^k x^{-2d} dx, \quad k = 0, 1, 2.$$

Now, solve for s_{11} , s_{12} and s_{22} when $d = 0$, and show that $s_{22} < 0$, contradicting the positive semidefiniteness of AA^* .

Exercise 9.8* When $p = 2$, show that the condition in Proposition 9.3.22 can be reduced to $C_{12} \leq 1$, where

$$C_{12} = \frac{\Gamma(H_1 + H_2 + 1)^2}{\Gamma(2H_1 + 1)\Gamma(2H_2 + 1)} \frac{\rho_{12}^2 \sin^2((H_1 + H_2)\frac{\pi}{2}) + \eta_{12}^2 \cos^2((H_1 + H_2)\frac{\pi}{2})}{\sin(H_1\pi) \sin(H_2\pi)}.$$

The constant C_{12} can be interpreted as the coherence function of the continuous-time stationary increment process $X_{H,\delta}(t) = B_H(t) - B_H(t - \delta)$, $t \in \mathbb{R}$; that is,

$$C_{12} = \frac{|f_{12}(\lambda)|^2}{f_{11}(\lambda)f_{22}(\lambda)},$$

where $(f_{jk}(\lambda))$ is the spectral density matrix of the stationary process $\{X_{H,\delta}(t)\}$. (The ratio above does not depend on λ or δ .) *Hint:* See Amblard, Coeurjolly, Lavancier, and Philippe [26], Proposition 9.

Exercise 9.9* Let $\{Z_n\}_{n \in \mathbb{Z}}$ be an $\mathbb{R}^p$ -valued white noise, satisfying $\mathbb{E}Z_n = 0$ and $\mathbb{E}Z_n Z'_n = I$. Let also $\{\Psi_m = (\psi_{jk,m})_{j,k=1,\dots,p}\}_{m \in \mathbb{Z}} \subset \mathbb{R}^{p \times p}$ be such that $\psi_{jk,m} = L_{jk}(m)|m|^{d_j-1}$, $m \in \mathbb{Z}$, where $d_j \in (0, 1/2)$ and $L(m) = (L_{jk}(m))_{j,k=1,\dots,p}$ is an $\mathbb{R}^{p \times p}$ -valued function satisfying $L(m) \sim A^+$, as $m \rightarrow \infty$, and $L(m) \sim A^-$, as $m \rightarrow -\infty$, for some $A^+ = (\alpha_{jk}^+)_{j,k=1,\dots,p} \in \mathbb{R}^{q \times q}$, $A^- = (\alpha_{jk}^-)_{j,k=1,\dots,p} \in \mathbb{R}^{q \times q}$. Show that the time series $\{X_n\}$ given by a two-sided linear representation

$$X_n = \sum_{m=-\infty}^{\infty} \Psi_m Z_{n-m},$$

is LRD in the sense of (9.4.3) in condition II-v with

$$r_{jk} = \frac{\Gamma(d_j)\Gamma(d_k)}{\Gamma(d_j + d_k)} \left(c_{jk}^1 \frac{\sin(d_k\pi)}{\sin((d_j + d_k)\pi)} + c_{jk}^2 + c_{jk}^3 \frac{\sin(d_j\pi)}{\sin((d_j + d_k)\pi)} \right),$$

where

$$c_{jk}^1 = \sum_{t=1}^p \alpha_{jt}^- \alpha_{kt}^- = (A^-(A^-)^*)_{jk},$$

$$c_{jk}^2 = \sum_{t=1}^p \alpha_{jt}^+ \alpha_{kt}^- = (A^+ (A^-)^*)_{jk},$$

$$c_{jk}^3 = \sum_{t=1}^p \alpha_{jt}^+ \alpha_{kt}^+ = (A^+ (A^+)^*)_{jk}.$$

Hint: This is Proposition 3.1 in Kechagias and Pipiras [551], noting that a different convention (9.1.9) is used therein for the autocovariance function.

Exercise 9.10 Show that the OFBF is (9.5.58) is well-defined; that is,

$$\int_{\mathbb{R}^q} \left| (e^{i\langle x, t \rangle} - 1) \prod_{j=1}^q |x_j|^{-H_j - \frac{1}{2}} \right|^2 dx < \infty,$$

when $0 < H_j < 1$.

Exercise 9.11 Show that the function g_0 can indeed be written as in (9.5.38) where $\tilde{g}$ is $-E^*/(H + 3\text{tr}(E^*)/2)$ -homogeneous.

Exercise 9.12 Verify that the conditions on the parameters found around (9.6.8)–(9.6.12) ensure that the corresponding spectral densities are integrable around $\lambda = 0$.

Exercise 9.13 Prove the statement in Example 9.5.15. *Hint:* See Corollary 2.12 in Biermé et al. [151].

Exercise 9.14 Check that the function $f(r) = (1 + r)^{-\beta/2}$, $\beta > 0$, appearing in Example 9.6.4 is completely monotone.

Exercise 9.15** Formulate a definition of general spatial anisotropic LRD, even if it may not be necessarily practical as indicated in the beginning of Section 9.6.

9.8 Bibliographical Notes

Section 9.2: Operator self-similarity for stochastic processes finds its origins in a related notion, that of operator stability for probability distributions. A probability distribution μ on $\mathbb{R}^p$ is called *operator stable* if, for any $t > 0$,

$$\mu^t = t^B \mu * \delta(b(t)), \quad (9.8.1)$$

where B is a matrix, $b : (0, \infty) \rightarrow \mathbb{R}^p$ is a function, and $*$ denotes the convolution.¹⁴ Operator stable distributions arise as limit laws of operator normalized and centered sums of i.i.d. vectors. See, for example, the excellent monographs on the subject by Jurek and Mason [542], and Meerschaert and Scheffler [710].

¹⁴ If ξ is a random vector with the probability distribution μ , then the right-hand side of (9.8.1) is the probability distribution of the random variable $t^B \xi + b(t)$. The left-hand side μ^t is the probability distribution whose characteristic function is $(\hat{\mu})^t$, where $\hat{\mu}$ is the characteristic function of μ .

On the one hand, the theories of operator stable distributions and operator self-similar processes bear lots of similarities (e.g., Mason [695]). On the other hand, first examples of operator self-similar processes were constructed as having independent increments whose distributions are operator stable (e.g., Hudson and Mason [499], Theorem 7). The latter is analogous to the connection between stable Lévy motions (which are self-similar) and stable distributions in one dimension.

In the vector setting $q = 1$, $p \geq 1$ (see (9.2.1)), the operator self-similarity was first considered in depth by Laha and Rohatgi [587], and Hudson and Mason [499]. In the multi-parameter setting $p = 1$, $q \geq 1$, the mathematical foundations of the operator self-similarity were laid in Biermé et al. [151].

For more recent fundamental work in the vector setting, see Meerschaert and Scheffler [709], Becker-Kern and Pap [111], Cohen, Meerschaert, and Rosiński [255]. This was preceded by influential work of Maejima and Mason [667], Maejima [666]. Operator self-similarity is also discussed in Meerschaert and Scheffler [710] (Section 11), Embrechts and Maejima [350] (Chapter 9).

Section 9.3: The material of this section is largely based on Didier and Pipiras [307, 308]. Operator fractional Brownian motions were also considered in Bahadoran, Benassi, and Debicki [72], Pitt [824], with the latter early paper apparently little known in the area (while published before Laha and Rohatgi [587], and Hudson and Mason [499]). Sample path properties are studied in Mason and Xiao [696]. Vector fractional Brownian motions were also considered in Lavancier et al. [597], Amblard and Coeurjolly [25], Amblard et al. [26]. Convergence in law to operator fractional Brownian motion is studied in Dai [279]. Finally, for wavelet-based estimation of operator fractional Brownian motions, see Abry and Didier [5], Coeurjolly, Amblard, and Achard [252].

Section 9.4: Section 9.4.1 follows part of the work of Kechagias and Pipiras [551]. See also Kechagias and Pipiras [550]. One interesting problem raised in Remark 9.4.5 and addressed in Kechagias and Pipiras [551] is the construction of vector LRD series with one-sided linear representations (9.4.32) which have general phases. It is shown in Kechagias and Pipiras [551] that general phases could be obtained by taking the entries in the coefficient matrices ψ_m as linear combinations of what the authors called *trigonometric power-law coefficients*

$$m^{d_0-1} \cos(2\pi m^a), \quad m^{d_0-1} \sin(2\pi m^a),$$

for suitable parameter values d_0 and a .

In the bivariate context $p = 2$, vector LRD time series in the sense 1 of Definition 9.4.1 with a general phase, and their estimation using an extension of the local Whittle method (see Section 10.6.1) were considered by Robinson [855]. A number of other authors considered vector LRD models and their estimation in the special case of zero phases $\phi_{jk} = 0$ (e.g., Lobato and Robinson [643], Lobato [642], Velasco [987], Robinson and Marinucci [857], Christensen and Nielsen [238], Nielsen [747, 748]). The case of the phases (9.4.30), as in one-sided FARIMA models, is another special case and was considered in, for example, Lobato [641], Robinson and Yajima [858], Shimotsu [905], Nielsen and Frederiksen [750], Nielsen [746]. The case of general ϕ is considered in Robinson [855] as indicated above, and without referring to ϕ explicitly in Robinson [852].

A more systematic study of fractional cointegration (Section 9.4.4) started with Marinucci and Robinson [691], Robinson and Yajima [858], though the notion itself has originated much earlier (e.g., Sowell [917], Dueker and Startz [331], Cheung and Lai [232], Baillie and Bollerslev [86]). This was followed by several fundamental papers by Chen and Hurvich [222, 223], and many other studies. To name but a few recent works, see Hualde and Robinson [497], Nielsen [749], Johansen and Nielsen [534, 535], Shimotsu [906], Łasak and Velasco [594]. Polynomial fractional cointegration is considered in Avarucci and Marinucci [57].

Multivariate LRD models also appear in Achard, Bassett, Meyer-Lindenberg, and Bullmore [14], Wendt, Scherrer, Abry, and Achard [1003] in the context of the so-called *fractal connectivity*. According to (9.4.6) in condition IV-v of the definition of vector LRD, the cross-spectral density $f_{jk}(\lambda)$ between the components j and k of the series satisfies

$$f_{jk}(\lambda) \sim G_{jk} \lambda^{-(d_j+d_k)},$$

as $\lambda \rightarrow 0$. Fractal connectivity postulates a more refined model where

$$f_{jk}(\lambda) \sim C_{jk} \lambda^{-d_{jk}}$$

with $d_{jk} \leq d_j + d_k$. The component j and k series are called *fractally connected* if $d_{jk} = d_j + d_k$ (and $C_{jk} \neq 0$). See also Sela and Hurvich [897], Kristoufek [579].

Section 9.5: The material here is largely based on Biermé et al. [151], and Baek, Didier, and Pipiras [70]. See also Clausel and Vedel [247].

Special cases of multiparameter OFBFs are found in the monographs by Leonenko [612], Cohen and Istas [253]. In particular, fractional Brownian sheet of Example 9.5.20 and related processes were studied in Ayache, Leger, and Pontier [63], Wang [995], Wang, Yan, and Yu [997], Wang [996], Makogin and Mishura [674]. For extensions, see Herbin and Merzbach [471], Clausel [246], Biermé, Lacaux, and Scheffler [152], Polisano, Clausel, Perrier, and Condat [827]. Interesting applications are considered in Benson, Meerschaert, Baeumer, and Scheffler [119], Roux, Clausel, Vedel, Jaffard, and Abry [871]. Estimation is considered by Lim, Meerschaert, and Scheffler [633]. Sample path properties were studied in, e.g., Kamont [548], Mason and Xiao [696], Xiao [1017], Li, Wang, and Xiao [626].

A number of other issues for fractional Gaussian fields are discussed in a survey paper by Lodhia, Sheffield, Sun, and Watson [644].

Section 9.6: The definition of isotropic multiparameter LRD follows Lavancier [595], Leonenko [612], Leonenko and Olenko [610]. Estimation in isotropic LRD models is considered in Anh and Lunney [36], Ludeña and Lavielle [650], Frías, Alonso, Ruiz-Medina, and Angulo [375].

Several references for anisotropic multiparameter LRD were given in Section 9.4.1. Estimation in anisotropic LRD models is considered in Boissy et al. [168], Beran et al. [125], Guo et al. [440], Wang [994]. Invariance principles for anisotropic fields are considered in Lavancier [596], Puplinskaitė and Surgailis [835]. As noted around (9.6.13)–(9.6.14), the concept of scaling transition for random fields was considered by Puplinskaitė and Surgailis [835]. See also Pilipauskaitė and Surgailis [810], Puplinskaitė and Surgailis [836].

Construction of multiparameter stationary and other fields with explicit covariance functions (cf. Example 9.6.4) is a well-studied problem in spatial statistics (e.g., Chilès and

Delfiner [237], Sherman [904], Schlather [889]). With some focus on LRD, this problem was considered in, for example, Gneiting and Schlather [412].

Other notes: The vector and multiparameter settings were considered separately in Chapter 9. In the joint setting (both $p \geq 1$ and $q \geq 1$), the operator self-similar random fields and the analogues of fractional Brownian motion are studied in Li and Xiao [625], Baek et al. [70], Didier et al. [309], Didier, Meerschaert, and Pipiras [310]. See also Ma [658], Tafti and Unser [942].

Another interesting class of random fields are spatio-temporal models $\{X(t, u)\}$, where t is interpreted as a time variable and u as a spatial variable, e.g., in $\mathbb{R}^2$ (X itself takes values in $\mathbb{R}$). LRD spatio-temporal models are considered in, for example, Frías, Ruiz-Medina, Alonso, and Angulo [374, 376].